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Pre-trained language models in medicine: A survey ☆

Xudong Luo [a,b,c,*], Zhiqi Deng [a,b,c], Binxia Yang [a,b,c], Michael Y. Luo [d]

[a] School of Computer Science and Engineering, Guangxi Normal University, Guilin 541004, China
[b] Guangxi Key Lab of Multi-source Information Mining, Guangxi Normal University, Guilin 541004, China
[c] Key Laboratory of Education Blockchain and Intelligent Technology, Ministry of Education, Guangxi Normal University, Guilin 541004, China
[d] Emmanuel College, Cambridge University, Cambridge, CB2 3AP, UK

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ABSTRACT

With the rapid progress in Natural Language Processing (NLP), Pre-trained Language Models (PLM) such as BERT, BioBERT, and ChatGPT have shown great potential in various medical NLP tasks. This paper surveys the cutting-edge achievements in applying PLMs to various medical NLP tasks. Specifically, we first brief PLMS and outline the research of PLMs in medicine. Next, we categorise and discuss the types of tasks in medical NLP, covering text summarisation, question-answering, machine translation, sentiment analysis, named entity recognition, information extraction, medical education, relation extraction, and text mining. For each type of task, we first provide an overview of the basic concepts, the main methodologies, the advantages of applying PLMs, the basic steps of applying PLMs application, the datasets for training and testing, and the metrics for task evaluation. Subsequently, a summary of recent important research findings is presented, analysing their motivations, strengths vs weaknesses, similarities vs differences, and discussing potential limitations. Also, we assess the quality and influence of the research reviewed in this paper by comparing the citation count of the papers reviewed and the reputation and impact of the conferences and journals where they are published. Through these indicators, we further identify the most concerned research topics currently. Finally, we look forward to future research directions, including enhancing models' reliability, explainability, and fairness, to promote the application of PLMs in clinical practice. In addition, this survey also collect some download links of some model codes and the relevant datasets, which are valuable references for researchers applying NLP techniques in medicine and medical professionals seeking to enhance their expertise and healthcare service through AI technology.

1. Introduction

In recent years, Pre-trained Language Models (PLMs) have become increasingly popular in studying and applying Natural Language Processing (NLP) [46]. Models such as BERT [47], ChatGPT [48], ClinicalBERT [49], and BioBERT [50] have demonstrated remarkable efficacy in domain-specific NLP tasks by learning generalised language representations from large-scale text corpora. PLMs are deep neural network models that are initially pre-trained (unsupervised) on large text corpora and then fine-tuned for downstream NLP tasks [51]. Some of the most renowned PLMs include BERT (Bidirectional Encoder Representations from Transformers), GPT [52] (Generative Pre-trained Transformer), and RoBERTa [53] (Robustly Optimised BERT Pre-training Approach).

Pre-training and transfer learning [54] are not new concepts in machine learning. However, recent PLMs are unprecedented in scale and effectiveness. For example, during pre-training, BERT introduced Masked Language Modelling (MLM), allowing bidirectional contextual modelling. GPT-3 [55] has demonstrated remarkable few-shot learning capabilities across various NLP tasks using massive pre-training data. More recently, multitask learning models like T5 [56] and mT5 [57] have been proposed.

The popularity of PLMs stems from several advantages. Firstly, they reduce training costs and data requirements for downstream tasks. The knowledge encoded in pre-trained weights can be transferred instead of training large models from scratch. Secondly, PLMs have significantly enhanced performance in various NLP tasks such as text classification, question answering, and summarisation, achieving state-of-the-art results in benchmark tests like GLUE [58] and SQuAD [59]. Lastly, PLMs exhibit improved contextual handling, disambiguation of polysemous words, and generalisation capabilities. Owing to these

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* Corresponding author at: School of Computer Science and Engineering, Guangxi Normal University, Guilin 541004, China.
E-mail addresses: luoxd@mailbox.gxnu.edu.cn (X. Luo), dzq@stu.gxnu.edu.cn (Z. Deng), binxiay@stu.gxnu.edu.cn (B. Yang), myl41@cam.ac.uk (M.Y. Luo).

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Table 1
Conferences where papers published.

Conference		Medical NLP task										
Name	Rank	TS	QAS	MT	SA	NER	IE	ME	RE	TM	Non-Eng	Others
International Conference on Computational Linguistics	Top	[1]	–	–	–	–	–	–	–	–	–	–
Machine Learning for Healthcare Conference	Ordinary	[2]	–	–	–	–	–	–	–	–	–	–
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases	Top	–	[3]	–	–	–	[4]	–	–	–	–	–
ACM International Conference on Intelligent Computing and its Emerging Applications	Ordinary	–	[5]	–	–	–	–	–	–	–	–	–
Machine Translation	Major	–	–	[6]	–	–	–	–	–	–	–	–
Clinical Natural Language Processing	Ordinary	–	–	[7]	–	–	–	–	[8]	–	–	–
International Joint Conference on Neural Networks	Major	–	–	–	–	[9]	–	–	–	–	–	–
International Conference on Asian Language	Ordinary	–	–	–	–	[10]	–	–	–	–	[10]	–
IEEE International Conference on Emergency Science and Information Technology	Ordinary	–	–	–	–	[11]	–	–	–	–	–	–
Health Text Mining and Information Analysis	Ordinary	–	–	–	–	–	[12]	–	–	–	–	–
The International AAAI Conference on Web and Social Media	Top	–	–	–	–	–	[13]	–	–	–	–	–
AMIA Annual Symposium	Major	–	–	–	–	–	–	–	[14]	–	–	–
Language Resources and Evaluation Conference	Ordinary	–	–	–	–	–	–	–	–	[15]	[15]	–
Findings of the Association for Computational Linguistics: EMNLP	Top	–	–	–	–	–	–	–	–	–	–	[16]

The references are from the sections of Text Summarisation (TS), Question Answering System (QAS), Machine Translation (MT), Sentiment Analysis (SA), Named Entity Recognition (NER), Information Extraction (IE), Medical Education (ME), Relation Extraction (RE), Text Mining (TM), Non English (Non-Eng), and Others.

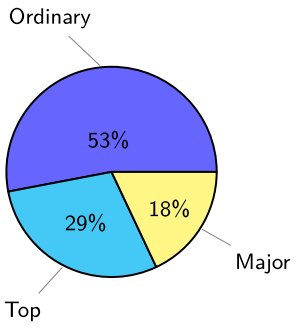
Table 2
Journals where papers published.

Journal		Medical NLP task										
Name	Reputation	TS	QAS	MT	SA	NER	IE	ME	RE	TM	Non-Eng	Others
Knowledge-Based Systems	JCR Q1, IF 8.8	[17]	–	–	–	–	–	–	–	–	–	–
Applied Soft Computing	JCR Q1, IF 8.7	–	–	–	–	–	–	–	–	–	–	[18]
Computer Methods And Programs in Biomedicine	JCR Q1 IF 6.1	[19]	–	–	–	–	–	–	–	–	–	–
IEEE Transactions on Multimedia	JCR Q1, IF 7.3	[20]	–	–	–	–	–	–	–	–	–	–
Journal of the American Medical Informatics Association	JCR Q1, IF 6.4	–	–	–	–	[21]	–	–	–	–	–	–
Bioinformatics	JCR Q1, IF 5.8	–	–	–	–	–	–	–	–	–	–	[22]
Scientific Reports	JCR Q2, IF 4.6	–	–	–	–	–	–	–	–	–	[23]	–
Journal of Biomedical Informatics	JCR Q1, IF 4.5	[24,25]	–	–	–	–	–	–	[26]	–	–	[27]
Data Intelligence	JCR Q1, IF 3.9	–	–	–	–	[28]	–	–	–	–	[28]	–
JMIR Medical Education	JCR Q1, IF 3.6	–	–	–	–	–	–	[29,30]	–	–	–	–
BMC Medical Informatics and Decision Making	JCR Q1, IF 3.5	–	–	–	–	[31]	–	–	–	–	–	–
Arabian Journal for Science and Engineering	JCR Q2, IF 2.9	–	[32]	–	–	–	–	–	–	–	–	–
Applied Sciences	JCR Q2, IF 2.7	–	[33]	–	–	–	–	–	–	–	–	–
Irish Journal of Medical Science	JCR Q3, IF 2.1	–	–	–	[34]	–	–	–	–	–	–	–
International Journal of Environmental Research and Public Health	JCR Q1, IF 0	–	–	–	[35]	–	–	–	–	–	–	–
PLoS Digital Health	–	–	–	–	–	–	–	[36]	–	–	–	–
Database	–	–	–	–	–	–	–	–	[37]	–	–	–
arXiv/bioRxiv Preprint	–	[38]	[39]	–	–	–	–	–	[40]	[41,42]	[42–44]	[45]

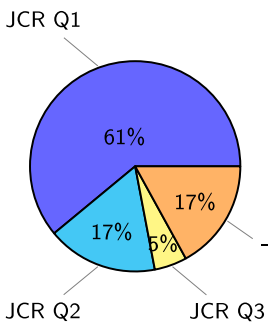
advantages, there is a growing interest in using PLMs in professional domains such as healthcare and medicine.

On the other hand, the application of Artificial Intelligence (AI) and NLP technologies in the medical field has been expanding in recent years [60]. There is an urgent need to make healthcare more proactive, personalised, and convenient. AI can automate mundane tasks, such as mining useful information from medical literature, to improve clinical decision-making [61]. However, the development of medical AI faces critical challenges, such as the lack of large, high-quality datasets, specialised vocabulary, ethical considerations, and regulatory bottlenecks [62–64]. This is where PLMs can play a significant role. They can transfer general capabilities learned from large corpora to downstream medical applications. Fine-tuning them on smaller datasets can also enhance the data efficiency of PLMs. From clinical documentation and doctor-patient dialogue to drug discovery and epidemic prediction, PLMs can be applied in various healthcare applications.

Although recent studies have shown that PLMs are highly effective in medical NLP tasks [65] such as named entity recognition, question answering, electronic health record generation, and phenotype extraction, there are still unresolved issues in training, evaluating, and integrating these models for application in high-impact clinical areas [66–68]. For example, in Oncology, cancer care involves complex decision-making regarding diagnosis, treatment options, and follow-up care. NLP can assist in extracting relevant information from unstructured data in medical records, literature, and research findings to support personalised medicine and evidence-based treatment plans [69]. Furthermore, more research is needed to assess the performance of existing PLMs in the medical field and to adjust methods accordingly [70].



(a) Distribution of references' conference ranks



(b) Distribution of references' journal ranks

Fig. 1. Rank distributions of the conferences and journals where the studies reviewed in this paper were published.

Therefore, a survey of the application of PLMs in medicine is beneficial to advance such research. However, such a comprehensive survey is currently lacking.

Thus, we conducted a systematic survey of the application of PLMs in the medical field. Specifically, we provided an overview of training language models in tasks such as text summarisation, question answering, machine translation, sentiment analysis, named entity recognition, information identification, medical education, relation extraction, and text mining. The overview includes explanations of the basic concepts; main methods used in the tasks; the importance of applying PLMs to tasks; how to apply PLMs to tasks; and various indicators used to evaluate the performance of PLMs in tasks. Moreover, we have specially summarised the recent important research works, analysing their motivations, commonalities, differences, strengths and weaknesses, and limitations. Finally, we discuss future research directions.

The main contributions of this paper are as follows. (1) It is the first comprehensive survey of the application of PLMs in healthcare. To date, there has been no survey on this topic. (2) This survey will help to synthesise the current state-of-the-art methods, key applications, datasets, evaluation results, and limitations of using PLMs in medical NLP in different studies. (3) Analysing the trends in the use of PLMs will reveal unique challenges faced in the medical field compared to general NLP, enabling us to highlight best practices and considerations. (4) Discussion of unresolved issues will drive new research to address the limitations of existing methods and unleash the full potential of PLMs in healthcare. (5) Analysis of the capabilities and limitations of PLMs in practical clinical and healthcare tasks will provide valuable insights for NLP and healthcare researchers, as well as medical professionals seeking to adopt these cutting-edge AI technologies. (6) This survey can guide NLP researchers new to the biomedical field, helping them understand useful data resources, evaluation metrics, ethical considerations, and future directions for their work.

In summary, this paper will serve as a comprehensive reference in the emerging field of medical NLP with PLMs, summarising the current state of knowledge and providing impetus for advancing research in this area. These insights may assist in transforming PLMs into practical AI applications, thereby improving healthcare. Tables 1–2 list the conferences and journals where the researches covered in this paper wre published, and Fig. 1 provides the proportions of these conferences (Fig. 1(a)) and journals (Fig. 1(b)).

The rest of this paper is organised as follows. Section 2 provides an overview of PLMs and their applications in the medical field. Section 3 reviews the research on text summarisation in medicine using PLMs. Section 4 explores medical question-answering systems based on PLMs. Section 5 briefly outlines the study of machine translation in the medical field using PLMs. Section 6 overviews research on sentiment analysis in the medical field using PLMs. Section 7 summarises the work on medical named entity recognition using PLMs. Section 8 describes the work on information identification in the medical field using PLMs. Section 9 reviews the research on medical education using PLMs, focusing on educational examinations. Section 10 discusses the research on extracting relationships from medical documents using PLMs. Section 11 discusses medical text mining based on PLMs. Section 12 discusses the work on NLP of non-English medical documents using PLMs. Section 13 outlines other applications of PLMs in the medical field. Section 14 discusses the challenges and future research directions in applying PLMs in the medical field. Finally, Section 15 concludes the paper.

2. Background

In this section, we will brief PLMs and provide an overview of their applications in the medical field.

2.1. Brief PLMs

PLMs are a type of AI model that undergoes pre-training on large-scale text datasets. This training enables the models to learn statistical relationships between words and phrases, which can be applied to various downstream NLP tasks [71] such as text generation, language translation, and question answering. Typically, PLMs are trained using a technique known as self-supervised learning. In self-supervised learning [72], the model is not explicitly told what to learn. Instead, it is given a set of tasks to perform without any assistance. For instance, the model might be asked to predict the next word in a sentence or to determine whether two sentences are semantically similar. By performing these tasks, PLMs learn to represent the meanings of words and phrases in a useful way for downstream tasks. These representations are known as embeddings [73], which are vectors composed of numbers that signify the meaning of a word or phrase. These vectors are usually long, with hundreds or even thousands of dimensions. Once a PLM has been trained, it can be fine-tuned [74] for specific tasks. Fine-tuning involves training the model on a small dataset of labelled data for that task. For example, PLMs can be fine-tuned using a dataset of text prompts and their corresponding outputs, thus enhancing their performance on specific tasks.

More details about the structure and training methods of PLMs such as GPT, BERT, and RoBERTa are as follows.

1. Structure:
    - GPT is mainly based on the decoder part of the Transformer architecture and does not use an encoder–decoder structure.
    - BERT and RoBERTa are based on the Transformer architecture and use self-attention rather than recursion to model long-term dependencies in text.
    - The Transformer use a multi-head self-attention mechanism to capture the associations between words and performs feature extraction through a forward-propagation network. At the same time, positional encoding is used

Table 3
Multilingual PLMs.

Model	Pre-training task	Institution	Corpus	Params	URL
BERT	MLM, NSP	Google Research	BooksCorpus (800M), English Wikipedia (2500M)	110M–340M	https://github.com/google-research/bert
Falcon 40B	MLM, NSP	Technology Innovation Institute	Public Internet Data (4T)	40B	https://huggingface.co/tiiuae/falcon-40b
Galactica	MLM, NSP	Meta AI	Public web pages, books, papers and other data (1.3T)	200B	https://github.com/paperswithcode/galai
mBART	Seq2Seq Denoising Autoencoder	Facebook AI	Web text in 25 languages	410M	https://github.com/pytorch/fairseq/tree/master/examples/mbart
Orca	MLM, NSP	Microsoft	Public web pages, books and other data	13B	–
XLNet	AR	Google Research	BooksCorpus, English Wikipedia, ClueWeb, Common Crawl	340M	https://github.com/zihangdai/xlnet
WMT21b	MLM	Facebook AI	CC-100, OSCAR, WikiMatrix, ParaCrawl dataset (1.5 TB)	–	–
Vicuna 33B	MLM, NSP	LM-SYS	Public Internet data (3.3T)	33B	https://github.com/lm-sys/FastChat

to encode the positional information of the words. The Transformer Encoder is stacked in multiple layers, forming a deep network to model language's syntactic and semantic relationships.

- BERT employs bidirectional encoding to represent the left and right context of words using masked language modelling.
- RoBERTa is based on BERT but has undergone some modifications, such as the removal of the next sentence prediction task.
- GPT is unidirectional, it only predicts text from left to right, unlike BERT, which performs bidirectional predictions.

2. Training:

- GPT is trained using a large amount of text data during the pre-training phase. Its objective is to predict the next word in a text sequence from left to right, thereby learning vocabulary, grammar, and some logic. During the fine-tuning phase, GPT can be adjusted to suit specific tasks, such as chatbots, text classification, or sentiment analysis. GPT's training adopts an autoregressive mode, where the model predicts the next word in a sentence without masking or deleting. This approach allows it to generate coherent and fluent text, whereas BERT and RoBERTa are more suited for understanding tasks due to their bidirectional nature.
- In the training process, the datasets used for large language models are typically unlabelled, large-scale text corpora. The model is trained in an autoregressive manner, aiming to predict the probability of the next word in a sentence. Given a word sequence $(x_1, \ldots, x_n)$, the model undergoes autoregressive training by maximising the probability of the next word:

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid x_1, \ldots, x_{i-1}).$$

- During the pre-training phase, the text is preprocessed, tokenised, and formatted into sequences before input into the model.
- The self-supervised goal enables the model to learn the relationships between words based on context without manual annotation.
- Masking a portion of the input tokens (for example, 15%) and predicting them enables the model to learn robust contextual representations.
- The model parameters are updated through backpropagation over multiple pre-training epochs to minimise the loss of the masked PLM.

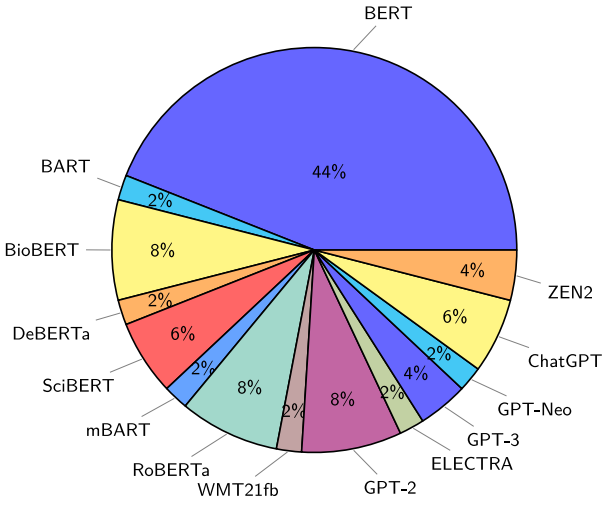


Fig. 2. Usage percentages of various PLMs in medicine.

- Effective pre-training strategies include gradual unfreezing of layers, differential fine-tuning rates, and stepwise unfreezing.
- After pre-training, PLMs can be fine-tuned for downstream tasks by undergoing supervised fine-tuning on datasets specific to those tasks.

The following are tasks where applying PLMs is highly effective: (1) *Natural Language Understanding tasks*: Sentiment Analysis, Named Entity Recognition, Question Answering, Relation Extraction. (2) *Natural Language Generation tasks*: Text Summarisation, Dialogue Generation, Machine Translation, Text Generation. (3) *Text Classification tasks*: Topic Classification, Spam Detection, Sentence Classification. (4) *Information Retrieval tasks*: Document Ranking, Paragraph Retrieval, Query-Document Matching. (5) *Structured Prediction tasks*: Syntactic Parsing, Coreference Resolution, Semantic Role Labelling.

Overall, PLMs such as BERT, GPT, T5 have achieved state-of-the-art or competitive results in a variety of NLP tasks including text classification, information extraction, question answering, machine translation, and dialogue systems. Their transfer learning capabilities enable them to quickly adapt to new tasks with limited labelled data. The context representations learned during pre-training encode crucial linguistic knowledge and can be fine-tuned for downstream tasks. Tables 3 to 5 lists the PLMs discussed in this paper and other important PLMs.

Table 4
English PLMs.

Model	Pre-training task	Institution	Corpus	Params	URL
ALBERT	MLM	Google Research	BooksCorpus, English Wikipedia	18M	https://github.com/google-research/ALBERT
SciBERT	MLM	AI2	BERT training corpus, Semantic Scholar corpus (1140M)	110M	https://github.com/allenai/scibert
RoBERTa	MLM	Facebook AI	BERT training data, additional data (160 GB)	125M	https://github.com/pytorch/fairseq/tree/master/examples/roberta
DeBERTa	MLM	Microsoft	–	140M	https://github.com/microsoft/DeBERTa
BioMedLM	LM	Stanford University	Subparts of The Pile(34.6B)	2.7B	https://github.com/stanford-crfm/BioMedLM
BART	DAE	Facebook AI	CNN/DailyMail dataset	140M	https://github.com/pytorch/fairseq/tree/master/examples/bart
CLaude	MLM, NSP	Anthropic	–	200B	–
CLIP	Contrastive Pre-training	Open AI	OpenAI's diverse set	–	https://github.com/openai/CLIP
CoHere	MLM, NSP	Cohere Technologies	Public text data such as web pages, social media, books. (0.46T)	20B	–
ELECTRA	RTD, MLM	Google Research	BooksCorpus (800M), English Wikipedia (2500M)	110M	https://github.com/google-research/electra
GPT-2	LM	Open AI	WebText (40 GB)	117–1542M	https://github.com/openai/gpt-2
GPT-3	ALM	Open AI	Common Crawl (570 GB)	175B	–
GPT-Neo	ALM	Open AI	The Pile (800 GB)	2.7B	https://github.com/EleutherAI/gpt-neo
ChatGPT	ALM	Open AI	Diverse dataset (GPT-3)	175G	–
LaMDA	MLM, NSP	Google Research	Dialogue, web data (0.1T)	137B	–
Llama	MLM, NSP	Meta AI	Public Internet data (1.4T)	7B–65B	https://github.com/facebookresearch/llama
PALM	MLM, NSP	Google Research	Internet-based data set (1.5T)	540B	–
Phi-1	MLM, NSP	Microsoft	Medical dataset (derived 2.1T)	1.3B	https://huggingface.co/microsoft/phi-1_5
StableLM	MLM, NSP	Stability AI	Medical dataset	3B–7B	https://github.com/Stability-AI/StableLM
T5	Seq2Seq, MLM	Google Research	C4 (Common Crawl's Cleaned Crawled Corpus)	–	https://github.com/google-research/text-to-text-transfer-transformer

Table 5
Chinese PLMs.

Model	Pre-training task	Institution	Corpus	Params	URL
BERT-wwm-ext Chinese	MLM	Yiming Cui	Chinese Wikipedia (4.5B)	108M	https://github.com/ymcui/Chinese-BERT-wwm
ERNIE	MLM, NSP, DAE	Baidu	Diverse Chinese sources, enhanced with Baike	–	https://github.com/PaddlePaddle/ERNIE
ZEN2	MLM with n-gram masking	HUAWEI	Chinese web pages, Baike, news (1.5 TB)	668M	https://github.com/sinovation/zen2
ERNIEBot	MLM, NSP	Baidu	Chinese web text (4 TB)	–	–
WuDao-WenYuan	WWM	Beijing Academy of Artificial Intelligence	Chinese Wikipedia, Baike, LTP (100 GB)	2.6B	https://github.com/BAAI-WuDao/Model
PLUG	MLM	Alibaba DAMO Academy	Diverse Chinese web text (1 TB)	27B	–
PanGu-α	MLM, NSP	Huawei Cloud	Chinese web pages, common crawl data (1.1 TB)	200B	https://openi.pc.ac.cn/PCL-Platform_Intelligence/PanGu-Alpha

2.2. PLMs in the medical field

PLMs have a wide range of potential applications in the medical field. (1) *Generating Medical Texts*: PLMs can generate medical texts, such as patient reports, discharge summaries, and clinical trial reports. (2) *Translating Medical Texts*: PLMs can translate medical texts from one language to another, which highly benefits doctors and patients who speak different languages. (3) *Answering Medical Questions*: PLMs can answer medical questions, which could be extremely valuable for doctors, patients, and researchers. (4) *Diagnostics*: PLMs can diagnose diseases by analysing medical images and texts. This can assist doctors in making more accurate diagnoses. (5) *Developing New Treatment Methods*: PLMs can develop new treatment methods by analysing medical data and helping doctors discover new ways to treat diseases. (6) *Personalised Medicine*: PLMs can achieve personalised medicine by considering the individual characteristics of patients, which can help enhance the effectiveness of treatments. PLMs are still relatively new technology, but they have the potential to transform the field of medicine fundamentally. As these models continue to evolve, they are likely to become more powerful and versatile.

Table 6
Usage of various PLMs in various medical NLP tasks.

Model	Medical NLP task										
	TS	QAS	MT	SA	NER	IE	ME	RE	TM	Non-Eng	Others
BERT	[19,20,38]	[5,32,33]	–	[34,35]	[10,31]	[13]	–	[8,14,26,37,40]	[15,41]	[10,15,23,43,44]	[18,22,27]
AlBERT	–	–	–	–	–	–	–	–	–	–	[16]
BioBERT	[17]	[3]	–	–	[9,31]	–	–	–	–	–	[16]
SciBERT	[24,25]	–	–	–	[31]	–	–	–	–	–	–
RoBERT	–	–	–	–	[28]	[12,13]	–	–	[42]	[28,42]	–
DeBERTa	–	–	–	–	–	[13]	–	–	–	–	–
BART	[1]	–	–	–	–	–	–	–	–	–	–
mBART	–	–	[6]	–	–	–	–	–	–	–	–
WMT21fb	–	–	[7]	–	–	–	–	–	–	–	–
ELECTRA	–	–	–	–	[11]	–	–	–	–	–	–
GPT-2	[38]	[33,39]	–	–	[21]	[4]	–	–	–	–	–
GPT-3	[2]	–	–	–	–	[4]	–	–	–	–	–
GPT-Neo	–	–	–	–		[4]	–	–	–	–	[45]
ChatGPT	–	–	–	–	–	–	[29,30,36]	–	–	–	[45]
ZEN2	–	–	–	–	[10]	–	–	–	–	[10]	–

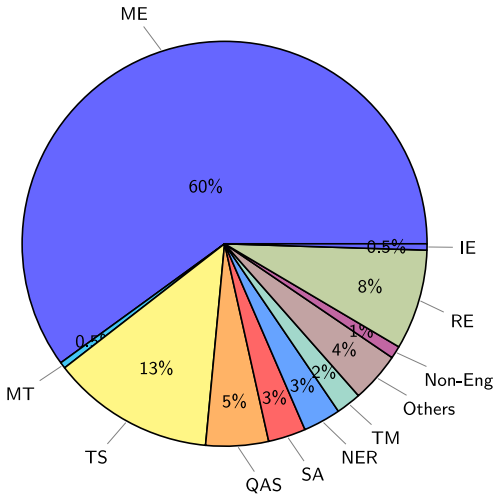


Fig. 3. Citation percentages of PLM-based methods for various medical NLP tasks.

The various PLMs used in the research covered in this paper are shown in Table 6, and Fig. 2 shows their usage rates. From Fig. 2, we can see:

- BERT is the most popular choice (see Fig. 2) because of its reliable contextual representation and the ability to capture semantic relationships between words. Recognising named entities in text effectively improves the accuracy and performance of named entity recognition.
- RoBERTa, BioBERT, and GPT-2 are the second most widely used models. RoBERTa addresses challenges in the medical field such as entity boundary ambiguity, domain adaptation, data scarcity, and named entity recognition. It is an optimised version of BERT, trained for longer periods using more data and larger-scale unlabelled text data for pre-training. This enables RoBERTa to capture contextual information better, handle entity boundary ambiguity, and exhibit improved domain adaptation and generalisation capabilities; BioBERT, through pre-training on large-scale biomedical corpora, offers a powerful tool for solving various medical NLP tasks. It not only understands complex biomedical texts but also excels in multiple tasks (especially in named entity recognition, relation extraction, and question answering), surpassing other state-of-the-art models. As a cross-lingual PLM, GPT-2, with its hundreds of millions of parameters and powerful generative capabilities, excels not only in specific domains (such as medicine) through pre-training and fine-tuning but also in zero-shot learning, adapting to various new medical NLP tasks. Compared to other models, GPT-2 has a distinct advantage in generative tasks, such as generating medical reports, medical record summaries, and research papers.
- SciBERT and ChatGPT rank third in usage. SciBERT is a BERT-based PLM specifically trained for texts in the biomedical domain, thus better understanding and representing medical terminology and concepts. SciBERT can be used to enhance the performance of tasks such as medical information extraction, medical terminology standardisation, and medical text classification; ChatGPT can generate fluent and natural language texts is suitable for intelligent dialogues, medical text writing, and data analysis report generation. ChatGPT can improve user interaction experience and satisfaction while enhancing work efficiency and quality.

Figs. 3–4 displays the citations of various PLM-based medical NLP methods covered in this survey according to Google Scholar (the total citation is 2867 times). As shown in Fig. 4, the paper by Kung et al. [36] on the application of ChatGPT in medical education has received the highest number of citations at 1073, accounting for 37.43% of the total. In contrast, the paper by Fu and Bu [11] on medical NER using PLMs has zero citations. Excluding the minimum and maximum, the average citation count is 40, with only 31.81% of the papers reaching or exceeding the average citation count. This reflects a significant disparity in citation distribution. On the other hand, papers on PLMs in medical education [29,30,36] have accumulated 1738 citations in total, accounting for 60% of the total citations. Despite an average citation count of 579, these papers represent only 6.8% of the total studies, suggesting that research in this field is still in its early stages but has received widespread attention.

Apart from medical education, the second most cited study is by Chintagunta et al. [2] on text summarisation in the medical field with PLMs, which has 90 citations. They developed an algorithm based on GPT-3 for creating synthetic training datasets that explicitly capture relevant medical information. Papers on PLMs in medical text summarisation [1,2,17,19,20,24,25,38] received 371 citations in total, accounting for 13%, further confirming its central position. Papers on relation extraction [8,14,26,37,40] have a more uniform number of citations, totalling 8% of the overall citations, indicating another area of widespread interest. It should be noted that papers published in 2023 generally have lower citation counts, possibly due to their recent publication and not having accumulated enough citations yet.

2.3. Development of PLMs

In recent years, the development of PLMs has advanced rapidly. Early models were trained on relatively small text datasets. However, recent models have been trained on much larger datasets, significantly improving performance.

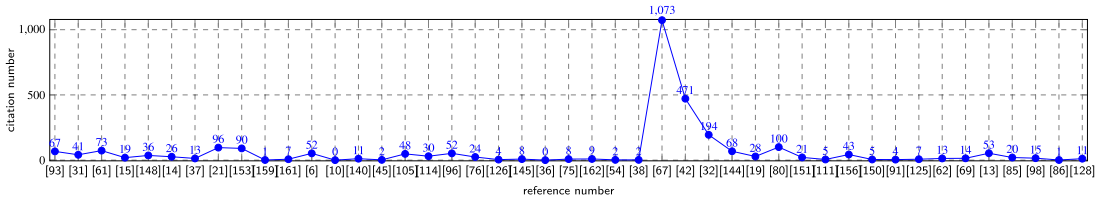


Fig. 4. Citations of the various PLM-based methods for medical NLP tasks.

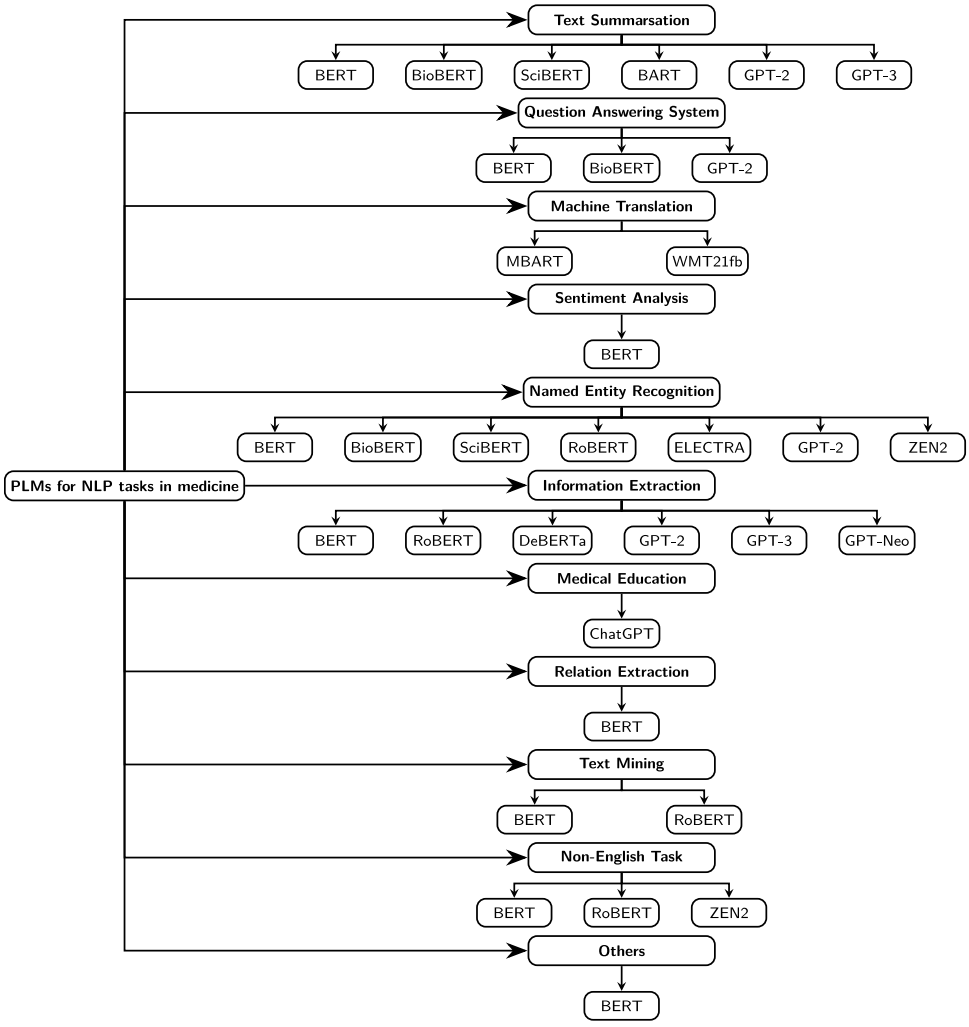


Fig. 5. PLMs for NLP tasks in the medical field.

The introduction of the Transformer model [51] is one of the most important developments in the field of PLMs. The Transformer is a type of neural network that excels in learning long-term dependencies, making it more effective in tasks that require understanding the context of sentences, such as machine translation. Another key development is the use of transfer learning, where models are trained on large datasets for general tasks and then fine-tuned for specific tasks, enhancing their performance in these specific areas.

The trend in training PLMs in recent years has been to use larger text datasets. This is due to the increasing availability of large-scale text datasets, such as Common Crawl and Wikipedia. Larger datasets enable PLMs to understand the statistical relationships between words and phrases more fully, thus performing better in downstream tasks. Another trend in the field of PLMs is the use of more complex architectures. For example, the BERT model uses a Transformer architecture with 12 layers and 110 million parameters, while the GPT-3 model uses a Transformer architecture with 17 layers and 175 billion parameters. These more complex architectures allow PLMs to learn more intricate relationships between words and phrases, leading to better performance in downstream tasks.

Finally, the trend of applying PLMs to more complex tasks is also increasing. For example, PLMs have been used to generate realistic dialogues for chatbots, achieve high-precision language translation, and answer questions in an information-rich manner. As PLMs continue to evolve, they are expected to be widely applied to increasingly complex tasks in the future. The application of PLMs in the medical field's NLP tasks covered in this paper is shown in Fig. 5.

3. Text summarisation

This section reviews the PLM-based research on text summarisation in medicine.

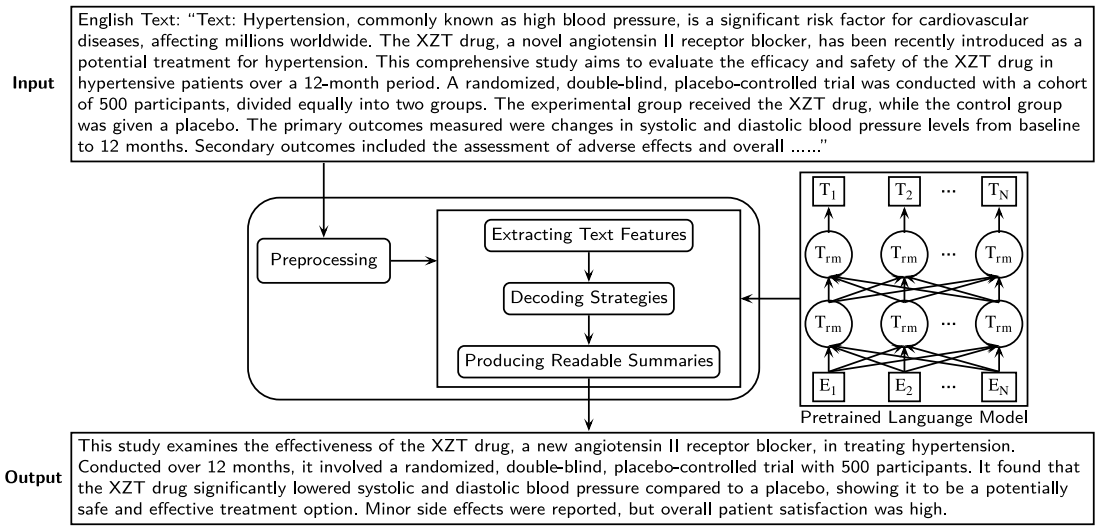


Fig. 6. PLMs-based model diagram for text summarisation in medicine.

3.1. Basic concepts

Text summarisation (TS) [75] is the process of using AI technology to condense lengthy texts into their key points. The main purpose of TS is to generalise the text document, highlighting its main ideas and points [76]. This makes the text more concise and easy to read, allowing readers to understand the essence of the lengthy text without reading it fully. By selecting the most prominent points, reducing details and redundancies, TS fundamentally distills the core essence of the text, saving the reader's time.

3.2. Main methods

Some of the main methods used in TS are as follows. (1) *Extractive Summarisation*: This involves selecting key sentences, important phrases, and paragraphs from the original text to create a summary. It relies on the statistical and linguistic characteristics of sentences. (2) *Abstractive Summarisation*: This uses advanced NLP technologies to analyse text and generate new phrases and sentences, filtering and conveying the most central information of the text. This method, compared to the extractive approach, is more complex. (3) *Multi-document Summarisation*: This creates a summary by finding key information common across multiple documents on the same topic. (4) *Query-based Summarisation*: This involves creating a summary based on user queries or questions about the text, summarising information relevant to the query.

Fig. 6 depicts a specialised framework for automated TS in the medical domain, using a PLM exemplified by BERT. It details the process from the input of raw English medical text to the output of a concise summary, encompassing stages such as preprocessing the data, extracting text features, applying decoding strategies, and ultimately producing an intelligible summary. This process highlights the role of PLMs, which can selectively be employed to execute these functions to streamline the information distillation process.

3.3. Advantages

Due to the domain adaptability, knowledge extraction ability, and text generation skills of PLMs, they play a significant role in medical summarisation tasks, as demonstrated in the following areas: (1) Medical documents contain a large amount of domain-specific vocabulary and terminology. PLMs with prior training can learn these terms and concepts from large medical corpora. (2) PLMs can accurately extract vital clinical concepts, test results, diagnoses, and medications from patient health records and medical journals. For example, the generative capabilities of models like GPT-3 enable them to create abstract summaries using medical ontologies without losing key information. (3) Since PLMs are initially trained on general text, fine-tuning them on medical summarisation datasets can adapt them to medical language. (4) Summarising research papers and clinical trial reports helps doctors stay informed about the latest developments. (5) The multi-document summarisation functionality helps to integrate information from multiple patient health record documents. (6) Query-based summarisation can extract relevant information from lengthy documents based on users' specific information needs. Summarising medical information can improve understanding for a general audience without medical knowledge. Moreover, consistent high-quality summaries can be generated through transfer learning even from scarce medical training data.

3.4. Basic application steps

Applying PLMs to medical texts involves the following steps: (1) Select an appropriate PLM, such as BERT, RoBERTa, or GPT-3, which have been trained on large general corpora. (2) Fine-tune the model on a medical summarisation dataset specific to the domain to adapt it to medical language, using datasets like PubMed, Medline, or medical research papers. (3) During the fine-tuning process, train the model to perform extractive and abstractive summarisation tasks related to the medical field. (4) Use transfer learning methods to Enhance the model's capabilities on low-resource medical summarisation datasets. (5) Integrate medical knowledge bases, ontologies, and terminologies into the model training to deepen clinical language understanding. (6) Establish a medical summarisation pipeline, including preprocessing text, extracting text features, summarising key information from the text, decoding strategies, and producing readable summaries. (7) For multi-document input, use algorithms like TextRank [77] to identify common themes and core concepts within documents. (8) Evaluate the model-generated medical summaries against benchmark data using metrics such as ROUGE, BERTScore, and manual evaluation. (9) Deploy the model in applications like personalised health record summaries, medical literature summaries, and clinical trial reports. (10) Continuously update the model's training data with the latest medical terminology to improve its robustness and accuracy.

Table 7
Datasets for medical TS.

Dataset	Used by	Language	Domain	Sample size	Download link
BioMed Central	[19]	English	Biomedical	–	https://www.biomedcentral.com/
PubMed	[17]	English	Biomedical	Over 36 million citations and abstracts from biomedical literature	https://pubmed.ncbi.nlm.nih.gov/
CORD-19	[24,38]	English	COVID-19	More than 1 million papers	https://github.com/allenai/cord19
BioASQ Task-6a	[3]	English	Biomedical	The training set/validation set/test set is 12M/1M/0.1M, with an average of 7.33 sentences in each document	http://participants-area.bioasq.org/datasets/
CNN/Daily Mail	[2,20]	English	General	286,817 training pairs, 13,368 validation pairs and 11 487 test pairs	https://paperswithcode.com/dataset/cnn-daily-mail-1
S2ORC	[25]	English	General	Over 136M papers	https://github.com/allenai/s2orc
OPEN-I	[20]	English	Biomedical Imaging	7471 chest X-ray images and 3955 reports	https://openi.nlm.nih.gov/faq#collection
MIMIC-CXR	[20]	English	Radiology	371,920 chest X-rays and 227,943 imaging studies	https://physionet.org/content/mimic-cxr/2.0.0/
MIMIC-III	[1]	English	Biomedical	67,830 hospital ICU hospitalisation records	https://github.com/datasets/mila-datasets--mimiciii

3.5. Evaluation metrics

Some common metrics for evaluating medical domain summarisation systems based on PLMs are as follows. (1) *Recall-Oriented Understudy for Gisting Evaluation (ROUGE)* [78]: Measures the overlap between model summaries and reference summaries by comparing n-grams, word sequences, and word pairs, widely used in summary quality analysis. (2) *BERTScore* [79]: Calculates the similarity between model summaries and reference summaries based on context embeddings, offering better semantic matching than ROUGE. (3) *Metric for Evaluation of Translation with Explicit Ordering (METEOR)* [80]: Compares similarity based on explicit word-to-word matching and synonyms, suitable for medical semantics. (4) *Consensus-based Image Description Evaluation (CIDEr)* [81]: mainly designed for image caption evaluation but also applicable to MT assessment. It measures the consensus between a machine-generated translation and multiple reference translations. The higher the CIDEr score, the better the translation quality. (5) *Semantic Textual Similarity (STS)* [82]: Calculates the semantic similarity between model summaries and reference summaries, better capturing specific domain meanings. (6) *Flesch Reading Ease (FRE)* [83]: Assesses the readability of model summaries, higher scores indicate lower reading difficulty. (7) *Recall* [84]: The percentage of original information retained in the summary, checking completeness. (8) *Precision* [84]: The percentage of relevant information in the summary relative to irrelevant information, checking conciseness. (9) *F1 Score* [84]: The harmonic mean of precision and recall, combining relevance and completeness. (10) *Accuracy* [85]: Compares the correctly predicted summary sentences against the total sentences. (11) *Bilingual Evaluation Understudy (BLEU)* [86]: Calculates matching degree and score by comparing machine-translated outputs with human-translated reference texts, scores are calculated based on the overlap of n-grams between generated and reference translations. (12) *Concept F1 Score* [87]: Based on the traditional F1 score, it is used to measure the performance of information extraction systems, especially in identifying and classifying concepts (such as entities, terms, or specific types of phrases) in text. (13) *Negation F1 Score* [88]: Based on the traditional F1 score, used to assess the model's ability to handle negation information. (14) *Manual Evaluation* [89]: Manually assesses the coherence, readability, conciseness, and information retention of the summarised results. A combination of objective metrics and manual evaluation provides the best quality assessment for medical summarisation models.

3.6. Datasets

Table 7 summarises some datasets used in medical TS, including language, domain, sample count, and download links.

(1) BioMed Central is an open-access publishing organisation that publishes more than 280 peer-reviewed journals in life sciences and medicine, with nearly 180 journals indexed in SCI and 175 journals having impact factors. The BioMed Central dataset includes papers and metadata from these journals, suitable for text mining and NLP research.

(2) PubMed [90] is a free biomedical literature search system developed by the National Center for Biotechnology Information at the National Library of Medicine in the USA, containing literature from over 4000 biomedical journals from more than 70 countries and regions. The PubMed dataset includes citations and abstracts from these publications, along with some images and tables suitable for image and text analysis research.

(3) CORD-19 [91] is an academic paper corpus about COVID-19 and related coronaviruses, curated and maintained by the team of semantic scholars at the Allen Institute for AI to support text mining and NLP research. The CORD-19 dataset includes over one million papers from multiple data sources, including full texts, metadata, citations, abstracts, and some images and tables.

(4) BioASQ [92] is a biomedical semantic indexing and question answering (QA) challenge that includes multiple tasks such as hierarchical text classification, machine learning, information retrieval, question answering with text and structured data, and multi-document summarisation. BioASQ provides various benchmark datasets.

(5) CNN/Daily Mail [93] is a TS dataset consisting of news papers and summaries from the CNN and Daily Mail websites. The dataset's task involves a hidden entity in the summary, forming a cloze-style question, with the answer being the hidden entity. The CNN/Daily Mail dataset contains nearly 1.4 million questions and can be used to train machine reading comprehension abilities.

(6) S2ORC [94] is a general corpus for NLP and text mining, containing 81.1 million English academic papers across multiple disciplines. The dataset includes rich metadata, paper abstracts, parsed references, and 8.1 million open-access papers structured full texts. The full texts are annotated with automatically detected citations, figures, tables, inline mentions, and their corresponding paper objects. Developed and maintained by the team of semantic scholars at the Allen Institute for AI, the S2ORC dataset is currently the largest publicly available machine-readable collection of academic texts.

(7) OPEN-I [95] is a service of the United States National Library of Medicine, providing access to open-source literature and biomedical image collections' abstracts and images, including charts, graphs, clinical images and so on. The OPEN-I dataset contains metadata and links for these literatures and images, suitable for image and text retrieval research.

(8) MIMIC-CXR [96] is a significant radiological dataset specifically focused on chest X-rays. Developed through a collaboration between

the Massachusetts Institute of Technology and Philips Healthcare, it is one of the largest publicly available datasets of its kind. The dataset encompasses an extensive collection of 371,920 chest X-ray images and 227,943 imaging studies. Each image in the MIMIC-CXR dataset is paired with corresponding radiological reports, offering a rich source of both visual and textual medical data. These reports provide detailed clinical annotations and insights.

(9) MIMIC-III [97] is a large, free dataset built by the Computational Physiology Laboratory at MIT, containing health data information of over 40,000 patients from the ICU of Beth Israel Deaconess Medical Center, spanning from 2001 to 2012. The MIMIC-III dataset includes patients' admission records, discharge records, diagnostic records, prescription information, lab results, microbiology test results, reports texts suitable for clinical decision support, survival analysis, quality improvement in intensive care, and other research.

3.7. Biomedical literature summarisation

3.7.1. Important research works

Biomedical TS is crucial for understanding the vast and ever-growing biomedical literature, and must be completed in an appropriate environment with relevant context. To address this challenge, in 2020, Moradi et al. [19] developed a TS model using BERT-generated contextual embeddings. They combined different versions of BERT with clustering techniques to extract the most relevant and informative sentences from input documents. Using BERT-large, their model achieved ROUGE-1 and −2 scores of 75.04 and 33.12, respectively, outperforming a group of domain-specific and domain-general methods. Their work provided a starting point for exploring how deep contextual PLMs can be applied to biomedical TS.

Deep neural networks can significantly enhance the performance of extractive TS. However, domain-aware external knowledge and document structure features must be considered when applying these methods to extractive summarisation in the biomedical field. To address these issues, in 2020, Du et al. [17] developed a TS model based on BioBERT, called BioBERTSum. BioBERT, a member of the BERT family, was trained as an encoder on a large-scale biomedical corpus and further fine-tuned for extractive TS. Notably, they embedded sentence positional information into the model to learn about the positions of sentences. Their model performed better than previous models on the PubMed dataset in ROUGE-1, ROUGE-2, and ROUGE-L assessment metrics, achieving scores of 37.45, 17.59, and 29.58, respectively. This research is a beneficial example of applying deep context-aware PLMs to biomedical TS.

Since the sudden outbreak of COVID-19, it has been a challenge for researchers to keep up with the rapidly growing literature about the virus. To address this challenge, in 2020, Kieuvongngam et al. [38] developed a TS model based on BERT and GPT-2 [98]. Specifically, they first converted each sentence in the document into a 768-dimensional vector using BERT. Then, they clustered these vectors using K-medoid, extracted summaries matching the keywords and trained them with GPT-2 to generate summarisation results. Their model was able to infer more accurate summaries from keywords.

Diagnosing, treating, and preventing COVID-19 have always been important focus areas for researchers. When publishing the latest research findings, researchers need to write paper summaries, greatly increasing their workload personally. To address this problem, in 2022, Cai et al. [24] developed a language-rich TS model called COVID-Sum based on SciBERT, capable of automatically generating reliable summary drafts and saving time and effort for researchers. The model first extracted prominent sentences from source papers and built a co-occurrence graph. Then, it encoded sentences and the co-occurrence graph using a SciBERT-based sequence encoder and a graph attention network encoder. Finally, it merged the two encodings to generate summaries for each scientific paper. Their model outperformed other TS models on the publicly available COVID-19 open research dataset, achieving ROUGE-1, -2, and -L scores of 44.56, 18.89, and 36.53, respectively.

Existing TS models based on PLMs usually fine-tune the PLM directly on the target corpus but overlook how fine-grained domain knowledge can help generate more coherent summaries by identifying the required context. To address this issue, in 2022, Xie et al. [25] developed KeBioSum, a framework focused on knowledge-injected training. To fine-tune the model for extractive TS tasks based on PLMs, they used adapter fusion technology to inject domain knowledge into the PLM. Their experiments showed that including the KeBioSum knowledge adapter can improve existing PLMs (such as BERT, RoBERTa, BioBERT, and PubMed-BERT) on extractive summarisation tasks on three biomedical literature datasets, performing better than strong baseline models, with ROUGE-1, -2, and -L scores improving by 1, 1.5, and 3 points, respectively.

3.7.2. Main similarities and differences

Similarities between these methods are as follows. (1) All use deep learning technology to complete the task of biomedical TS, a challenging and important task in biomedical information retrieval and analysis. (2) They all base their methods on PLMs, using large corpora to learn general language representations, which are then fine-tuned for specific tasks and domains. (3) They evaluate their methods on standard benchmark datasets such as PubMed, BioASQ, and CORD-19, which contain annotated biomedical text summaries. (4) They compare their methods with other existing advanced methods based on PLMs, such as BERTSUM, SciBERT, and BioBERT.

The differences between these methods are as follows. (1) *Input Text Type*: The first two papers focus on summarising biomedical research literature, while the third and fourth papers concentrate on summarising COVID-19 scientific papers. The fifth paper covers both types of input texts. (2) *Output Summary Type*: The first, second, and fifth papers produce extractive summaries, i.e., selecting the most important sentences from the original text to form the summary. The third and fourth papers produce abstractive summaries, i.e., creating new sentences using natural language generation technology to capture the essence of the original text. (3) *Methods Used*: The first paper uses deep contextual embedding to quantify the information content of sentences and sort them for extraction. The second paper encodes sentences using a PLM in the biomedical domain, selecting sentences based on their similarity to queries. The third paper encodes and decodes sentences using the BERT and GPT-2 models for summary generation. The fourth paper uses a language-rich SciBERT model to generate abstract summaries with syntactic and semantic features. The fifth paper improves the performance of extractive summarisation using a PLM with domain knowledge.

3.7.3. Limitations

These methods also have some limitations. (1) They fail to capture clinical context when ranking summaries, which may result in omitting important information relevant to user queries or purposes. (2) They do not consider checking the quality of documents before summarising, potentially leading to summaries generated from low-quality or unreliable sources. Moreover, they are unable to identify context and implicit information in biomedical texts that may not explicitly match user queries, posing difficulties in understanding the meaning and significance of texts. (3) There is a lack of purpose-driven paragraph ranking and selection, potentially affecting the coherence and informativeness of summaries. (4) The annotated corpus for biomedical TS is limited, which might hinder the development and evaluation of new methods and technologies.

Table 8
Comparison of TS methods based on PLMs.

Reference	Publication venue	Focus	Pro	Con	Code
Moradi et al. (2020) [19]	Computer Methods and Programs in Biomedicine (JCR Q1, IF 6.1)	Create an appropriate environment and relevant context for biological TS	Use the context-embedded TS model generated by BERT	Unable to capture clinical data when sorting abstracts	https://github.com/BioTextSumm/BERT-based-Summ
Du et al. (2020) [17]	Knowledge-Based Systems (JCR Q1, IF 8.8)	Improve the performance of extractive summarisation in the biomedical field	Have domain-aware external knowledge, taking into account document structural characteristics	Longer sentences may be truncated by the padding mechanism	–
Kievuongngam et al. (2020) [38]	arXiv Preprint	Helps researchers keep up with the rapidly growing literature on the virus	Can infer more precise summaries from literature keywords	Cannot identify contextual and implicit information present in biomedical texts	https://github.com/VincentK1991/BERT_summarization_1
Cai et al. (2022) [24]	Journal of Biomedical Informatics (JCR Q1, IF 4.5)	Help researchers generate the abstract of the paper	Automatically provide researchers with reliable abstract drafts	No consideration is given to checking the quality of the document	–
Xie et al. (2022) [25]	Journal of Biomedical Informatics (JCR Q1, IF 4.5)	No consideration of fine-grained domain knowledge	Helps generate coherent abstracts by identifying required context	Limited annotated corpus of biomedical text abstracts	–
Cai et al. (2021) [20]	IEEE Transactions on Multimedia (JCR Q1, IF 7.3)	Existing text summary models cannot be directly applied to radiology reports	A PLM specifically for chest radiology was developed	Lack of specificity in the model Domain knowledge and terminology	–
Gao et al. (2022) [1]	International Conference on Computational Linguistics (Top)	Optimises the generation of abstracts for medical progress records	Generates rich and coherent abstracts	May lose important information or introduce redundancy and inconsistency in the abstract	–
Chintagunta et al. (2021) [2]	Machine Learning for Healthcare Conference (Ordinary)	Solve the difficulty obtaining labelled data	Ability to generate high-quality training data and merge it with human-labelled data	May not fully reflect the diversity and complexity of real-world healthcare conversations	–

Table 9
Comparison of TS methods' result based on PLMs.

Reference	Model	Evaluation dataset	R-1	R-2	R-L	BERT score	BLEU -1/-2/-3	Concept F1	Negation F1
Moradi et al. (2020) [19]	BERT-based summariser (large)	Homemade	75.04	33.12	–	–	–	–	–
Du et al. (2020) [17]	BioBERTSum	CNN/DailyMail	43.13	19.71	37.47	–	–	–	–
		PubMed	37.45	17.59	29.58	–	–	–	–
Kievuongngam et al. (2020) [38]	BERT	CORD-19	–	–	–	–	–	–	–
	GPT2		–	–	–	–	–	–	–
Cai et al. (2022) [24]	COVIDSum	CORD-19	44.56	18.89	36.53	–	–	–	–
Xie et al. (2022) [25]	PubMedBERT-all-full	CORD-19	32.04	12.61	29.1	53.36	–	–	–
		PubMed-Long	36.39	16.27	33.28	59.96	–	–	–
		PubMed-Long	37.44	16.72	34.08	56.81	–	–	–
		s2orc	43.98	18.27	39.93	64.94	–	–	–
		Heart disease	49.7	37	21.8	–	–	–	–
Cai et al. (2021) [20]	ChestXRayBERT	OPEN-I /MIMIC-CXR	41.3	28.6	41.5	–	28.5/14.4/6.1	–	–
Gao et al. (2022) [1]	T5	UMLS (Objective)	–	–	14.91	50.13	–	–	–
		UMLS (Subjective)	–	–	10.91	50.09	–	–	–
	BART	UMLS (Objective)	–	–	11.61	48.86	–	–	–
		UMLS (Subjective)	–	–	9.55	48.95	–	–	–
Chintagunta et al. (2021) [2]	PEGASUS	Human labelled	–	–	55.59	–	–	35.96	21.09
		GPT-3-ENS	–	–	58.44	50.09	–	40.58	28.89
	DRSUM	Human labelled	–	–	52.7	–	–	39.95	26.75
		GPT-3-ENS	–	–	26.66	–	–	39.47	50.85

3.8. Medical report summarisation

3.8.1. Important research works

Summarising as much crucial information as possible in the results section of radiological reports can facilitate more effective communication between radiologists and referring physicians. However, due to specific terminology in radiology, existing TS models usually struggle to be directly applied to radiological reports. To address this issue, in 2021, Cai et al. [20] developed a PLM specifically for chest radiology, named ChestXRayBERT. They first collected scientific papers related to radiology and pre-trained ChestXRayBERT on these papers. Subsequently, they developed a TS model that integrates ChestXRayBERT with a Transformer decoder. Finally, they fine-tuned the model on chest X-ray reports. On the publicly available OPEN-I and MIMIC-CXR datasets, they demonstrated through experiments that their model outperforms other neural network-based TS models, achieving ROUGE-1, -2, and -L scores of 41.3, 28.6, and 41.5, respectively.

Gao et al. [1] proposed using pre-trained sequence-to-sequence (seq2seq) models to generate abstract summaries of medical progress notes. They used BART and PEGASUS models and fine-tuned them on a newly created dataset named "Progress Note Summarisation (PNS)", containing about 10,000 progress notes with question summaries. The evaluation of the models included ROUGE scores and human assessment. The fine-tuned BART model exhibited outstanding performance, surpassing previous methods in ROUGE scores. The human evaluation showed that the summaries generated by the model were informative and coherent, but they might lack crucial details compared to human-written summaries. Moreover, the model occasionally generated content not present in the original notes. Overall, pre-trained seq2seq models can generate useful summaries from progress notes, but there is room for improvement in accurately capturing key details. The PNS dataset offers potential for future research.

3.8.2. Main similarities and differences

The two papers share similarities in the following aspects. (1) They both use Transformer-based architectures, a powerful and popular model for NLP tasks, capable of capturing long-distance dependencies and semantic relationships in text. (2) They both use standard automatic evaluation metrics to measure the performance of their methods, including ROUGE, BLEU, and BERTScore, which assess the quality of summaries by comparing them with human-written literature summaries. (3) They both compare their methods with existing state-of-the-art approaches such as Pointer-Generator, Bi-LSTM, and ClinicalBERT, which are also based on PLMs or neural networks.

The main differences between these publications are as follows. (1) *Input Text Type*: The first paper focuses on summarising chest radiology reports written by radiologists to describe chest X-ray images. The second paper focuses on summarising medical progress notes written by physicians to record daily progress and care plans for patients during hospital stays. (2) *Output Summary Type*: The first paper generates concise summaries of chest radiology reports' main observations and conclusions. The second paper produces a list of issues, which are diagnostic and problem lists in patients' daily care plans. (3) *Methods Used*: The first paper uses ChestXRayBERT, a PLM designed explicitly for the field of chest radiology, fine-tuned for the summarisation task. The second paper employs T5 and BART, general domain PLMs, adjusted with data augmentation and domain adaptation techniques for summarisation tasks. (4) *Datasets Used*: The first paper uses a dataset from the Indiana University chest X-ray collection containing 1000 chest radiology reports. The second paper uses a dataset of 2000 medical progress notes from the MIMIC-III Intensive Care Medical Information Market.

3.8.3. Limitations

These methods also have some limitations. (1) Memory consumption and performance degradation of PLMs in handling long clinical texts are common issues in the biomedical field. This might limit the models' ability to capture the global context and structure of texts, impacting the quality of summaries. (2) To handle long texts, some methods adopt truncation or sliding window strategies, which might lead to the loss of crucial information or introduce redundancy and inconsistency in generating summaries. (3) The lack of domain-specific knowledge and terminology in PLMs affects the accuracy and readability of summaries.

3.9. Medical dialogue summarisation

3.9.1. Important research works

In 2021, Chintagunta et al. [2] developed an algorithm based on GPT-3 to address the challenge of developing effective medical dialogue generation models due to the difficulty in obtaining annotated data. They adopted an innovative approach by creating a synthetic training dataset to capture relevant medical information explicitly. Specifically, they expanded the original 210 human-annotated examples to be comparable with an algorithmic dataset using 6400 human-annotated examples. The experimental results showed that their model could generate high-quality training data and produced more accurate and convincing summaries when combined with human-annotated than those trained solely on human data. The model scored 53.39 on the ROUGE-L evaluation metric.

3.9.2. Limitations

However, this approach also has some limitations. Firstly, their evaluation used a small, limited dataset containing 210 human-annotated examples, which might not fully reflect the diversity and complexity of real-world medical dialogues. Secondly, the paper did not compare its method with other data augmentation techniques or baseline models. They mainly relied on GPT-3 as the core component of their data generation algorithm. However, GPT-3 is not specifically trained for medical domain expertise and might introduce errors or inconsistencies in the generated synthetic data. For instance, GPT-3 could generate incorrect or outdated medical terminology, diagnoses, or treatment methods. Moreover, the study did not thoroughly analyse or evaluate the quality and efficacy of the synthetic data generated by GPT-3, nor did it address the ethical and legal issues involved in applying synthetic data to medical dialogue summarisation. Questions such as how to use synthetic data in a responsible and transparent manner, how to safeguard the privacy and informed consent rights of patients and healthcare personnel, and how to mitigate or prevent potential biases or harms that might arise from synthetic data were not addressed.

This section's methods comparison and the code download links are shown in Table 8, and the comparison of experimental evaluation results is presented in Table 9. In their evaluation, it is important to note that Gao et al. [1] did both ordinary and subjective-part-included assessments for the regular model and the data-augmented trained model. For ease of presentation, the evaluation results listed in Table 9 represent the average scores obtained by both the regular model and the data-augmented trained model.

4. Question answering systems

This section explores the PLM-based medical question-answering systems.

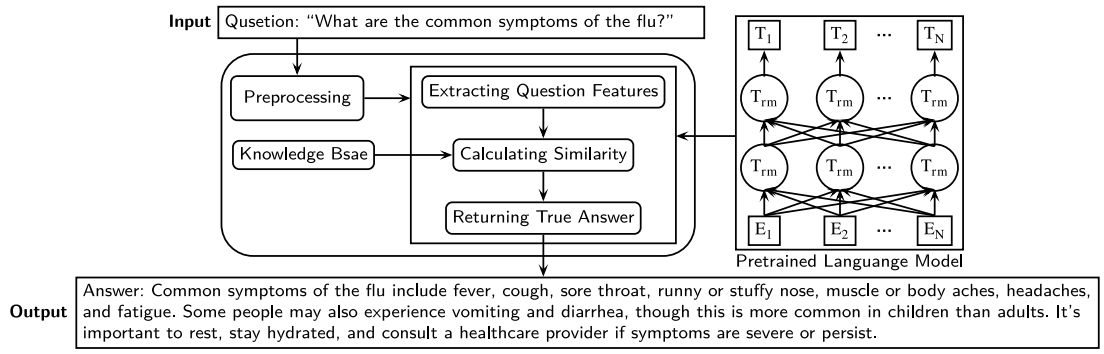


Fig. 7. PLMs-based model diagram for question answering system in medicine.

Table 10
Datasets for medical QAS.

Dataset	Used by	Language	Domain	Sample size	Download link
BioASQ	[3,39]	English	Biomedical	4719 questions	https://github.com/dmis-lab/bioasq-biobert
NCBI	[3]	English	Biomedical	793 papers, 2783 annotations	https://www.ncbi.nlm.nih.gov/datasets/
CHIP-STS	[5]	Chinese	Biomedical	6750 samples, 1821 annotations	https://tianchi.aliyun.com/dataset/95414
MEDIQA-2019	[33]	English	Medical	Diverse, from multiple sources	https://github.com/yuanh2015/diseaseBERT
TRECQA-2017	[33]	English	General	53,417/11,177/1442 Q&A in train/development/test set	https://trec.nist.gov/data/qa.html
CORD-19	[32]	English	COVID-19	Over 100,000 papers	https://github.com/allenai/cord19
SQuAD	[32]	English	General	10,570 papers and 100,000 Q&A pairs	https://rajpurkar.github.io/SQuAD-explorer/
MedMCQA	[39]	English	Medical	183,822/4183/6150 Q&A in train/development/test set	https://medmcqa.github.io/
PubMedQA	[39]	English	Medical	450/50/500 Q&A in train/development/test set	https://pubmedqa.github.io/
MedQA	[39]	English	Medical	10,178/1272/1273 Q&A in train/development/test set	–

4.1. Basic concepts

Question Answering Systems (QAS) [99] are computer programs or AI models designed to understand natural language questions and provide accurate, concise answers. QAS aims to mimic human understanding and comprehension of questions, generating answers based on existing knowledge or data sources. QAS can be divided into two main categories: (1) *Retrieval-based systems*: These systems rely on predefined question–answer pairs or databases to retrieve the most relevant answer to a given question. They use techniques like keyword matching, information retrieval, and pattern matching to identify the best match from a set of predefined answers. (2) *Generation-based systems*: These systems generate answers by understanding the meaning of questions and creating responses from scratch. They typically use NLP techniques such as semantic parsing, syntactic analysis, and machine learning algorithms to understand questions and generate coherent, context-appropriate responses.

QAS are widely used in various fields including general knowledge, specific domains (like medicine, law and finance), and even personalised domains (like customer support). They can be implemented as standalone applications, integrated into search engines or virtual assistants, or deployed as chatbots.

4.2. Main methods

Some of the main methods used in QAS include (1) *Rule-based methods*: The most traditional approach, where the system parses and answers questions based on preset rules. These rules might involve keyword identification, syntactic analysis, and pattern matching. (2) *Retrieval-based methods*: Here, the system searches for information in a large collection of documents (like the internet or specific databases) to find answers relevant to the posed question. This often involves keyword search, relevance assessment, and document ranking. (3) *Knowledge graph-based methods*: These methods use structured knowledge sources, like knowledge graphs or databases, to provide precise answers. These systems understand the relationships and attributes between entities, thereby offering more accurate answers. (4) *Machine learning and deep learning methods*: The most common approaches in modern QAS. By training models on large datasets, systems learn how to understand and answer questions. Deep learning is particularly adept at extracting information from complex data patterns. (5) *Hybrid methods*: Many modern systems combine several of the above methods to improve accuracy and efficiency. For example, a system might use both retrieval-based methods and deep learning to understand and respond to questions. (6) *Conversational systems and interactive Q&A*: These systems not only answer individual questions but can maintain a dialogue, offering more relevant and personalised answers based on context or conversational history.

Fig. 7 shows a general architecture of a question-answering system in the medical context, emphasising using a PLM. It details a multistage process where a medical question is inputted and then subjected to preprocessing. Subsequent steps include question features extracted and analysed for similarity against a knowledge base, ultimately leading to an accurate answer. While the diagram features a PLM examplified by BERT to facilitate these tasks, the framework also allows alternative methods to perform these tasks. It implies the flexibility of the framework. The PLM, if employed, synthesises the processed information to return the most accurate answer.

4.3. Advantages

Large PLMs like BERT, GPT-3 have the following advantages in building QAS for the medical field: (1) Exposure to a vast array of text data during the pre-training process, endowing them with rich language knowledge and deep understanding of natural language, aiding in better comprehension of medical questions expressed in natural language. (2) Models like BERT use bidirectional context during pre-training, enabling a deep understanding of the context of questions and paragraph text, thereby inferring correct answers, crucial for medical domain Q&A. (3) PLMs provide excellent sentence and word representations, aiding in retrieving relevant medical paragraphs to answer questions and capturing semantics and semantic relations. (4) Fine-tuning PLMs on medical text data such as research papers, electronic records, textbooks can adapt them to the language style of medical subjects and acquire specific domain knowledge. (5) The availability of large annotated medical Q&A datasets enables effective fine-tuning

of PLMs through supervised learning for QA tasks. (6) Pre-trained representations can be adapted effectively to medical QA tasks with less data than training a model from scratch, overcoming the challenge of limited specific domain training data. (7) PLMs fine-tuned for medical QA tasks have achieved state-of-the-art results on benchmarks, proving their ability to understand complex health-related questions and provide accurate answers.

4.4. Basic application steps

When applying PLMs to QAS in the medical domain, the following steps can be taken:

(1) Choose an appropriate PLM as the base model, selecting models that have been pre-trained on large-scale text corpora, such as BERT, RoBERTa, or ALBERT, and preferably considering larger-scale models. (2) Collect high-quality medical text data, such as research publications, clinical trial reports, medical textbooks, doctor-patient dialogues as datasets. (3) Preprocess the datasets, including cleaning, tokenisation, converting to lowercase to transform them into the format required by PLMs. Then, perform supervised fine-tuning on medical text datasets using annotated Q&A pairs, enabling the model to learn domain-specific knowledge. (4) During fine-tuning, add a question–answer output layer on top of the PLM architecture for predicting answers. Train over multiple epochs using techniques such as cyclic learning rates, warm-up, and early stopping to fine-tune the model effectively. (5) Evaluate the fine-tuned model's QA performance on unseen medical questions, iterating through adjustments in hyperparameters or training processes. To improve accuracy, integrate multiple fine-tuned PLMs, or supplement training data with additional knowledge resources like ontologies, research summaries. (6) Deploy the fine-tuned QAS into applications like chatbots, search engines, and virtual assistants, enabling it to answer real-world medical questions. (7) Regularly retrain the model with new data to ensure it keeps up with the latest medical knowledge and terminology.

4.5. Evaluation metrics

The following are key metrics for evaluating QAS based on PLMs in the medical domain: (1) Accuracy. (2) F1 Score. (3) Precision. (4) Recall. (5) *Mean Reciprocal Rank (MRR)* [100]: The average of the reciprocal ranks of the first correct answer, focusing on ranking performance. (6) *Mean Average Precision (MAP)* [101]: The mean of average precision for each question, assessing overall precision. (7) *Confidence* [102]: The model's certainty level in its predicted answers. Ideally, confidence should be calibrated. (8) *Exact Match (EM)* [103]: Measures the degree of the exact match between the model's output and the correct answer. (9) Human Evaluation [104]: Compares the performance of machine learning models to human-level performance on specific tasks.

4.6. Datasets

Table 10 summarises some datasets used in medical QAS, including language, domain, sample count, and download links.

(1) BioASQ has been briefed in Section 3.6

(2) NCBI [105,106] is a crucial resource in the biomedical field, known for its array of databases and tools, including prominent ones like PubMed, GenBank, and BLAST. Based in the United States, NCBI (National Center for Biotechnology Information) offers extensive data and tools that are pivotal for various biomedical text mining and analysis applications. These applications encompass named entity recognition, relation extraction, and the construction of knowledge graphs. Additionally, NCBI provides a specialised Biomedical Named Entity Recognition (NER) dataset through its NCBI Disease Corpus, dedicated to annotating disease entities in PubMed papers, further emphasising its role as a comprehensive resource for biomedical research and data analysis.

(3) CHIP-STS [107] is a Chinese medical text semantic similarity judgment dataset provided by Ping An Healthcare Technology. The dataset includes question pairs related to 5 major disease categories, with output results being 0/1 labels, indicating whether the two questions are semantically similar. This dataset is part of the Chinese Medical Information Processing Challenge (CBLUE), which aims to promote the development of Chinese medical NLP technology and community.

(4) MedIQA-2019 [108] is a shared task held at the ACL-BioNLP 2019 workshop, aimed at advancing Natural Language Inference (NLI), Question Entailment (RQE), and their application in Medical Question Answering (QA). MEDIQA-2019 includes three subtasks: NLI, RQE, and QA. The NLI task involves determining the inference relationship (entailment, neutral, or contradiction) between two sentences, using the MedNLI dataset, which consists of 14,049 sentence pairs from MIMIC-III clinical records. The RQE task involves determining the entailment relationship between two medical questions, using the RQE dataset with 8588 medical question pairs. The QA task involves filtering and re-ranking answers automatically retrieved by the medical QA system CHiQA, using the MedQuAD dataset, which contains 47,000 question–answer pairs.

(5) TRECQA-2017 [109] is a dataset and evaluation task used in the TREC 2017 Live QA Track to assess real-time question-answering systems. It comprises questions and answers from Yahoo! Answers, categorised into medical and non-medical groups. The medical questions and answers were sourced and validated from NLM databases by professional medical information retrievers, while the non-medical ones were retrieved and verified from the web by TREC evaluators. The dataset is divided into training, development, and test sets with the ratio of 53,417/1117/1442 question–answer pairs. Originating from TREC's QA tracks from 1999 to 2004, TRECQA-2017 has two versions: the original and a 'clean' version. The clean version, often preferred, omits questions without answers or with only positive/negative answers in its development and test sets. It includes 1229/65/68 questions in its training/development/test sets, respectively, alongside the same number of question–answer pairs as the original.

(6) CORD-19 has been briefed in Section 3.6.

(7) SQuAD [59] is a reading comprehension dataset proposed by Stanford University, consisting of crowdsourced questions and corresponding answers based on Wikipedia papers, where each question's answer is a text fragment or interval from the related paper. SQuAD has two versions: 1.1 and 2.0. Version 1.1 includes over 100,000 Q&A pairs, while version 2.0 adds over 50,000 unanswerable questions to the 1.1 version, to test the models' resistance to interference.

(8) MedMCQA [110] is a multiple-choice question dataset specially designed for the medical field, aimed at testing and improving machines' understanding and application of medical knowledge through NLP technology, especially automatic question-answering systems and deep learning models. This dataset encompasses questions from various medical sub-fields, such as biochemistry, anatomy, and pharmacology, to cover a broad range of medical knowledge. It includes 182,822 training, 4183 development, and 6150 testing questions from AIIMS PG and NEET PG questions found online and in books.

(9) PubMedQA [111] is a question-answering dataset focused on biomedical science. It comprises questions and answers from PubMed literature covering various medical topics, including diseases, drugs, and treatment methods. The dataset aims to serve as a benchmark for evaluating the ability of natural language processing models to comprehend biomedical texts and answer related questions. The dataset consists of 450/50/500 training/development/testing questions constructed by combining PubMed article abstracts with altered article titles and answers related to the abstracts.

(10) MedQA [36] is a question-answering dataset focused on the medical domain. It comprises questions from various medical subfields covering clinical, laboratory, and imaging aspects. It consists of 10,178 training, 1272 development, and 1273 testing questions sourced from USMLE questions found online.

4.7. Important research works

4.7.1. Summary

In 2020, Yoon et al. [3] used the bidirectional Transformer model BioBERT, pre-trained on biomedical texts, to address challenges in biomedical QA, such as complex terminology and limited training datasets. The model encoded questions and context texts as representational data, then deployed a question-answering output layer on these encodings to predict answer spans' start/end positions. The model was fine-tuned on two biomedical question-answering datasets (BioASQ and NCBI), and results showed that their question-answering system model significantly outperformed previous methods based on BERT and Recurrent Neural Network (RNN) models, with an absolute error rate reduction of 8%–19%. This work highlighted the potential of domain-specific pre-training in enhancing technical domain question-answering performance.

In 2021, addressing the deficiency of retrieval-based question-answering systems in not fully using sentence semantic information, Zhao et al. [5] developed a BERT-based deep text matching model, namely Potential Topic Extraction Medical BERT (PT-McBERT). This model could deeply capture semantic and syntactic information in natural language. They also introduced a potential topic extraction module that could extract keywords or phrases from questions and answers, using them as additional features for text matching.

In 2021, addressing the issue of limited samples in medical QA, Zhou and Zhang [33] developed a question-answering system that could answer questions online about various diseases, particularly COVID-19. The model first used BERT to obtain medical word embeddings, then used GPT-2 to generate additional questions, followed by T5-small to extract topics from questions, and subsequently computed the cosine similarity between the extracted topics and potential answers. Finally, answers were predicted based on similarity using XGBoost.

In 2023, to enable medical service experts, policymakers, and clinical scientists to track the latest significant research findings and assist the public in obtaining accurate and reliable information about the latest COVID-19 developments, Alzubi et al. [32] developed the COBERT question-answering system. It answered complex queries by retrieving from the publicly available CORD-19 research dataset, containing about 59,000 coronavirus-related literature. Initially, it used a TF-IDF vectorise to capture the top 500 most relevant papers. Then, it used a BERT model fine-tuned on the COVID-19 dataset to distil sentences from the filtered documents. Finally, based on sentence scores, it presented users with brief answers, relevant paper titles, and source papers.

In 2024, Bolton et al. [39] developed BioMedLM, a GPT-style PLM with 2.7 billion parameters, tailored for the biomedical field by training exclusively on PubMed articles. Designed to test if smaller, focused models could rival the performance of larger, general-purpose models like GPT-4 in biomedical tasks, BioMedLM demonstrates strong capabilities in multiple-choice biomedical question-answering and free-response medical inquiries. Achieving competitive results, such as 57.3% on MedMCQA and 69.0% on MMLU Medical Genetics, it presents an efficient, domain-specific, and transparent alternative to larger language models, addressing their issues of cost, computational demand, and opaque training data.

4.7.2. Main similarities and differences

The similarities among these papers are as follows. (1) They all used PLMs for biomedical QA, a practical and challenging task in medical information management and analysis. (2) They all employed Transformer-based architectures. (3) They evaluated their methods using standard evaluation metrics (such as F1, MRR, and Accuracy), which assess the quality of summaries by comparing generated summaries with human-written reference summaries.

The main differences between these papers are as follows. (1) *Type of Input Questions*: The first paper focused on answering factual questions, i.e., questions with one or a series of factual answers. The second paper mainly addressed non-factual questions, requiring more complex reasoning and analysis. The third paper concentrated on answering questions related to the COVID-19 pandemic. The fourth paper focused on answering medical questions involving medical knowledge and terminology. The fifth paper supported input of various medical problems. (2) *Type of Input Data*: The first and fifth papers used biomedical literature, such as PubMed abstracts and full-text papers, as the information source. The second paper used medical Q&A pairs from online health forums like WebMD and HealthTap. The third paper used COVID-19 scientific literature from the CORD-19 dataset. The fourth paper employed medical Q&A pairs from online health forums and biomedical literature. (3) *Methods Used*: The first paper used a pre-trained biomedical question-answering model (BioBERT), which was fine-tuned on multiple biomedical question-answering datasets. The second paper employed a deep text matching model that used convolutional neural networks and attention mechanisms to learn semantic representations of questions and answers. The third paper used a BERT-based COVID-19 question-answering system (COBERT), adapted from BioBERT and augmented with COVID-19 domain knowledge. The fourth paper employed a medical question-answering solution based on data augmentation and transfer learning (DATLMedQA), which used synthetic data and PLMs to enhance biomedical question-answering performance. The fifth paper employed an autoregressive, decoder-only Transformer which is similar in architecture to GPT-2, and used learned absolute positional embeddings for each position in the sequence. The comparisons of their methods with their code download links are shown in Table 11, and the comparisons of experimental evaluation results are in Table 12.

4.7.3. Limitations

Each of these methods has certain limitations:

The approach by Yoon et al. [3] was trained and evaluated only on a small biomedical question-answering dataset (BioASQ). Therefore, its performance might be challenging to apply to other biomedical datasets. Moreover, they did not compare their model with other state-of-the-art (SOTA) PLMs.

The method by Zhao et al. [5] focused only on question matching rather than end-to-end QA. Their evaluation was limited to a small dataset specific to consumer health questions and did not compare with other deep matching models.

The approach by Alzubi et al. [32] was restricted to questions related to COVID-19. The model architecture was relatively simple (BERT fine-tuning). They did not conduct ablation studies to understand the various components of the model deeply, and the scope of their evaluation was relatively limited.

The method by Zhou and Zhang [33] relied on data augmentation techniques, which could introduce noise, and it was evaluated only on a single biomedical question-answering dataset. They did not compare other domain transfer learning methods with training directly on in-domain data, and their depth of analysis was limited.

The model built by Bolton et al. [39], despite being smaller than models like GPT-4, training a 2.7 billion parameter model still requires substantial computational resources, which might be a barrier for some organisations or researchers.

5. Machine translation

This section provides a concise overview of research on using PLMs for machine translation in the medical domain.

Table 11
Comparison of QAS methods based on PLMs.

Reference	Publication venue	Focus	Pro	Con	Code
Yoon et al. (2020) [3]	European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (Top)	Optimising biomedical QA	Can effectively handle the scarcity of biomedical training data	Performance may not generalise to other biomedical datasets	https://github.com/dmis-lab/bioasq-biobert
Zhao et al. (2021) [5]	ACM International Conference on Intelligent Computing and its Emerging Applications (Ordinary)	Retrieval-based QAS do not fully use semantic information in sentences	Can capture the semantic and syntactic information of natural language	Not an end-to-end QAS	–
Zhou et al. (2021) [33]	Applied Sciences (JCR Q2, IF 2.7)	Samples in medical QA is limited	Can answer questions about various diseases (especially COVID-19) online	Limited to COVID-19 questions	https://github.com/ShuohuaZhou-NLPer/Question_Answering/
Alzubi et al. (2023) [32]	Arabian Journal for Science and Engineering (JCR Q2, IF 2.9)	Automatically tracks the latest major research results	Can help users obtain accurate and reliable COVID-19 latest information	Relies on data enhancement technology that may introduce noise	–
Bolton et al. (2024) [39]	arXiv Preprint	To address the issue of models being large and computationally costly	More accessible for researchers with limited resources	Performance on general NLP tasks or topics outside biomedicine might be limited	https://huggingface.co/stanford-crfm/BioMedLM

Table 12
Comparison of QAS methods' result based on PLMs.

Reference	Model	Evaluation dataset	Avg F1	Mac F1	MRR	Accuracy	Precision	Exact Match (EM)
Yoon et al. (2020) [3]	Unnamed	BioASQ	43.16	75.87	48.41	–	–	–
Zhao et al. (2021) [5]	PT-McBERT	CHIP-STS	89.42	–	–	–	–	–
		cMedQA	96.8	–	–	–	–	–
Zhou et al. (2021) [33]	Unnamed	MedIQA-2019	–	–	92.17	80.23	84.31	–
		TRCEQA-2017	–	–	58.49	80.5	63.71	–
Alzubi et al. (2023) [32]	COBERT	CORD-19	87.3	–	–	–	–	81.6
		SQuAD 1.1	87.5	–	–	–	–	81.6
Bolton et al. (2024) [39]	BioMedLM	BioASQ	–	–	–	95.7	–	–
		PubMedQA	–	–	–	74.4	–	–
		MedMCQA	–	–	–	57.3	–	–
		MedQA	–	–	–	54.7	–	–

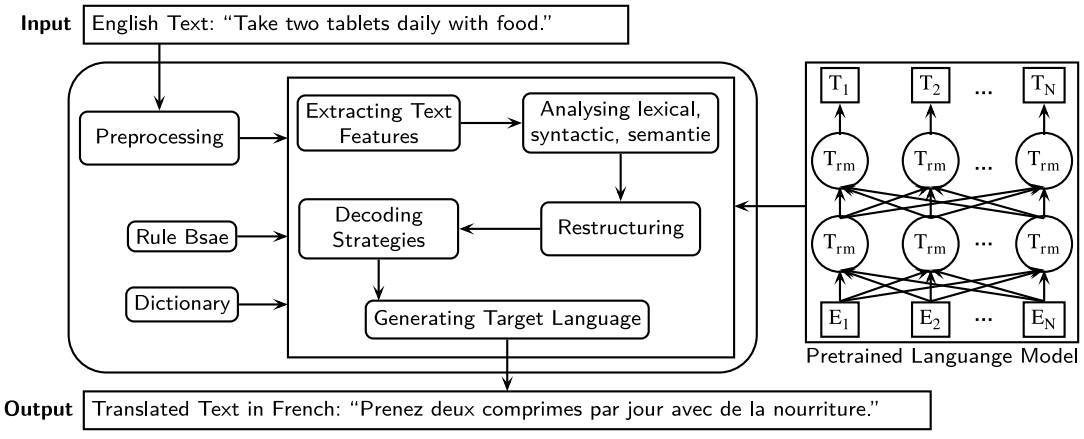


Fig. 8. PLMs-based model diagram for machine translation in medicine.

5.1. Basic concepts

Machine Translation (MT) [112] is the technology of using computers to automatically translate text or speech from one language into another. It is based on NLP technology and relies on statistical or neural network models trained on large bilingual corpora, thereby learning how to transform input text into output text in another language. MT aims to overcome language barriers and achieve cross-linguistic communication. The quality of MT can be assessed by metrics such as BLEU. With the development of neural network technology, its effectiveness is continually improving. Despite this, accurately translating ambiguous words, jargon, and metaphors remains a challenge. Current

research focuses on how to make MT more powerful and closer to the effect of human translation, to meet the needs of large-scale automatic translation in various practical scenarios.

5.2. Main methods

Some of the main methods used in MT are as follows. (1) *Rule-Based MT (RBMT)*: This method relies on a large number of grammar and vocabulary rules written by linguists. The translation process follows these rules to convert the source language into the target language. Although this method can be highly accurate, it requires much manual work and may not be flexible enough for complex language structures. (2) *Statistical MT (SMT)*: SMT uses large bilingual text corpora to "learn" how to translate. It generates translations by analysing the statistical relationships of words and phrases in these texts. This method is more flexible than RBMT but may not be as accurate in terms of grammar and sentence structure. (3) *Example-Based MT (EBMT)*: This method uses previously translated texts as "examples" or references. When faced with a new translation task, the system looks for similar examples and generates translations based on them. EBMT is particularly suitable for areas with highly consistent and specialised corpora. (4) *Neural MT (NMT)*: This is the most advanced method, using deep learning and neural networks to simulate how the human brain processes language. NMT systems can learn complex language patterns and generate fluent, natural translations. They are particularly good at handling long sentences and maintaining contextual coherence. (5) *Hybrid methods*: Some systems combine several of the above methods to leverage their advantages. For example, a system might combine SMT and NMT to improve translation accuracy and naturalness. With technological advancements, MT methods continue progressing, especially NMT, which has become the preferred method for many modern MT applications due to its outstanding performance.

Fig. 8 illustrates a framework for machine translation in medical applications, incorporating a PLM exemplified by BERT. It outlines a workflow encompassing preprocessing, feature extraction, lexical, syntactic, and semantic analysis, and the subsequent decoding and restructuring phases that culminate in generating a target language output. Each process phase is represented as a potential function of the PLM. Moreover, the framework also allows for integrating rule-based systems and dictionaries, highlighting PLM's versatility in either augmenting or supplanting conventional translation methodologies in medicine.

5.3. Advantages

PLMs show efficiency in building MT systems in the medical field for the following reasons. (1) Models like BERT, pre-trained on a vast amount of text data, have accumulated rich language knowledge, laying a solid foundation for medical language translation. (2) Models pre-trained using masked language modelling techniques can understand bidirectional context and relationships between words, which is crucial for language conversion. (3) Pre-trained representations capture semantic and syntactic information translatable across languages, establishing a model base for translation tasks. (4) Language models pre-trained on multilingual corpora can improve the translation quality of limited medical data through cross-linguistic information sharing and transfer learning. (5) Medical terminology databases (such as UMLS, ICD-10) can provide supplementary structured knowledge for fine-tuning. (6) Using PLMs is more efficient than training large translation models from scratch, which requires extensive data support. (7) PLMs fine-tuned on medical texts (like mBART and mT5) have achieved state-of-the-art results, further confirming their translation capabilities. (8) Pre-trained representations also provide initial settings for training custom medical translation models.

5.4. Basic application steps

MT in the medical field using PLMs can follow these general guidelines: (1) Select an appropriate multilingual PLM (mBART, XLM-R, or mT5) as the base model. (2) Collect parallel medical text data in the source and target languages, such as doctor-patient dialogues, medical research papers, drug labels. (3) Clean, tokenise, and normalise the data for preprocessing to fit the PLMs better. (4) Fine-tune the PLMs on parallel medical texts using techniques like back-translation, making them more suitable for translation tasks. (5) Effectively fine-tune large PLMs using techniques such as cyclic learning rates, differential fine-tuning, early stopping. (6) Evaluate translation quality on reserved medical test sets using metrics like BLEU, METEOR, and others. (7) Address issues like overfitting and underfitting by adjusting hyperparameters or using techniques like noise injection to enhance training data. (8) Deploy fine-tuned translation models in medical applications, such as medical record translation and multilingual symptom checker. (9) Continuously improve the model by retraining on the latest parallel medical text data. (10) For low-resource languages, transfer learning from related high-resource medical translation models.

5.5. Evaluation metrics

Key metrics for evaluating MT systems based on PLMs in the medical field are as follows. (1) Bilingual Evaluation Understudy (BLEU). (2) *SacreBLEU* [113]: A variant of the BLEU metric used to evaluate MT performance, aiming to provide a more consistent, standardised way of calculating BLEU scores. SacreBLEU addresses some issues in standard BLEU evaluation methods, such as different researchers using different preprocessing steps or reference corpora, making BLEU scores difficult to compare across different experiments. By standardising the evaluation process, SacreBLEU makes BLEU scores more comparable across different studies and systems. (3) *Translation Edit Rate (TER)* [114]: TER measures the number of edits (insertions, deletions, and substitutions) needed to convert a machine-generated translation into a reference translation. The lower the TER score, the higher the translation quality. (4) Metric for Evaluation of Translation with Explicit Ordering (METEOR). (5) Recall-Oriented Understudy for Gisting Evaluation (ROUGE). (6) *NIST Evaluation* [115]: Refers to a series of evaluation activities organised or participated in by the National Institute of Standards and Technology (NIST), calculating the weighted geometric mean of n-gram accuracy scores. It emphasises higher-order n-grams and assigns different weights based on their importance. The higher the NIST score, the better the translation quality. (7) Consensus-based Image Description Evaluation (CIDEr). (8) *Crosslingual Optimised Metric for Evaluation of Translation (COMET)* [116]: A neural network-based metric using machine learning to assess translation quality more accurately. It considers translations' fluency, fidelity, and semantic accuracy, providing a more comprehensive and in-depth evaluation of translation quality. COMET is designed to be cross-linguistic, universal, and adaptable to evaluating various language pairs.

5.6. Datasets

Table 13 summarises some datasets used in medical MT, including language, domain, sample count, and download links.

(1) WMT[1] is the main event for MT and MT research, with sample counts varying depending on the year and language pair. For example, the WMT 2021 Chinese–English MT training set contains 25 million bilingual sentence pairs, and the WMT 2017 English–German MT training set contains 4.5 million bilingual sentence pairs. The WMT dataset is provided by the Workshop on MT (WMT), which holds an annual MT evaluation task involving various language pairs and text domains.

[1] The Conference on MT https://machinetranslate.org/wmt.

Table 13
Datasets for medical MT.

Dataset	Used by	Language	Domain	Sample size	Download link
WMT	[6,7]	Multilingual	Translation	Dataset-specific	https://github.com/wmt-conference
ClinSpEn2022	[7]	Spanish	Biomedical	–	https://temu.bsc.es/clinspen/datasets/

Table 14
Comparison of MT methods based on PLMs.

Reference	Publication venue	Focus	Pro	Con	Code
Wang et al. (2021) [6]	Machine Translation (Top)	Enhancing biomedical translation tasks	Achieved good results in translation tasks from German/French/Spanish to English	Did not explicitly address issues of domain adaptability or terminological consistency	–
Han et al. (2023) [7]	Clinical NLP Workshop (Ordinary)	Applying large-scale multilingual PLMs to MT in the clinical domain	Demonstrated capability in translating different subtasks (such as clinical cases, clinical terms)	Only focused on one domain (clinical) and one data source (electronic health records)	https://github.com/facebookresearch/fairseq/tree/main/examples/wmt21

Table 15
Comparison of MT methods' result based on PLMs.

Reference	Model	Evaluation dataset	BLEU	BLEU-HF	ScareBLEU	METEOR	COMET	R-1
Wang et al. (2021) [6]	TMT	WMT2021 (EN→DE/EN←DE)	38.16/23.32	–	–	–	–	–
		WMT2021 (EN→FR/EN←FR)	48.05/43.90	–	–	–	–	–
		WMT2021 (EN→ES/EN←ES)	52.99/41.57	–	–	–	–	–
		WMT2021 (EN→RU/EN←RU)	29.98/25.43	–	–	–	–	–
Han et al. (2023) [7]	WMT21fb	ClinSpEn2022 (Task1)	–	0.3266	34.3	0.5868	0.3448	59.27
		ClinSpEn2022 (Task2)	–	0.2431	24.39	0.584	0.8584	66.99
		ClinSpEn2022 (Task3)	–	0.3859	40.71	0.5686	0.9908	71.99

The WMT dataset aims to promote the research and application of MT to help users access relevant knowledge.

(2) ClinSpEn2022 [117] is from the WMT22[2] Biomedical-MT conference's ClinSpEn2022 shared task. The goal of ClinSpEn2022 is to develop MT for the clinical domain in the Spanish-English language pair.

5.7. Important research works

5.7.1. Summary

In 2021, Wang et al. [6] applied the strategy of pre-training and fine-tuning to the task of biomedical translation. The PLM is initially self-trained on large-scale monolingual data and then fine-tuned on downstream bilingual data. Specifically, they used mBART for pre-training and achieved good results in translation tasks from German, French, and Spanish to English, with an average increase of 4.37 BLEU points.

In 2023, Han et al. [7] explored the potential application of Massively Multilingual Pre-trained Language Models (MMPLMs) in the field of clinical MT, particularly for languages not covered in the pre-training stage, to see if they could be adapted through transfer learning. They used Meta-AI's MMPLMs, pre-trained on seven language pairs and fourteen translation directions, including English, Czech, German, Hausa, Icelandic, Japanese, Russian, and Chinese. Then, they fine-tuned these models using carefully selected clinical domain data for the English–Spanish translation task, which was not included in the pre-training corpus. They evaluated the fine-tuned models on three subtasks (clinical cases, clinical terminology, and clinical outcomes), showing great potential for transfer learning MT in the clinical domain for completely uninvolved languages.

5.7.2. Main similarities and differences

The main similarities are as follows. (1) Both used neural MT models based on the Transformer architecture, the mainstream technology in the field of MT. (2) Both used PLMs (like BERT and mBART) with generic or domain-specific knowledge to enhance neural MT models. (3) Both applied data augmentation techniques, like back-translation and paraphrasing, to increase the quantity and diversity of training data for neural MT models. (4) Both focused on biomedical texts, a challenging yet significant area for MT applications.

The main differences between the two papers are as follows. (1) *The type of input and output languages*: the first paper focuses on translating biomedical texts between English and various European languages (such as German, French, Spanish, and Russian), while the second focuses on clinical text translation between English and other languages (such as Chinese, Italian, and Basque). (2) *The type of PLMs*: The first paper used a MT system developed by Tencent AI Lab based on different Transformer architectures, like mBART, a multilingual encoder–decoder model pre-trained on large-scale monolingual data. The second paper used Meta-AI's MMPLMs, also based on the Transformer architecture, but pre-trained on multilingual parallel data from different domains and sources. (3) *The type of transfer learning strategy*: The first paper used pre-training and fine-tuning strategies to adapt the PLM to the biomedical domain. The second paper used data augmentation and domain adaptation strategies, using synthetic data and domain-specific knowledge to improve the performance of the PLMs in the clinical domain. (4) *The type of evaluation datasets*: The first paper used the WMT21 biomedical translation task dataset, covering biomedical abstracts from multiple sources (like Scielo, EDP Sciences, Cochrane, and NHS). The second paper used the clinical NLP workshop dataset, including clinical cases, clinical terminology, and ontological concepts from different sources (like MedlinePlus, SNOMED CT, and UMLS). The comparison and code download links of their methods are shown in Table 14, and their experimental evaluation results are compared in Table 15.

5.7.3. Limitations

Moreover, there are some limitations to these methods. Wang et al. [6] used a single model for both directions, which may not capture the asymmetry and diversity of languages. They also did not explicitly address domain adaptability or terminological consistency issues, which could affect the translation quality of specific sub-domains or terminologies. Han et al. [7] focused only on one domain (clinical)

[2] The 7th Conference on MT https://www.statmt.org/wmt22/.

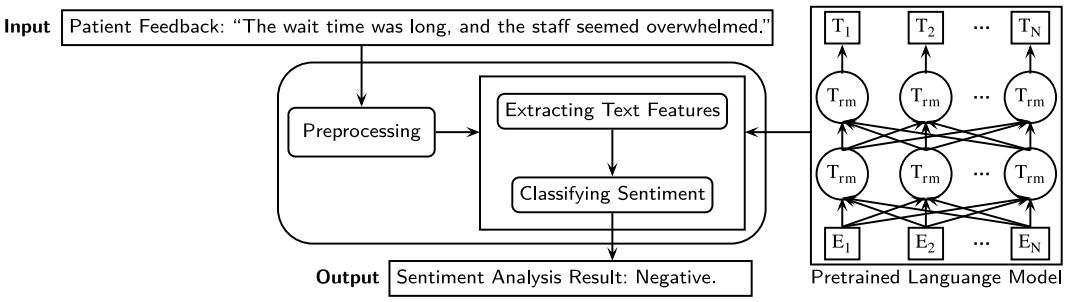


Fig. 9. PLMs-based model diagram for sentiment analysis in medicine.

and one data source (electronic health records), potentially limiting the applicability of their research findings to other areas and data sources in the biomedical field. They also did not assess the impact of transfer learning on other aspects of translation quality, such as fluency, adequacy, or readability.

6. Sentiment analysis

This section provides a review of work on PLMs for sentiment analysis of medical texts.

6.1. Basic concepts

Sentiment Analysis (SA) [118] employs NLP and text analysis techniques to identify and extract subjective information from textual data, such as opinions, emotions, and attitudes [119]. Its main functions include detecting whether the text expresses positive, negative, or neutral sentiments; differentiating factual statements from opinion statements; identifying the themes and targets of emotions; analysing sentiments about specific aspects of products or services; and determining the intensity of emotions. SA is widely applied in areas like brand monitoring, product feedback analysis, social media analysis, and public health monitoring, with the advancement of NLP technology continually enhancing its performance.

6.2. Main methods

Some main methods used in SA are as follows. (1) *Lexicon-based methods*: These methods use a predefined sentiment lexicon, where each word is assigned a positive, negative, or neutral sentiment label. The system determines the overall sentiment of the text by analysing the emotional tendency and intensity of the vocabulary in the text. This approach relies on accurate and comprehensive lexicons but may lack flexibility in handling complex language usage. (2) *Machine learning-based methods*: These typically involve training classifiers, such as Support Vector Machines (SVM), Naive Bayes classifiers, or Random Forests, using annotated datasets. During training, the model learns to identify emotions based on text features (e.g., word frequency, and n-grams). This approach can handle more complex linguistic structures but requires extensive annotated data for training. (3) *Deep learning methods*: Recently, deep learning methods, especially those based on neural networks, have become very popular in SA. For instance, Convolutional Neural Networks (CNN) and RNN, particularly Long Short-Term Memory (LSTM) networks and Transformer architectures have proven highly effective in capturing complex emotional patterns in text. These methods can automatically learn and extract features, usually providing higher accuracy and flexibility. (4) *Hybrid methods*: Some systems combine the above mentioned methods to enhance the accuracy and robustness of SA. For example, a system might use both lexicon-based methods and deep learning models to leverage the strengths of both approaches. (5) *Transfer learning and PLMs*: Using large PLMs like BERT, GPT, and XLNet for transfer learning has become a powerful approach in SA. These models are pre-trained on large-scale textual data and then fine-tuned for specific SA tasks to improve performance.

Fig. 9 provides a schematic for conducting sentiment analysis using a PLM exemplified by BERT. It outlines the process from input, such as patient feedback, through preprocessing and feature extraction to the final task of sentiment classification. This workflow demonstrates the PLM's potential to streamline the analysis by directly managing these stages, although alternative methods can also be applied within this flexible framework.

6.3. Advantages

PLMs are highly effective for SA in the medical field for reasons such as: (1) Training on large text corpora has endowed these models with a rich linguistic knowledge, enabling them to understand emotionally coloured language. (2) Contextual representations from PLMs can capture semantic relationships crucial for identifying emotions, sentiments, and opinions in text. (3) Transfer learning from general domain SA allows the models to broadly understand emotionally coloured words and phrases. (4) Fine-tuning on specific domain texts like patient feedback and doctor reviews enables better adaptation to the nuances of medical language and emotional expression. (5) Compared to traditional word-level embeddings, pre-trained contextual embeddings significantly enhance performance in SA. (6) Self-supervised pre-training enables SA with fine-tuning in limited annotated medical domain data. (7) Multilingual PLMs provide cross-linguistic transfer learning capabilities for low-resource language medical SA. (8) PLMs can be combined with medical ontological knowledge, enhancing understanding of contextually relevant emotions. (9) PLMs offer strong initialisation capabilities for training custom medical SA models.

6.4. Basic application steps

Some general guidelines for using PLMs for SA in the medical field are as follows. (1) Choose a suitable PLM that has been trained on a large textual corpus, like BERT or RoBERTa. (2) Collect domain-specific data, such as patient comments, drug reviews, doctor ratings, social media posts related to healthcare topics. (3) Perform preprocessing operations like text cleaning, tokenisation, padding to prepare for model training and fine-tuning. (4) Employ sentiment classification techniques, like adding a classification layer on top, to fine-tune the PLM on domain data. (5) Utilise strategies like different learning rates, gradual unfreezing, early stopping for effective model fine-tuning. (6) Evaluate the fine-tuned model on a reserved medical SA dataset using metrics like accuracy, F1 score. (7) Analyse examples of classification errors to gain valuable insights and iteratively improve by adding more training data, adjusting hyperparameters. (8) For aspect-level SA, provide aspect words like 'doctor', 'treatment' as additional inputs to the model. (9) Continuously collect new domain data to recalibrate the model, keeping it up-to-date.

Table 16
Datasets for medical SA.

Dataset	Used by	Language	Domain	Sample size	Download link
Twitter dataset	[35]	Multilingual	Biomedicine, Epidemiology	Over 152 million tweets	https://www.lero.ie/sites/default/files/journal%20copy%20of%20scraped%20and%20cleaned%20data.xlsx
Reviews	[34]	Multilingual	Biomedicine, Epidemiology	1287 user reviews	https://github.com/HSEIreland/covid-tracker-app/pulse

Table 17
Comparison of SA methods based on PLMs.

Reference	Publication venue	Focus	Pro	Con	Code download link
Quyen et al. (2021) [35]	International Journal of Environmental Research and Public Health (JCR Q1, IF 0)	Detecting and understanding anti-vaccine negative sentiment	Using BERT to identify negative sentiment in tweets	May not reflect views and sentiments in other languages and cultures	–
Rekanar et al. (2022) [34]	Irish Journal of Medical Science (JCR Q3, IF 2.1)	Increase app usage by recognising and addressing user concerns	Can make emotional judgments on new user feedback	May not capture the complexity and diversity of user emotions and perspectives	–

6.5. Evaluation metrics

The following are some common metrics used for evaluating SA in the medical field: (1) Accuracy. (2) Precision. (3) Recall. (4) F1 Score. (5) *Area Under the Receiver Operating Characteristic Curve (AUC-ROC)* [120]: Measures the model's ability to differentiate between positive and negative sentiments at different classification thresholds, providing a single numeric representation of the model's overall performance. (6) *Mean Squared Error (MSE)* [121]: Commonly used in regression-based SA tasks, MSE measures the average squared difference between predicted and actual sentiment values. Lower MSE values indicate better performance. (7) *Mean Absolute Error (MAE)* [122]: Also a common metric for regression-based SA tasks, MAE measures the average absolute difference between predicted and actual sentiment values. Lower MAE values indicate better performance. (8) *Fleiss' Kappa* [123]: Measures the consistency between model-predicted sentiments and actual sentiments, accounting for consistency that could occur by chance. It is mainly used to assess the scoring consistency between two or more raters. (9) *Confusion Matrix* [124]: Provides a clear view of the model's correct and incorrect classifications for each category.

6.6. Datasets

Table 16 summarises some datasets used in medical SA, including their languages, domains, sample size, and download links.

(1) The Twitter dataset [35] is collected and maintained by researchers at Panacea Lab, fetching tweets related to COVID-19 daily from the Twitter API and providing metadata, text, images, and videos of the tweets. It contains over 152 million tweets. The Twitter dataset aims to facilitate research and applications on the social impact and spread dynamics of COVID-19, helping scientists and the public to gain relevant knowledge.

(2) The Reviews [34] are collected from the Google Play Store and the iTunes App Store, which is provided by the Health Service Executive (HSE) of Ireland, containing user reviews of the COVID Tracker Ireland app from the Google Play Store and iTunes App Store. COVID Tracker Ireland is an application designed to help mitigate the spread of COVID-19, tracking user contacts and providing health advice and the latest information. The goal of the Google Play Store and the iTunes App Store dataset is to assess user satisfaction, functionality, and areas for improvement of the COVID Tracker Ireland app, helping developers and health authorities gain relevant feedback.

6.7. Important research works

6.7.1. Summary

Since the outbreak of COVID-19 in 2019, SA has been a focal point in the medical domain. Vaccination is an effective method to prevent coronavirus infection. However, the rapid development and dissemination of COVID-19 vaccines have led to distrust and anti-vaccination sentiments among some of the public, posing significant challenges to public health. To address this, in 2021, Quyen et al. [35] collected anti-vaccination tweets from Twitter and used BERT to identify negative sentiments in these tweets. Identifying and analysing public sentiment on social media helps devise anti-vaccination strategies, thus strengthening vaccination regimes and controlling the spread of COVID-19.

Effective control strategies during infectious disease outbreaks include identifying contacts and isolating them quickly. Promoting contact tracing programs can significantly shorten the time to identify and notify contacts. Therefore, health departments need to understand and address users' concerns to encourage using such trackers. Rekanar et al. [34] conducted SA-based research on all reviews of the HSE COVID-19 contact tracing app from the Google Play store and Apple Inc. to understand public attitudes. They used BERT to extract semantic representations of user feedback texts to initialise a text classification model. They then fine-tuned the BERT model on the user feedback comment dataset for sentiment classification (positive, negative, neutral), followed by connecting the fine-tuned BERT model as a text feature extraction module with a subsequent classification layer, creating an end-to-end SA model. Finally, the end-to-end model using fine-tuned BERT was used to perform sentiment judgment on new user feedback, outputting positive and negative classification results. This helps the developers of such tracking applications address user feedback, thereby improving application usage rates and achieving control over the epidemic.

6.7.2. Main similarities and differences

The similarities between these two papers are as follows. (1) Both used machine learning and NLP techniques to analyse social media data related to the COVID-19 pandemic. (2) Both aimed to provide insights and recommendations for public health agencies and policymakers to address the challenges and opportunities presented by the pandemic. (3) Both considered SA as a key method for measuring and understanding user or public opinions and emotions.

The main differences between these two papers are the following. (1) *Focus on different aspects of the COVID-19 pandemic*: one on attitudes towards vaccination and the other on contact tracing applications. (2) *Use different data sources*: one from Twitter and the other from app stores. (3) *Have different goals*: one is to detect and understand anti-vaccine negative sentiments, and the other to assess and improve user

Table 18
Comparison of SA methods' result based on PLMs.

Reference	Model	Evaluation dataset	F1	Accuracy	Precision	Recall	AUC	Fleiss'Kappa
Quyen et al. (2021) [35]	Bi-LSTM-128	Twitter dataset	45.5	89.8	44	47.2	85.8	–
	BERT		95.5	91.6	93.4	97.6	84.7	–
	SVM-linear		31.2	92.3	19.5	78.6	85.6	–
	Complement NB		27.1	88.8	23	32.8	62.7	–
Rekanar et al. (2022) [34]	BERT	App Reviews	–	–	–	–	–	0.693

satisfaction with the application. (4) *Employed different SA methodologies*: one used a three-class scheme (*pro-vaccine*, *anti-vaccine*, and *neutral*), while the other used a four-class scheme (*strong positive*, *weak positive*, *strong negative*, and *weak negative*). Their methodologies are compared in Table 17, and the experimental evaluation results are compared in Table 18. Neither paper provided related experimental codes.

6.7.3. Limitations

There are some limitations to these methods. Rekanar et al. [34] focused only on English tweets, which may not reflect the perspectives and emotions of other languages and cultures. They used a keyword-based method to collect tweets, potentially missing some anti-vaccine-related tweets not containing the keywords or including irrelevant tweets that do contain them. Moreover, they relied on PLMs and lexicons for SA and topic modelling, which may not capture the nuances and specificities of anti-vaccination discourse. They did not conduct a detailed evaluation of their models' and lexicons' performance or accuracy in anti-vaccine tweets, nor did they compare them with other methods or baselines.

Quyen et al. [35] analysed feedback only from the Google Play store, which may not adequately represent the views and experiences of users from other platforms or sources. They used a simple rule-based method to categorise feedback into positive, negative, and neutral, which may not fully capture the complexity and diversity of user emotions and opinions.

7. Named entity recognition

This section brief summarises the application of PLMs towards named entity recognition on medical.

7.1. Basic concepts

Named Entity Recognition (NER) [125] is an NLP technique used to identify and extract defined entities from unstructured text, such as people, organisations, locations, numbers, dates [126]. The process of NER includes text tokenisation, detecting entity sequences using rules or machine learning models, classifying detected entities into predefined categories, and disambiguating entities with the same name. NER is widely applied in healthcare, e-commerce, finance, and other domains for tasks ranging from chatbots to knowledge management. Advances in machine learning and NLP technologies, particularly those based on trained annotated text corpora, have continually improved the accuracy of NER systems.

7.2. Main methods

Some main methods used in NER are as follows. (1) *Rule-based methods*: These rely on predefined rules and patterns to identify entities. For example, regular expressions can be used to recognise entities that fit specific patterns, like phone numbers or email addresses. The advantage of this method is its high accuracy, but it lacks flexibility and scalability, requiring extensive manual work. (2) *Dictionary-based methods*: This involves using specific domain dictionaries or lists of entities to identify entities in the text, for example, using a list containing all known city names to recognise locations. The limitation of this method is that it cannot recognise new entities not in the dictionary. (3) *Machine learning-based methods*: These use annotated datasets to train machine learning models to recognise and classify entities automatically. These methods typically include traditional machine learning techniques like SVM, Random Forests, Conditional Random Fields (CRF). (4) *Deep learning-based methods*: In recent years, deep learning technologies have made significant progress in the NER domain. Neural networks, especially RNN and LSTM networks, are more effective in handling sequential data and capturing contextual information in text. More advanced models like Bidirectional Encoder Representations from Transformers (BERT) and other Transformer models have demonstrated superior performance. (5) *Transfer learning and PLMs*: Using PLMs (such as BERT, GPT) trained on large datasets and then fine-tuning them for specific tasks can significantly improve NER. This approach leverages the powerful language understanding capability of PLMs to enhance adaptability to new domains and low-resource scenarios. (6) *Ensemble methods*: Sometimes, combining multiple methods (like rules, machine learning, and deep learning) can improve the overall performance of the system, especially when dealing with diverse and complex datasets.

Fig. 10 outlines an architecture for NER in medical texts, detailing a process that begins with preprocessing, extracting text features, recognising entities, and classifying them. Though the structure also allows for incorporating alternative approaches to accomplish these tasks, it can be processed more effectively by a PLM exemplified by BERT, which is indicated on the right.

7.3. Advantages

PLMs excel in medical domain NER tasks for a number of reasons. (1) PLMs have a broad understanding of real-world entities (like people, places, organisations), which can be leveraged for medical entity recognition. (2) Contextual word representations in pre-training capture syntactic and semantic information useful for entity detection and disambiguation. (3) Models trained with masked language models can learn contextual relationships between words and entities, providing information for entity classification. (4) Large PLMs encode a vast amount of world knowledge in their parameters, compensating for limited domain-specific training data. (5) Fine-tuning on domain-specific texts like medical journals, patient records, and clinical notes adapts models to medical entity types, writing styles, and terminologies. (6) Compared to traditional word embedding methods, pre-trained embeddings provide better representations, improving entity detection in the medical domain. (7) For low-resource languages, multilingual PLMs allow cross-linguistic transfer learning. (8) PLMs like BioBERT and ClinicalBERT have achieved state-of-the-art results in biomedical NER tasks. (9) PLMs offer a strong starting point for training custom medical NER models for specific tasks.

7.4. Basic application steps

Some general guidelines for using PLMs for NER in the medical domain are as follows. (1) Choose a suitable PLM trained on a large textual corpus, such as BERT, RoBERTa, or BioBERT. (2) Collect domain-specific text data containing entities, such as clinical records, research papers, medical textbooks. (3) Annotate the dataset, marking the types of entities of interest, such as drugs, diseases, symptoms, treatments. (4) Preprocess the text data, converting it to a format accepted

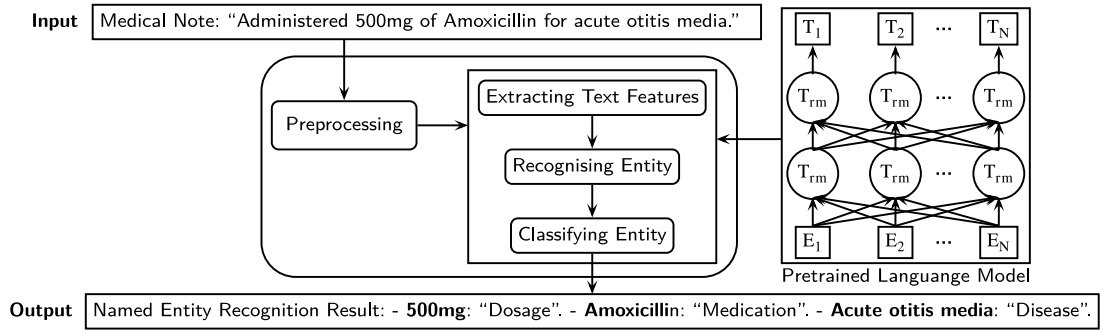


Fig. 10. PLMs-based model diagram for named entity recognition in medicine.

Table 19
Datasets for medical NER.

Dataset	Used by	Language	Domain	Sample size	Download link
NCBI	[9]	English	Biomedical	793 mentions, 2783 instances	https://www.ncbi.nlm.nih.gov/datasets/
BC5CDR	[9]	English	Biomedical	1500 with PubMed IDs	https://biocreative.bioinformatics.udel.edu/resources/corpora/biocreative-v-cdr-corpus/
BC4CHEMD	[9]	English	Chemical	10,000 instances	https://biocreative.bioinformatics.udel.edu/resources/corpora/chemdner-corpus/
BC2GM	[9]	English	Biomedical	20,000 instances	https://biocreative.bioinformatics.udel.edu/resources/corpora/biocreative-ii-corpus/
JNLPBA	[9]	English	Biomedical	2000 instances	https://www.geniaproject.org/shared-tasks/bionlp-jnlpba-shared-task-2004
LINNAEUS	[9]	English	Biomedical	–	https://linnaeus.sourceforge.net/
Species-800	[9]	English	biology	800 PubMed abstracts	https://species.jensenlab.org/
ChiMed	[10]	Chinese	Medical	200,744 pages Q&A	https://github.com/yuanzhe/YuanTian_ChiMed
Yidu-N7K	[28]	Chinese	Medical	–	https://opengk.cn/dataset/yidu-n7k
CCKS-2020-CEM	[28]	Chinese	Medical	26,414/5706 texts in train/test set	https://gitee.com/gongmingwang/ccks_ner
China Judgments Online	[11]	Chinese	Poison	Over 500 first-instance verdicts on drug-related cases	https://wenhsut.court.gov.cn/
EliIE	[31]	English	Biomedical	230 instances	https://clinicaltrials.org/
Covance	[31]	English	Disease	470 instances	In-house by Covance
Chia	[31]	English	Disease	1000 instances	https://clinicaltrials.org/

by the PLM (tokens, sequences). (5) Fine-tune the PLM by adding NER predictions and training on annotated data. (6) Use strategies like differential learning rates for effective fine-tuning of the PLM. (7) Evaluate the model's predictions based on annotations using metrics like precision, recall, and F1 score. (8) Analyse errors to gain insights into model deficiencies and additional training data requirements. (9) Use ensemble methods or integrate medical ontologies like UMLS to enhance model performance. (10) Deploy the fine-tuned model in applications like clinical documentation improvement and health record analysis. (11) Regularly retrain the model using new annotated data to handle the evolving medical terminology, abbreviations.

7.5. Evaluation metrics

The following are some common metrics used for evaluating NER in the medical domain: (1) Precision. (2) Recall. (3) F1 Score. (4) Accuracy. (5) *True Positive Rate (TPR) or Sensitivity* [127]: Measures the proportion of true positive predictions among all actual positive entities, calculated as the ratio of true positive predictions to the sum of true positives and false negatives. (6) *False Positive Rate (FPR)* [127]: Measures the proportion of false positive predictions among all actual negative entities, calculated as the ratio of false positive predictions to the sum of true negatives and false positives. (7) *Specificity* [127]: Measures the proportion of true negative predictions among all actual negative entities, calculated as the ratio of true negative predictions to the sum of true negatives and false positives. (8)*Matthews Correlation Coefficient (MCC)* [128]: A measure of the quality of binary classifications, it considers true positives, true negatives, false positives, and false negatives, providing a balanced metric that takes into account all four outcomes. (9) *Average Precision (AP)* [129]: Calculates the average precision at different recall levels, commonly used in imbalanced datasets scenarios, and has unique advantages in evaluating multi-class NER models. (10) Bilingual Evaluation Understudy (BLEU).

7.6. Datasets

Table 19 summarises some datasets used in medical NER, including their languages, domains, sample size, and download links.

(1) NCBI has been briefed in Section 4.6.

(2) BC5CDR [130] is a dataset provided by the BioCreative V project, is designed for Biomedical NER. It focuses on annotating chemical substances, diseases, and chemical-disease interactions, extracted from 1500 PubMed papers. This comprehensive dataset includes detailed annotations of relationships between chemicals, diseases, and drugs, making it a valuable resource for biomedical research, particularly in understanding the intricate interactions among these entities.

(3) BC4CHEMD [131] is a collection of 10,000 PubMed abstracts, provided by the BioCreative IV shared task, totalling 84,355 chemical entities, manually annotated by chemical literature curators.

(4) BC2GM [132] is a biomedical NER dataset provided by the BioCreative II Gene Mention Recognition task, annotating gene mentions from PubMed papers. This dataset contains 150,00 training and 5000 test samples, each being a sentence with one or more gene mentions. The purpose of this dataset is to evaluate and enhance the gene mention recognition capability in the biomedical field. The evaluation

Table 20
Comparison of NER methods based on PLMs.

Reference	Publication venue	Focus	Pro	Con	Code download link
Naseem et al. (2021) [9]	International Joint Conference on Neural Networks (Major)	Reduce the cost of manual annotation of clinical free text data	Pre-annotation using active learning can reduce the required labelling time	Smaller data sets	https://github.com/usmaann/BioALBERT
Li et al. (2021) [21]	Journal of the American Medical Informatics Association (JCR Q1, IF 6.4)	Developing methods to generate synthetic clinical notes	Training using a corpus of clinical records generated by GPT-2	Synthetic data may not fully reflect the characteristics of real clinical texts	–
Tan et al. (2021) [10]	International Conference on Asian Language (Ordinary)	Extract valuable information from a large amount of clinical medical texts	Build a Chinese medicine NER model based on different PLMs and aggregate the results	The scope of evaluation is limited	–
Wen et al. (2021) [28]	Data Intelligence (JCR Q1, IF 3.9)	Make fuller use of unlabelled medical text to improve medical NER	Use unlabelled medical text to train PLMs within the domain	Reliance on text quality and accuracy can lead to errors in the model	–
Fu et al. (2021) [11]	IEEE International Conference on Emergency Science and Information Technology (Ordinary)	The corpus in the drug-related field is not rich	A large-scale drug-related text data set has been constructed	The quality and completeness of the drug-related text data set may be affected sexual influence	–
Li et al. (2022) [31]	BMC Medical Informatics and Decision Making (JCR Q1, IF 3.5)	Accurately extract meaningful information from complex and unstructured clinical trial documents	Utilise multiple data sources to complete clinical trial eligibility criteria text	There may be certain limitations in evaluation	–

metric for this dataset is the F1 score, a measure of overall performance based on precision and recall.

(5) JNLPBA [133] is a biomedical dataset from the GENIA 3.02 version corpus, created through controlled searches of MEDLINE. From this search, 2000 abstracts were selected and manually annotated according to a small taxonomy of 48 categories based on chemical classification. The GENIA corpus annotations used 36 terminal categories.

(6) LINNAEUS [134] is a general-purpose dictionary matching software capable of processing various document formats in the biomedical domain (MEDLINE, PMC, BMC, OTMI, text). It can generate various types of outputs (XML, HTML, tab-separated value files, or save to a database). It also contains methods to act as a server (including load balancing among multiple servers), allowing clients to request matches over the network.

(7) Species-800 [135] is a species entity corpus based on manually annotated abstracts. This corpus consists of 800 PubMed abstracts with identified biological mentions. To increase the taxonomic mentions' diversity in the corpus, 100 abstracts were selected from each of the following eight categories to compile these 800 abstracts: bacteriology, botany, entomology, medicine, mycology, protozoology, virology, and zoology. Annotations were made on 800 abstracts, focusing on the species level. However, higher taxonomic groups' mentions (like genus, family, and order) were also considered.

(8) ChiMed [136] is a dataset for Chinese medical question-answering, containing 200,744 pages of Q&A, each page being a medical question asked by a patient and one or multiple medical answers given by one or more professional doctors. The ChiMed dataset is provided by researchers at the Beijing Institute of Technology, containing Chinese medical QA data from Baidu Zhidao, with the task being to select the best answer from candidate answers given a question.

(9) Yidu-N7K is a data set of CHIP 2019[3] clinical terminology standardisation task using the Chinese language, with an unspecified sample number. The Yidu-N7K dataset is provided by Yidu Cloud, containing medical terminologies from electronic medical records in different hospitals, with the task being to match these terminologies with standard terminologies, outputting binary labels of 0/1.

(10) CCKS-2020-CEMR [137] is a dataset for clinical electronic medical record entity recognition, using Chinese language, with training set 26,414 texts and test set 5706 texts. The CCKS-2020-CEMR dataset is provided by CCKS 2020, containing texts from electronic medical records in different hospitals, with the task being to identify and classify medical entities in the texts, such as symptoms, diagnoses, drugs, outputting the entity type and location offset.

(11) China Judgments Online [11] is a dataset for legal text analysis. The China Judgments Online dataset provided by the China Judgments Online contains first-instance verdicts of drug-related cases from courts across the country, with the task being to extract relevant information from the verdicts, such as defendants, charges, sentences, drug types, drug quantities.

(12) EliIE [138] is a dataset specifically for Alzheimer's disease, containing 230 files from Clinicaltrials.org, each file annotated with information about the purpose, design, results of clinical trials.

(13) Covance [139] is a dataset covering various diseases, containing 470 files produced internally by Covance, each file annotated with information about the definition, symptoms, diagnosis, treatment of diseases.

(14) Chia [140] is a dataset covering various diseases, containing 1000 files from clinicaltrials.org, each file annotated with information about the purpose, design, results of clinical trials.

7.7. Important research works

7.7.1. Summary

In 2021, Naseem et al. [9] developed a PLM called bioALBERT for biomedical NER. The model is trained on a massive biomedical corpus to capture NER related to the biomedical background, thus improving the accuracy of biomedical NER. In 2021, Li et al. [21] used GPT-2 to generate a clinical notes corpus and then trained their medical NER model on this corpus. In 2021, Tan et al. [10] built Chinese medicine NER models based on different PLMs, then aggregated results from multiple models to obtain the final recognition results. In 2021, Wen et al. [28] used PLMs and domain dictionaries

[3] The 5th China Conference on Health Information Processing (CHIP) http://www.cips-chip.org.cn:8000/evaluation.

Table 21
Comparison of NER methods' result based on PLMs.

Reference	Model	Evaluation dataset	F1	Precision	Recall	BLEU-1/-2/-3/-4
Naseem et al. (2021) [9]	BioALBERT	NCBI	96.13 (Avg)	97.21 (Avg)	95.09 (Avg)	–
		BC5CDR	97.78 (Avg)	99.68 (Avg)	95.94 (Avg)	–
		BC4CHEMD	96.17 (Avg)	97.77 (Avg)	94.63 (Avg)	–
		BC2GM	96.42 (Avg)	98 (Avg)	94.89 (Avg)	–
		JNLPBA	83.37 (Avg)	85.63 (Avg)	81.2 (Avg)	–
		LINNAEUS	99.72 (Avg)	99.96 (Avg)	99.49 (Avg)	–
		Species-800	98.75 (Avg)	98.99 (Avg)	98.51 (Avg)	–
Li et al. (2021) [21]	GPT-2	Homemade	–	–	–	97.69/92.39/85.17/77.28
Tan et al. (2021) [10]	ZEN2-large + BiLSTM + CRF + Radical features	ChiMed	84.43	–	–	–
	ZEN2-base + BiLSTM + CRF + Radical features		84.14	–	–	–
Wen et al. (2021) [28]	Fusion model	CCKS-2020-CEMR	88.7/95.3 (strict/relax)	–	–	–
Fu et al. (2021) [11]	ELECTRA-BiLSTM-CRF	China Judgments Online	94	–	–	–
Li et al. (2022) [31]	PubMedBERT	Covance	71.5	–	–	–
		EliIE	83.2	–	–	–
		Chia	62.2	–	–	–

to build medical entity dictionaries. They trained a RoBERTa-based PLM using unlabelled medical text through masked language modelling. Then, they fine-tuned the PLM using a BiLSTM-CRF sequence labelling model and evaluated it on unannotated Chinese electronic health record datasets, showing excellent entity recognition effects. In 2021, Fu and Bu [11] built a large-scale drug-related text dataset and developed an ELECTRA-BiLSTM-CRF NER model that achieved 94% F1 score on the drug-related text dataset. In 2022, Li et al. [31] compared PLMs like BioBERT, BlueBERT, PubMedBERT, SciBERT, BERT and SpanBERT for NER with three existing clinical trial eligibility criteria corpora, finding that PubMedBERT outperformed other PLMs in specific domains.

7.7.2. Main similarities and differences

These papers also have some similarities: (1) They aim to leverage PLMs that capture contextual information from large corpora to improve the accuracy and generalisability of biomedical text NER. (2) They evaluate their approaches on standard benchmarks or datasets. (3) They report promising results demonstrating that their methods are more effective than existing methods or baselines.

These studies have differences in several aspects: (1) *Data sources used for training and evaluation*: some use synthetic clinical notes, some use real-world medical records, some use annotated corpora from different domains, and some use combinations of the above. (2) *Language used*: some models are specific to English, some Chinese, and some multi-lingual. (3) *PLMs used*: some use BERT, some ELECTRA, some RoBERTa, and some use variants or combinations. (4) *Additional features or methods*: some use active learning, some use domain dictionaries, some use other techniques like radical features, smart face image processing to improve performance or efficiency.

Their method comparisons and code download links are shown in Table 20, and the experiment evaluation comparisons in Table 21. Notably, for the study of Naseem et al. [9], the experimental evaluations of BioALBERT models trained on different versions (Base, Large) and different datasets (PubMed, PubMed+PMC) are performed. For convenient show, the results in Table 21 are the average scores of the four models on different datasets.

7.7.3. Limitations

These methods still have some limitations. Naseem et al. [9] compared several active learning methods on a small dataset of only 1500 sentences, so more extensive comparisons are needed to draw fuller conclusions. Li et al. [21] relied on synthetic data for experiments, which may not fully reflect the characteristics of real clinical text. Also, they only evaluated on a single dataset, lacking cross-dataset validation. Tan et al. [10] focused only on Chinese medicine NER, which has limited scope. Moreover, these studies did not compare their methods with current state-of-the-art NER methods. Wen et al. [28] used potentially noisy unlabelled datasets, which may affect model performance. Analysis of model components was also relatively insufficient. Fu and Bu [11] only evaluated on figure legends, so it is unclear if the results apply to real clinical text. Overall, the analysis in these studies was relatively limited. As for Li et al. [31], although they compared multiple corpus models, they did not perform ablation experiments or hyperparameter optimisation, so their evaluation may have some limitations.

8. Information extraction

This section delineates research on using PLMs for information extraction tasks in medicine.

8.1. Basic concepts

Information Extraction (IE) [141] encompasses various techniques, including NER, relation extraction, event extraction, attribute identification, co-reference resolution, and template filling. Its main goal is to automatically extract key information from unstructured text and transform it into structured forms like templates and databases. This function enables the conversion of unstructured text data into actionable information and knowledge, supporting a wide range of applications across different fields. Modern IE systems rely on machine learning and deep learning technologies, which are widely applied to large-scale annotated text corpora, providing technical support for the automatic extraction and integration of information.

8.2. Main methods

Some main methods used in IE include (1) *Text mining and NLP technologies*: Using NLP to parse and understand text data, including tasks like part-of-speech tagging, NER, dependency parsing. These technologies help systems understand text's linguistic structure and meaning, thereby identifying key information. (2) *Computer Vision technologies*: When IE involves image or video data, computer vision technologies are used to recognise and classify visual content. For example, image recognition can be used to detect objects or scenes in photos, or video

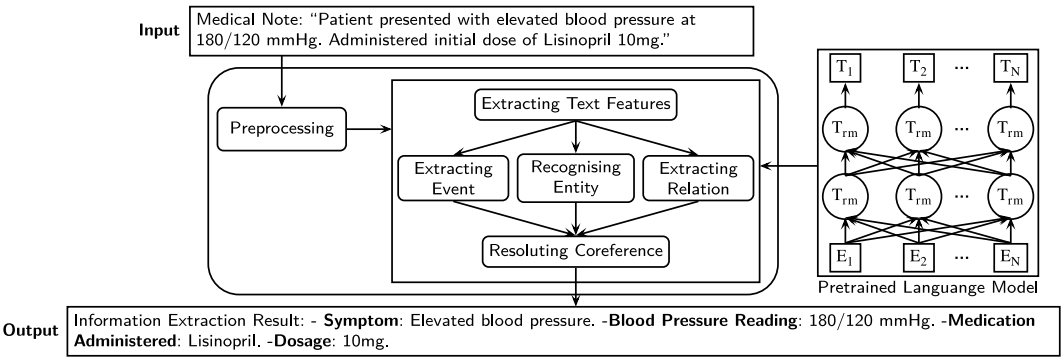


Fig. 11. PLMs-based model diagram for information extraction in medicine.

Table 22
Datasets for medical IE.

Dataset	Used by	Language	Domain	Sample Size	Download Link
Tweet Corpora	[12]	English	Healthcare	12,331 annotated healthcare-related tweets	https://github.com/medeffects/tweet_corpora
Twitter	[4]	English	COVID-19	12 million COVID-19 related tweets	https://twitter.com/
PreClude	[13]	English	Healthcare	3285 healthcare-related statements	–

analysis can be used to extract information from specific frames. (3) *Audio analysis technologies*: For audio data, IE might involve speech recognition (converting speech to text) or audio event recognition (like identifying specific sounds or background noise). (4) *Pattern recognition and machine learning*: These technologies are used to identify patterns and trends in data. Machine learning models, especially deep learning models, are widely used for IE tasks in various data types (text, images, audio). (5) *Signal processing technologies*: For audio and video data, signal processing techniques (like filtering, transformation) can be used to extract and enhance specific information in the data. (6) *Semantic analysis and knowledge graphs*: These methods are used to extract and understand high-level semantic information in text. Knowledge graphs help systems understand relationships between entities, while semantic analysis techniques are used to understand the implied meaning and context in the text.

Fig. 11 presents a framework for IE in the medical field, illustrating a multi-step process that includes preprocessing, text feature extraction, event extraction, entity recognition, relation extraction, and co-reference resolution, which a PLM exemplified by BERT can orchestrate. The workflow is designed to process medical text and extract structured data, such as symptoms, medication, and dosages, demonstrating the flexibility to employ PLM capabilities or alternative methods for each step.

8.3. Advantages

For several reasons, PLMs are highly effective for IE in the medical field. (1) Pre-training provides models with extensive language knowledge, including entities, relationships, events, which can be transferred to medical IE tasks. (2) Contextual representations in PLMs capture syntactic and semantic cues useful for recognising medical entities, relationships, and events. (3) Models pre-trained with masked language models can learn contextual links between words that are helpful for co-reference resolution. (4) Large PLMs encode vast amounts of world knowledge, supplementing medical text training data. (5) Fine-tuning PLMs with medical texts adapts the model to domain terminologies, writing styles, and information types. (6) Pre-trained contextual embeddings offer advantages over traditional embeddings for medical NER. (7) Transfer learning from general domains reduces the need for extensive annotated medical texts. (8) The latest research in medical IE confirms the effectiveness of PLMs in this field. (9) PLMs provide a solid baseline for training custom medical IE models.

8.4. Basic application steps

Some general guidelines for using PLMs for IE in the medical field are as follows. (1) Choose an appropriate PLM trained on large text corpora, such as BERT or ClinicalBERT. (2) Collect relevant medical text data, like clinical notes, discharge summaries, medical journals. (3) Annotate text data for entities, relationships, events, attributes, to generate training data. (4) Preprocess and tokenise text data to fit the input format of the PLM. (5) Fine-tune the PLM with annotated data for different IE tasks, such as NER, relation extraction. (6) Use strategies like differential learning rates and gradual unfreezing for effective fine-tuning of large PLMs. (7) Evaluate using metrics like precision, recall, F1 score, and analyse errors for insights and data improvement. (8) Incorporate medical ontologies like UMLS as additional knowledge sources during the fine-tuning process. (9) Add a template prediction module on top of the PLM for end-to-end information filling. (10) Train integrated models on different IE tasks to enhance overall performance, then deploy them for extracting information and populating knowledge bases from medical records, publications. (11) Regularly retrain the model with new annotated data to track constantly evolving medical terminologies and keep the model updated.

8.5. Evaluation metrics

The following are some common metrics used for evaluating IE in the medical field: (1) Precision. (2) Recall. (3) F1 Score. (4) Accuracy. (5) True Positive Rate (TPR). (6) Specificity. (7) False Positive Rate (FPR). (8) Matthews Correlation Coefficient (MCC). (9) Average Precision (AP). (10) Bilingual Evaluation Understudy (BLEU).

8.6. Datasets

Table 22 summarises some datasets used in medical IE, including their languages, domains, sample size, and download links.

(1) Tweet Corpora [142] is a dataset for analysing personal medical experiences. It contains 12,331 tweets with sentiment and topic annotations, each tweet published by Twitter users describing their personal experiences in the medical field, such as diseases, treatments, prevention.

(2) Twitter [4] is a dataset for COVID-19-related analysis, containing 12 million tweets related to COVID-19, collected from 22 January 2020 to 29 April 2020. Each tweet includes the tweet's text, metadata, images, and videos.

Table 23
Comparison of IE methods based on PLMs.

Reference	Publication venue	Focus	Pro	Con	Code download link
Zhu et al. (2020) [12]	Health Text Mining and Information Analysis (Ordinary)	Identifying personal experience tweets about drugs	Conducted a binary classification task on the dataset and continuously updated the dataset	Could not cover all possible aspects and outcomes of drug use	–
Jiang et al. (2021) [4]	European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (TOP)	Identifying symptom expressions in social media texts	COVID-19 symptoms can be identified from natural language texts using GPT-LM	May limit the generalisation and reliability of their models	–
Gatto et al. (2023) [13]	The International AAAI Conference on Web and Social Media (TOP)	Exploring the health conflict detection task	Used three PLMs	Cannot cover all possible health information types and sources	–

(3) PreCluDe [143] is a dataset for the health/medical field, containing 3285 health advice statements, each provided by professional doctors or health experts, covering different health topics like diet, exercise, sleep, psychology.

8.7. Important research works

8.7.1. Summary

In 2020, Zhu et al. [12] developed a method using RoBERTa and updated PLMs to identify personal drug effect experiences through tweets. They used a dataset of 2000 tweets mentioning the names and effects of drugs, such as side effects, benefits, and dosage, and categorised them into personal experiences and non-personal experiences. They performed binary classification on the dataset, fine-tuned the RoBERTa model, and continuously updated it with new tweets from Twitter using a query-based approach. The experimental results showed that the fine-tuned and updated model achieved high accuracy (over 95%), significantly outperforming the baseline model using a pre-trained BERT classifier.

In 2022, Jiang et al. [4] developed a novel few-shot learning method using generative pre-trained transformer language models (GPT-LM) to identify COVID-19 symptoms from natural language text. They used a dataset of 1000 tweets mentioning COVID-19 symptoms (such as fever, cough, and shortness of breath) and labelled them as COVID-19 positive or negative. They fine-tuned GPT-2, GPT-3, and GPT-Neo on a small subset (10%) of the dataset using a prompt-based few-shot text classification approach. The experiments demonstrated that the fine-tuned models achieved high accuracy (over 90%), outperforming the benchmark model using an unfine-tuned or updated pre-trained BERT classifier.

In 2023, Gatto et al. [13] explored the task of Health Conflict Detection (HCD), which involves identifying and classifying conflict types between two pieces of health advice. They used a dataset of annotated pairs of health advice from various online sources (like WebMD, Mayo Clinic, and Healthline) and developed three novel systems based on PLMs (such as BERT, RoBERTa, and DeBERTa) to perform HCD. Using various metrics, they evaluated the developed systems' performance on different conflict types (like contradictions, inconsistencies, and incompleteness). The results showed that their systems outperformed the baseline systems, with DeBERTa-v3 achieving the best results in all experiments, with an average F1 score of 0.68. Moreover, they introduced a human-machine interactive synthetic data augmentation method to expand the existing HCD dataset.

8.7.2. Main similarities and differences

The similarities between these papers are as follows. (1) They all used PLMs for health IE, tasks involving the use of NLP technology to identify valuable information from health-related text. (2) They all adopted a similar fine-tuning process and objective to BERT, namely training models by predicting masked words and the next sentence. This helps the model learn general language patterns and representations for transfer in subsequent tasks. (3) They all demonstrated the effectiveness and practicality of their models in various medical IE tasks, including binary classification, multi-class classification, multi-label classification, and TS. These tasks are crucial for extracting valuable information from health-related text and supporting various applications, such as clinical decision support, drug discovery, disease diagnosis, and health education.

The main differences between these methods are as follows. (1) *The tasks they used to fine-tune and evaluate models*: Zhu et al. [12] identified personal drug effect experiences, Jiang et al. [4] identified COVID-19 symptoms, and Gatto et al. [13] identified conflicting pairs of health information. (2) *The datasets they used for fine-tuning and evaluating models*: Zhu et al. used a dataset of tweets reporting personal medication experiences, Jiang et al. used a small dataset mentioning COVID-19 symptoms, Gatto et al. used a dataset related to health advice containing contradictory information about COVID-19. (3) *The base models for their PLMs*: Zhu et al. used RoBERTa, Jiang et al. used GPT-2, GPT-3, and GPT-Neo, and Gatto et al. used BERT and RoBERTa. (4) *The results they achieved in their tasks*: Zhu et al. improved performance by updating RoBERTa's parameters with new data, Jiang et al. achieved comparable or better results with less data and computation, and Gatto et al. outperformed other baseline methods in both binary and multi-class tasks.

Their methodologies are compared in Table 23, and the experimental evaluation results are compared in Table 24. It is noteworthy that Zhu et al. [12] evaluated models trained with different training steps (160k/106k/53k); Jiang et al. [4] assessed models trained with different small sample data (1-shot/5-shot/10-shot/15-shot); Gatto et al. [13] analysed the performance of models on specific labels for different conflict types. For ease of presentation, the results in Table 24 are the average scores of all training steps, the average scores of models trained with all small sample data, and the average scores for all specific labels. None of these three papers made their codes available publicly.

8.7.3. Limitations

These methods still have some limitations. Zhu et al. only used tweets reporting personal medication experiences, which may not cover all possible aspects and outcomes of drug use. They also did not compare their model with other PLMs (such as BERT or GPT-2), which might affect the validity and reliability of their results. They could consider using other types of texts describing drug effects (like reviews, comments, or papers) and comparing their model with other PLMs to validate their method's effectiveness further. Jiang et al. only used a small subset of data mentioning COVID-19 symptoms, which may limit the generalisability and reliability of their model. They also did not evaluate their model in other languages or domains, which might affect

Table 24
Comparison of IE methods' result based on PLMs.

Reference	Model	Evaluation dataset	F1	Precision	Recall	Accuracy	AUC
Zhu et al. (2020) [12]	RoBERTa (Updated)	Tweet Corpora	74.9 (Avg)	73.9 (Avg)	76.03 (Avg)	87.6 (Avg)	93.3 (Avg)
	RoBERTa (Twitter)		71.53 (Avg)	69.83 (Avg)	73.37 (Avg)	85.77 (Avg)	91.63 (Avg)
Jiang et al. (2021) [4]	GPT-2-125M	COVID-19 related tweets	4.1 (Avg)	2.4 (Avg)	14.73 (Avg)	–	–
	GPT-2-345M		10.07 (Avg)	7.73 (Avg)	15.67 (Avg)	–	–
	GPT-NeoX-20B		58.03 (Avg)	55.1 (Avg)	61.43 (Avg)	–	–
Gatto et al. (2023) [13]	Random Guess	Homemade	31 (Avg)	–	–	–	–
	TFIDF+RF		14 (Avg)	–	–	–	–
	GloVE		41 (Avg)	–	–	–	–
	SBERT		42 (Avg)	–	–	–	–
	BERT		54 (Avg)	–	–	–	–
	B+C BERT		54 (Avg)	–	–	–	–
	RoBERTa		63 (Avg)	–	–	–	–
	DeBERTa-v3		68 (Avg)	–	–	–	–

the portability and applicability of their model. They could consider using larger and more diverse health-related text datasets to improve their work and evaluate their model in different languages or domains (like Chinese or Spanish, cancer or diabetes). Gatto et al. only used online health information websites as sources for conflicting health information, which might not cover all possible health information types and sources. They also did not consider the context and credibility of online information, which might affect the detection of conflicting information. They could expand their work by using health information sources from other platforms (like news papers, blogs, forums, and social media posts) and incorporating context and credibility features into their model.

9. Medical education

This section reviews the literature on using PLMs for medical education, with an emphasis on their use for educational examinations.

9.1. Basic concepts

Medical education [144] is the process of training and guiding medical professionals, such as doctors, nurses, pharmacists, aiming to equip them with fundamental scientific understanding, clinical skills, professional ethics and conduct, communication skills, and clinical reasoning abilities. Medical education covers basic science education, clinical internships, simulation and model practice, communication training, medical ethics and legal knowledge, development of clinical reasoning abilities, and ongoing continuing education. Methods of medical education include classroom teaching, workplace learning, online education, and simulations dedicated to nurturing medical students' professional knowledge, skills, competence, teamwork abilities, and commitment to lifelong learning. In the process of medical education, students' knowledge, clinical skills, academic records, and professional behaviour are comprehensively assessed to ensure they possess the necessary capabilities and qualities required in the medical profession.

9.2. Advantages

For medical education, PLMs are extremely useful as they provide extensive language knowledge and contextual word representations, aiding in understanding medical concepts and terminology. After fine-tuning medical textbooks and journals, PLMs can reflect educational content effectively. These models can act as knowledge bases, covering real-world knowledge needed to supplement medical learning. Also, PLMs can be customised as personalised medical learning assistants, generate tailored quizzes, identify knowledge gaps, provide assessment feedback, and even answer medical science questions when adequately fine-tuned. Overall, the language abilities and knowledge gained through pre-training make PLMs highly valuable in supporting all aspects of medical education.

9.3. Basic application steps

Some suggestions for using PLMs in medical education are as follows. (1) Choose a suitable PLM, such as BERT or RoBERTa, trained on different texts. (2) Collect educational materials like textbooks, case studies, research papers, and classroom lectures. (3) Preprocess materials by tokenisation, cleaning, and format them as inputs for the PLM. (4) Fine-tune the model on medical education data for downstream tasks like paragraph ordering, question generation. (5) For personalised learning, add memory components to remember students' interactions. (6) Generate multiple-choice questions or summaries, focusing on key points to test understanding. (7) Enable clarifying dialogues when the model is uncertain to improve interpretability. (8) Analyse students' answers to questions to identify knowledge gaps and misunderstandings. (9) Use classification and generation functions to provide feedback on written assessments. (10) Continuously train new educational materials and student interactions for ongoing improvement. (11) Evaluate model-generated content through expert reviews and student surveys. (12) Deploy PLMs as virtual medical tutors, quiz applications to aid learning.

9.4. Important research works

9.4.1. Summary

In 2023, Kung et al. [36] trained ChatGPT in medical dialogues and assessed its performance in the United States Medical Licensing Examination (USMLE), a standardised test for medical students and graduates. They reported that ChatGPT achieved an average score of 67.8% across all four USMLE steps, comparable to human candidates' pass rate. They also analysed the strengths and weaknesses of ChatGPT and discussed the potential applications and impacts of using large PLMs for AI-assisted medical education. They concluded that ChatGPT is a useful tool for medical students and educators, but ethical and societal issues need to be addressed.

In 2023, Gilson et al. [29] analysed ChatGPT's performance on 350 questions from the USMLE to assess its medical knowledge. ChatGPT scored a total of 52.9%, performing worst on basic science questions and best on clinical management questions. Although ChatGPT showed some strengths in clinical reasoning, it performed poorly in visual interpretation, spatial reasoning, and applying basic science knowledge. The study concluded that ChatGPT has potential but limited medical knowledge, requiring further research to understand the impact of PLMs on medical education and assessment. Concerns were raised that information provided by PLMs might be incorrect, and efforts are underway to develop robust evaluation methods to distinguish between human knowledge and AI-generated knowledge.

In 2023, Eysenbach et al. [30] reviewed the potential applications of ChatGPT and AI in medical education. They used simulated dialogues to showcase ChatGPT's language generation capabilities while also pointing out its limitations in understanding complex medical concepts. They

Table 25
Analysis and comparison of the application of PLMs in medical education.

Reference	Publication venue	Focus	Pro	Con
Kung et al. (2023) [36]	PLoS Digital Health (Ordinary)	Evaluates ChatGPT Performance on the United States Medical Licensing Examination (USMLE)	Scores sufficient for medical licensure	Ethical and social issues that need to be addressed
Gilson et al. (2023) [29]	JMIR Medical Education (Non-SCI, Q1, IF 3.6)	Evaluates ChatGPT Performance on the United States Medical Licensing Examination (USMLE)	Demonstrated Strengths in Clinical Reasoning	Weak Performance in Visual Interpretation, Spatial Reasoning, and Applying Basic Science Knowledge
Eysenbach et al. (2023) [30]	JMIR Medical Education (Non-SCI, Q1, IF 3.6)	Review of ChatGPT and potential applications of AI in medical education	Use of simulated conversations to demonstrate	Possible risks of misinformation and plagiarism

discussed how AI, like ChatGPT, can contribute to medical education by providing personalised tutoring, generating study content, and offering feedback. However, they also emphasised the risk of misinformation and plagiarism if AI is used irresponsibly. The paper highlighted the need for more rigorous research to assess AI's educational effects, ethics, and impact on learning. They called for further research to integrate AI into medical curricula and develop appropriate usage guidelines. Moreover, they believed it necessary to differentiate and evaluate human knowledge and AI-generated knowledge.

9.4.2. Main similarities and differences

The similarities between these papers are as follows. (1) All papers focus on assessing ChatGPT's performance in medical education and the USMLE exam. (2) They aim to analyse the impact of ChatGPT and other similar PLMs on future medical education and assessment. (3) These papers unanimously consider AI systems like ChatGPT as a significant advancement in the field of medical education, potentially changing the way medical learning and testing are conducted. (4) All papers acknowledge the need for further research and evaluation to fully understand the strengths and limitations of ChatGPT in medical education. (5) They unanimously agree that given the rapid progress of AI as shown by ChatGPT, medical curricula and assessments may need to be adjusted.

The differences between these methods are as follows. (1) Kung et al. directly tested ChatGPT on USMLE exam questions. They found that ChatGPT could pass Step 1 and Step 2 CK (Clinical Sciences, with Steps 1, 2, and 3 representing different assessment stages) in the US medical practitioner assessment, achieving enough scores to apply for a medical license, but encountered difficulties in Step 3. (2) Gilson et al. did not directly test ChatGPT but evaluated its potential based on reviews of its capabilities. They believed that ChatGPT currently cannot reliably pass the USMLE exam but may be able to in the future. (3) Eysenbach et al. analysed the impact of PLMs like ChatGPT on medical education and assessment but did not conduct original tests on ChatGPT. (4) Kung et al. provided original data on ChatGPT's performance in the USMLE exam, while Gilson et al. relied on secondary reviews. Eysenbach et al. provided analysis but no original testing. (5) Kung et al. tested a broad sample of all USMLE exams, but Gilson et al. only analysed Step 1. (6) Kung et al. concluded that ChatGPT can currently pass Steps 1/2 CK, but Gilson et al. stated that it currently cannot reliably pass any USMLE exam. (7) Eysenbach et al. discussed the need to adjust medical education and assessment based on the progress of AI as demonstrated by models like ChatGPT. Their application analysis is compared in Table 25.

9.4.3. Limitations

There are several limitations to these methods. Firstly, the sample size of questions tested on ChatGPT (Kung et al.) is relatively small, so the results might not fully represent the breadth of medical knowledge. Moreover, these studies rely on published reviews of ChatGPT rather than direct testing (Gilson et al.), so their assessment process may not be robust enough.

A common limitation of these methods is the lack of in-depth understanding of ChatGPT's reasoning process; it is unclear exactly how it arrives at its responses. Testing was limited to standard multiple-choice questions and did not evaluate clinical reasoning skills. The peer-review and quality control mechanisms for ChatGPT are not clearly defined. Moreover, there may be potential biases in ChatGPT's training data that could affect its performance. The ethical implications of AI in medical education have not been thoroughly researched. If there is a lack of transparency in ChatGPT's system, it could lead to the emergence of incorrect assumptions.

10. Relation extraction

This section summarises research focused on PLMs for extracting relationships from texts in the medical domain.

10.1. Basic concepts

Relation extraction (RE) [145] involves identifying semantic relationships between entities in the text using techniques such as NER, entity linking, relation detection, classification, and knowledge base population. It aims to determine whether a specific relationship exists between specific entities and classify the relationship type. This enables the automatic extraction of structured information about entities and relationships from unstructured text, storing it in knowledge bases. RE is crucial in building knowledge bases, creating social networks, and question-answering systems. Machine learning is widely applied in modern methods, such as CNN and RNN trained on annotated text. Challenges include handling ambiguous, implicit, and indirect relationships. Overall, RE transforms unstructured text into structured facts about entities and their relationships.

10.2. Main methods

Some main methods in RE are as follows. (1) *Neural pattern learning*: This method uses deep learning networks to learn underlying patterns from the text directly. Compared to traditional feature-based methods, neural pattern learning can automatically learn complex features from data, improving the precision and recall of RE. (2)*End-to-end models*: End-to-end learning models minimise preprocessing and post-processing steps, learning directly from raw text to the final RE output. These models, often based on deep learning, can use information in the text more comprehensively. (3) *Multi-task learning*: In this approach, the RE task is combined with other NLP tasks (such as entity recognition, part-of-speech tagging). Through this, models can learn useful representations from related tasks, thus improving the performance of RE. (4) *Cross-language and cross-domain RE*: This method aims to develop RE models that can handle text in multiple languages or across different domains. This typically involves transfer learning and domain adaptation techniques. (5) *Attention mechanisms and context modelling*: Using attention mechanisms to capture key information in the text better, especially when dealing with long texts. Attention

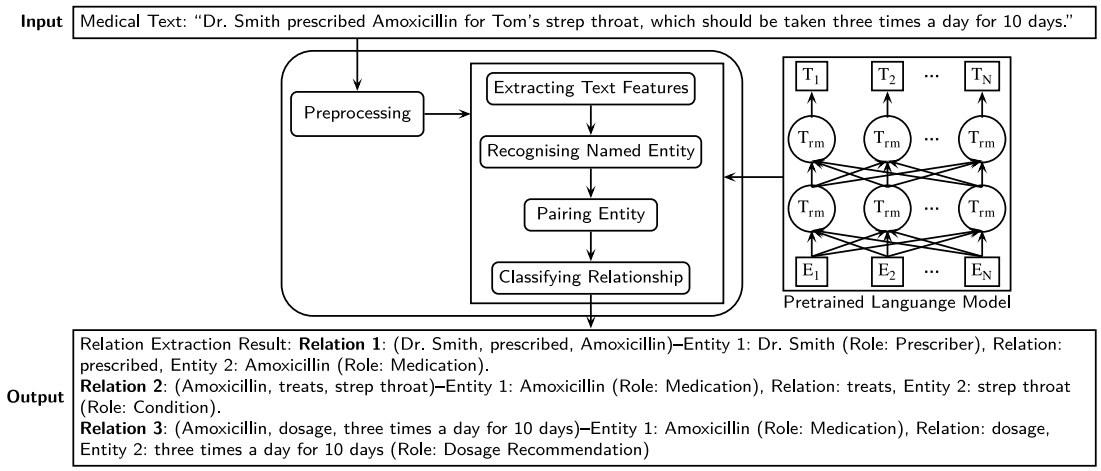


Fig. 12. PLMs-based model diagram for relation extraction in medicine.

Table 26
Datasets for medical RE.

Dataset	Used by	Language	Domain	Sample size	Download link
n2c2	[14,40]	English	Medical	505 discharge abstracts	https://portal.dbmi.hms.harvard.edu/projects/n2c2-nlp/
i2b2	[14,37]	English	Biomedical	310 de-identified discharge abstracts	https://www.i2b2.org/NLP/DataSets/
BC5CDR	[37]	English	Biomedical	1500 PubMed papers	https://biocreative.bioinformatics.udel.edu/resources/corpora/biocreative-v-cdr-corpus/
TCM	[37]	Chinese	Traditional Chinese Medicine	Abstracts of 106,150 papers	http://artertiminer.org/TCMRelExtr
MIMIC-III	[8]	English	Clinical Care	112,000 clinical report records	https://paperswithcode.com/dataset/mimic-iii
MADE1.0	[40]	English	Medical	10,000 questions	https://github.com/burgersmoke/MADE-CRF
CMED	[26]	Chinese	Medical	500 clinical notes	–

mechanisms help the model focus on text parts most relevant to specific relationships. (6) *Adversarial learning and data augmentation*: Using adversarial learning and data augmentation techniques to enhance the model's generalisability. These methods create challenging examples or increase the diversity of training data, helping models be more robust when faced with new or noisy data. (7) *Knowledge graph integration*: Integrating RE with knowledge graphs, using external knowledge to enrich the model's understanding. This approach is particularly suitable for complex entities and relationships, where additional knowledge can provide important contextual information.

Fig. 12 details a process flow for RE in the medical context, employing a PLM exemplified by BERT. It shows a procedure that begins with preprocessing, extracting text features, recognising named entities, pairing entities, and classifying their relationships to distil complex medical information from text. While this specialised task can be managed by a PLM, indicating its ability to execute the entire process, the depicted framework also supports using alternative techniques for each step.

10.3. Advantages

PLMs excel in RE in the medical field, thanks to the rich vocabulary, entity relationships, and dependency relations knowledge provided by the models, supporting accurate identification of relationships in the medical domain. The contextual word representations provided by PLMs encode semantic and syntactic cues, playing a key role in inferring relationships. Moreover, the understanding of contextual links between words by PLMs enhances their capability in RE. With the wealth of world knowledge embedded in large PLMs, they can effectively supplement the limited annotated data in the medical field. Fine-tuning on medical texts allows models to adapt to domain-specific terminologies and relationship recognition. Meanwhile, transfer learning from general domains reduces the dependence of RE tasks on large-scale annotated medical data. Overall, pre-training provides models with a strong foundation in NLP, while domain-specific fine-tuning ensures accuracy in extracting medical relationships.

10.4. Basic application steps

Suggestions for using PLMs for RE in the medical field are as follows. (1) Choose a suitable PLM, such as BioBERT or ClinicalBERT, trained on large corpora. (2) Collect relevant medical field text, such as research papers, clinical trials, case reports. (3) Annotate domain-relevant entities, relationships, and entity types in the text. (4) Preprocess the text, such as tokenisation, lowercasing, cleaning to ensure the text format meets the input requirements of the PLM. (5) Fine-tune the PLM on annotated data, then fine-tune the model on relation annotation tasks to predict relationships. Use strategies like differential learning rates, gradual unfreezing, for effective fine-tuning. Evaluate model performance using metrics like precision, recall, F1 score, and analyse errors to understand the model's limitations and additional data needs. (6) Deploy the fine-tuned model in applications such as building medical knowledge bases. (7) Regularly retrain the model with new annotated data to adapt to constantly changing terminologies and knowledge.

10.5. Evaluation metrics

Some key metrics for evaluating RE systems based on PLMs in the medical field are as follows: (1) Precision; (2) Accuracy; (3) Recall; (4) F1 Score; (5) Area Under the Receiver Operating Characteristic Curve (AUC-ROC); (6) Confusion Matrix; (7) Mean Reciprocal Rank (MRR); and (8) *Error rate* [146]: The proportion of incorrect relations extracted by the model (the lower the better).

10.6. Datasets

Table 26 summarises some datasets used in medical RE, including their languages, domains, sample size, and download links.

(1) n2c2 [147] involves 505 discharge summaries randomly extracted from the electronic medical record system of affiliated hospitals of Harvard Medical School. Each summary was de-identified and manually summarised by professional medical editors, generating a brief summary containing the patient's main problems, treatment processes, and outcomes.

(2) i2b2 [148] annotates entities and relationships related to the patient's medical history, diagnosis, and treatment in each summary. Entities include five types: Problem, Test, Treatment, Person, and Time, and relationships include four types: Problem–Test, Problem–Treatment, Test–Treatment, and Problem–Problem. Each relationship has a direction and a type, with direction indicating the dependency relationship between entities and type indicating the semantic relationship.

(3) BC5CDR has been briefed in Section 7.6.

(4) TCM [149] is provided by Tsinghua University's ArnetMiner project, containing abstracts of 106 150 papers, each with annotations of entities and relationships related to traditional Chinese medicine.

(5) MIMIC-III has been briefed in Section 3.6.

(6) MADE1.0 [150] is provided by the MADE project of the National Library of Medicine, USA, containing 10,000 questions, each drug-related, with one or more correct answers.

(7) CMED [26] is provided by the Institute of Computing Technology, Chinese Academy of Sciences, containing 500 clinical notes, each with annotations of entities and relationships related to the patient's condition, diagnosis, and treatment.

10.7. Important research works

10.7.1. Summary

In 2019, Wei et al. [14] explored state-of-the-art NLP technologies to extract meaningful relationships from unstructured clinical records. They developed a model based on BERT and fine-tuned BERT on an annotated corpus of clinical narratives to predict relationships between medical entities. Utilising pre-training of BERT, the model achieved an F1 score of 0.864, outperforming previous methods. This demonstrated the effectiveness of using large PLMs for transfer learning to extract information from electronic health records efficiently. The paper highlighted the potential of these advanced NLP technologies in unlocking knowledge from clinical texts, thereby achieving the goal of understanding patient data relationships in a wide range of clinical applications. Overall, the paper provided a paradigm for how modern AI can be applied to enhance the usation of electronic medical record data.

In 2019, Chen et al. [37] conducted research on improving RE from medical texts, employing general PLMs and task-specific fine-tuning. They developed a method that used the BERT model to initialise the weights of a bidirectional LSTM network, which was then fine-tuned on an annotated clinical record relation dataset. This framework combined the general language knowledge encoded in BERT with the specific needs of the target medical RE task. Their method achieved an F1 score of 0.882 on benchmark datasets, significantly better than baseline LSTM and traditional machine learning methods. The results indicated that combining PLMs with fine-tuning is an effective strategy for promoting deep learning techniques to extract information from electronic health records. This helps to construct knowledge from unstructured medical records better, supporting downstream clinical applications. In summary, the study provided a versatile framework to fully exploit the complementary advantages of pre-trained representations and task-specific customisation.

In 2019, Lin et al. [8] conducted research on using BERT to extract temporal relationships between medical events in clinical texts. They proposed fine-tuning a BERT-based neural network on annotated medical notes to identify temporal relationships within and between sentences. Their model achieved F1 scores of 0.824 and 0.632 for intra-sentence and inter-sentence temporal relationship extraction, respectively. This indicated the ability of BERT to learn context representations, capturing local relationships within sentences and longer-distance dependencies crossing sentence boundaries. The study emphasised the effectiveness of transfer learning using PLMs for clinical temporal IE. The process of generating timelines from medical records was simplified by handling intra-sentence and inter-sentence relationships in a single model. Overall, this research demonstrated the applicability of context language models like BERT in clinical temporal relationship extraction, better using the time-encoded information in patient notes.

In 2021, Yang et al. [40] conducted research on using the latest Transformer-based language models (such as BERT) to extract semantic relationships from electronic health records. They developed a BERT-based model, trained it on expert-annotated clinical relation datasets, and compared it with CNN and LSTM baseline models. The BERT-based model achieved a micro F1 score of 0.943, outperforming baseline models. This indicated that Transformer-based contextual representations are better suited for modelling relationships between medical entities than non-contextual CNNs and LSTMs. They further demonstrated that transfer learning from large PLMs has driven technological advancements in efficiently extracting information from unstructured electronic health records. Clinical RE can promote various downstream applications, such as knowledge base construction, question-answering, and decision support. Overall, this research highlighted the advantages of modern Transformer-based network models in fine-tuning for domain-specific NLP tasks in clinical fields.

In 2023, Ramachandran et al. [26] developed three novel BERT-based event detection models to extract medication changes and their attributes from clinical narratives. They used the Contextualised Medication Event Dataset (CMED), a corpus containing clinical records annotated with information about medication changes, such as type, initiator, temporality, possibility, and negation. They evaluated the performance of their proposed systems in identifying medication mentions and classifying attributes of medication changes. They experimentally showed that the developed systems outperformed baseline systems from previous work on CMED, with an overall average F1 score for medication change classification reaching 0.827.

10.7.2. Main similarities and differences

These studies have some similarities as follows. (1) Their common goal is to use deep learning methods to extract relations from unstructured clinical texts. (2) They all use transfer learning from large PLMs (such as BERT), rather than training neural networks from scratch. (3) Most methods fine-tuned BERT or BERT-based models for their target medical RE tasks. (4) Their research results show that PLMs perform better than baseline CNN, LSTM, and traditional machine learning methods. (5) These methods all emphasise the effectiveness of PLMs in representing clinical contexts. (6) They all focus on relation types critical to the clinical domain, such as semantic relations, temporal relations, and drug changes. (7) These methods in their evaluations all used expert-annotated relation corpora from clinical records and electronic health records. (8) All methods modelled clinical RE as a classification task based on pre-trained language encoder representations.

Despite these similarities, they also have some notable differences. (1) Wei et al. [14] focused on using BERT for RE, while Chen et al. developed a more general framework using PLMs for RE. (2) Wei et al.'s research emphasised evaluation of discharge summaries, Chen et al. [37] also used discharge summaries, while other papers covered a variety of data sources. (3) Lin et al. [8] emphasised extraction of temporal relationships, while other studies focused on semantic RE. (4) Yang et al. [40] compared BERT and other Transformer-based models with CNNs and LSTMs, while other studies focused on PLMs. (5) Ramachandran et al. [26] targeted specific relation types (drug changes), whereas other papers dealt more broadly with RE issues.

Table 27
Comparison of RE methods based on PLMs.

Reference	Publication venue	Focus	Pro	Con	Code download link
Wei et al. (2019) [14]	AMIA Annual Symposium (Major)	Extracting meaningful relationships from unstructured clinical records	Can predict relationships between medical entities	May not adequately represent the diversity of clinical narratives	–
Chen et al. (2019) [37]	Database (-)	Preventing overfitting of deep learning models used for medical RE	Combining the general linguistic knowledge encoded in BERT with the requirements of the target medical RE task	May result in limited generalisability of the model across the entire healthcare system limit	https://github.com/chentao1999/MedicalRelationExtraction
Lin et al. (2019) [8]	Clinical NLP Workshop (Ordinary)	Extracts the temporal relationships of medical events in clinical texts	Can identify temporal relationships within and between sentences in medical notes	Fails to fully meet all possible information needs	–
Yang et al. (2021) [40]	arXiv Preprint	Extracts semantic relationships from electronic health records	More suitable for modelling relationship dependencies between medical entities	May raise patient privacy issues	https://github.com/uf-hobi-informatics-lab/ClinicalTransformerRelationExtraction
Ramachandran et al. (2023) [26]	Journal of Biomedical Informatics (JCR Q1, IF 4.5)	Extracting medication changes and their attributes from clinical narratives	Employing contextual medication event datasets	Computational and data accessibility barriers remain for clinical practice	–

Table 28
Comparison of RE methods' result based on PLMs.

Reference	Model	Evaluation dataset	F1	Precision	Recall
Wei et al. (2019) [14]	Fine-Tuned BERT	n2c2	93.95 (Avg)	98.125 (Avg)	90.11 (Avg)
	Feature Combined BERT		91.75 (Avg)	96.49 (Avg)	87.48 (Avg)
	Fine-Tuned BERT	i2b2	76.79 (Avg)	75.33 (Avg)	76.76 (Avg)
	Feature Combined BERT		70.19 (Avg)	68.59 (Avg)	71.89 (Avg)
Chen et al. (2019) [37]	1d-CNN fine-tuning BERT with pre-trained model	BC5CDR	70.85	67.22	74.89
		TCM	92.82	95.18	90.61
		i2b2	70.85	67.22	74.89
Lin et al. (2019) [8]	BERT-MIMIC-TS	Clinical TempEval (colon cancer)	67.9	67.3	68.6
		Clinical TempEval (brain cancer)	55.8	45.7	71.5
Yang et al. (2021) [40]	BERT-clinical	2018 MADE1.0	88.31 (Avg)	90.16 (Avg)	86.52 (Avg)
	RoBERTa-clinical		89.33 (Avg)	91.09 (Avg)	87.63 (Avg)
	XLNet-clinica		89.1 (Avg)	91.1 (Avg)	87.18 (Avg)
	BERT-clinical	2018 n2c2	95.16 (Avg)	95.62 (Avg)	94.71 (Avg)
	RoBERTa-clinical		95.38 (Avg)	96.33 (Avg)	94.44 (Avg)
	XLNet-clinica		95.36 (Avg)	96.05 (Avg)	94.67 (Avg)
Ramachandran et al. (2023) [26]	FABLE	CMED (Span level)	91	87.4	95
	BioBERT		95.1	95.2	95.1
	Clinical BERT		94.5	93.4	95.7
	BlueBERT		93.1	92.5	93.8
	PubMedBERT		94.1	92.5	95.8

Their methodologies comparison and code download links are summarised in Table 27, with experimental evaluation results in Table 28. It should be noted that Wei et al. [14] evaluated FT-BERT (Fine-Tuned BERT) and FC-BERT (Feature Combined BERT) trained on different datasets (BERT, BioBERT, MIMIC BERT); Yang et al. [40] also used binary and multi-class classification strategies in their experiments. For ease of show, the results in Table 28 are the average scores of all FT-BERT and FC-BERT trained on different datasets [14] and the average scores of both strategies Yang et al. [40].

10.7.3. Limitations

These methods still have some limitations. (1) The datasets are relatively small in scale, with most methods using only a few hundred to a few thousand examples, which may not fully represent the diversity of clinical narratives. (2) Some methods are trained and evaluated on data from a single medical institution, which may limit the model's generalisability across the entire healthcare system. (3) Some methods only focus on specific types of relations, failing to fully meet all possible information needs. (4) Model performance largely depends on clean, annotated corpora, and in the real world, unlabelled or noisy data might reduce the model's performance. (5) Evaluation methods are generally limited to relation classification metrics like F1, and do not fully assess the model's practicality and utility in the real world. (6) These methods' models lack interpretability, making it difficult to understand how the models make decisions. (7) Data used for model development might raise patient privacy concerns, requiring careful handling. (8) These methods show unclear performance in dealing with complex and ambiguous types of relations, which might need further improvement. (9) Applicability to non-English languages may be limited, as most methods are trained on English-language corpora. (10) Computational and data accessibility barriers still exist in widespread clinical practice, which may limit the practical application scope of these methods.

11. Text mining

This section discusses the researches on medical text mining based on PLMs.

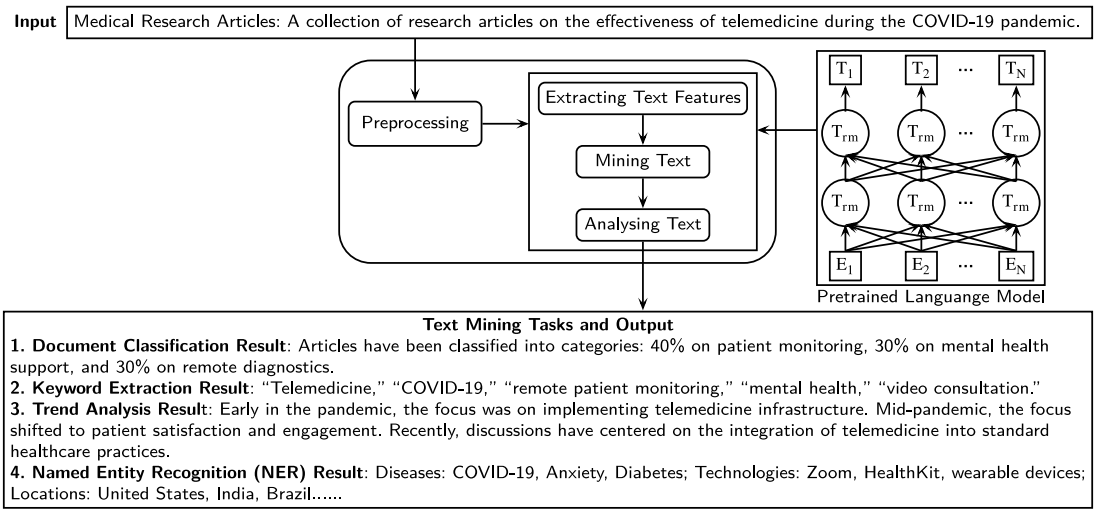


Fig. 13. PLMs-based model diagram for text mining in medicine.

11.1. Basic concepts

Text mining [151] involves using techniques like tokenisation, feature extraction, dimensionality reduction, and pattern recognition to extract valuable information from unstructured text data. This process includes transforming text into a structured format, extracting numerical features to represent text content, reducing redundant features, applying statistical and machine learning models to identify meaningful patterns, and ultimately interpreting results to gain deep insights. Text mining aims to transform raw text data into actionable knowledge applicable in various fields for analysing textual data. For example, it can be used to analyse customer feedback, social media content, survey responses, to reveal trends, sentiments, correlations, and value themes. Text mining plays a crucial role in data-driven decision-making, business intelligence, and insight discovery.

11.2. Main methods

Main methods in text mining are as follows. (1) *Text Preprocessing*: Fundamental steps in text mining, including stemming, tokenisation, removing stop words, normalisation which help transform the raw text into a more analysis-friendly format. (2) *Term Frequency-Inverse Document Frequency (TF-IDF) Analysis*: TF-IDF is a statistical method used to evaluate the importance of a word in a document set or a corpus. It helps understand the most meaningful words in a document. (3) *Topic Modelling*: Techniques like Latent Semantic Analysis (LSA), Latent Dirichlet Allocation (LDA) are used to identify topics or concepts from text collections. These methods assist in understanding hidden themes in large volumes of text data. (4) *Sentiment Analysis*: Determines the emotional tone (positive, negative, neutral) expressed in the text. It is widely used in social media analysis, product review analysis. (5) *Text Classification*: Automatically assigns text to one or multiple predefined categories using machine learning techniques (such as Naive Bayes, Support Vector Machine, Deep Learning). (6) *NLP*: NLP techniques are used to understand and interpret human language's intent and meaning, including NER, syntactic analysis, semantic analysis. (7) *Sequence Modelling and Deep Learning*: Using deep learning models, such as RNN, LSTM, and Transformer, to process sequence data, particularly suitable for understanding context and long-distance dependencies in text. (8) *Data Visualisation*: Transforming text data into visual forms, such as word clouds, trend graphs, to help understand data more intuitively.

Fig. 13 illustrates a text mining model tailored for extracting insights from medical research articles, employing a PLM exemplified by BERT. It delineates a structured approach encompassing preprocessing, text feature extraction, mining, and analysis to facilitate tasks like document classification, keyword extraction, trend analysis, and named entity recognition. The PLM can perform these steps more effectively and can be alternated with non-PLM techniques.

11.3. Advantages

PLMs are highly effective for text mining in the medical field for a number of reasons. (1) PLMs like BERT provide excellent representations of medical text, capturing semantic and contextual information crucial for feature extraction and pattern recognition. (2) Pre-training techniques like masked language modelling enable models to learn contextual relationships between words, aiding in uncovering hidden insights and associations in the text. (3) Transfer learning from general domains allows PLMs to learn associations between medical entities, events, treatments, for mining medical texts. (4) Fine-tuning PLMs on medical text corpora adapts them to the terminologies, idioms, and language styles used in medicine. (5) Pre-trained word embeddings excel in representing domain vocabularies and medical ontologies compared to traditional embeddings. (6) The availability of large medical knowledge bases like UMLS enhances the integration of expert domain knowledge into pre-training and fine-tuning. (7) Self-supervised pre-training reduces dependence on possibly scarce large annotated medical text datasets. (8) PLMs provide a beneficial initialisation for adapting and training custom text mining models, allowing for rapid convergence on small datasets.

11.4. Basic application steps

Some suggestions for using PLMs in medical text mining are as follows. (1) Choose appropriate PLMs like BioBERT, ClinicalBERT, trained on large general text corpora. (2) Collect relevant medical field text data, such as research papers, clinical trial reports, discharge summaries. Preprocess the text data through formatting, tokenisation, cleaning, to ensure the text format meets the input requirements of the PLM. (3) Fine-tune PLMs on medical text corpora to adapt to the domain's style and terminology. (4) Effectively fine-tune models using strategies like differential learning rates, gradual unfreezing, and early stopping. Evaluate text mining performance on reserved medical datasets using accuracy, F1 score, Area Under the Receiver Operating Characteristic Curve. (5) Enhance model capabilities by combining medical ontology knowledge and dictionary features. Use ensemble models trained with different text mining objectives to gain comprehensive insights. (6) Analyse incorrectly classified examples to gain insights and refine the model. (7) Deploy the fine-tuned model in applications like SA of drug reviews, summarisation of clinical trial reports.

Table 29
Datasets for medical text mining.

Dataset	Used by	Language	Domain	Sample size	Download link
PubMed	[41]	English	Biomedical	Over 36 million citations and abstracts of biomedical literature	https://pubmed.ncbi.nlm.nih.gov/
MIMIC-III	[41]	English	Clinical Care	112,000 clinical report records	https://physionet.org/content/mimic-iii/
CyberLeninka	[42]	Russian	Science	338,000 papers	https://cyberleninka.ru/
RuMedBench	[42]	Russian	Biomedical	1000 papers	https://github.com/sberbank-ai-lab/RuMedBench
OSCAR	[15]	Multiple	Medical	138 GB corpus	https://github.com/oscar-project
ViMQ	[15]	Vietnamese	Medical	34,984 entities and 10,027 sentences	https://github.com/tadeuguy/vimq
acrDrAid	[15]	Arabic	Medical	135 abbreviations and 424 extended texts	https://github.com/demdecuong/vihealthbert/tree/main/dataset/acrDrAid

11.5. Evaluation metrics

Some common metrics for evaluating text mining in the medical field are as follows: (1) Precision; (2) Recall; (3) F1 Score; (4) Accuracy; (5) Confusion Matrix; (6) Recall-Oriented Understudy for Gisting Evaluation (ROUGE); (7) Error rate; (8) *Model efficiency* [152]: Time and computational resource usation during training and inference; and (9) *Qualitative analysis* [153]: Human expert evaluation of the extracted information and insights.

11.6. Datasets

Table 29 summarises some datasets used in medical text mining, including their language, domain, sample number, and download links.

(1) PubMed has been briefed in Section 3.6.

(2) MIMIC-III has been briefed in Section 3.6.

(3) CyberLeninka [154] is a Russian scientific dataset containing 338,000 papers covering various scientific fields such as mathematics, physics, chemistry, biology, geography, social sciences, humanities.

(4) RuMedBench [155] provided by the AI Laboratory of Sberbank of Russia, is a Russian biomedical dataset containing 1000 papers, each annotated with entities and relationships related to drugs, diseases, genes, proteins.

(5) OSCAR [156] is a large-scale web corpus composed of multiple languages, which is obtained by using the goclassy architecture to perform language classification and filtering on the Common Crawl corpus. OSCAR is used to train multilingual deep learning models, such as BART, and it contains 138 GB of text. OSCAR currently has 166 different languages to choose from, and each language has both original and deduplicated forms.

(6) ViMQ [157], provided by the School of Computer Science and Engineering at Vietnam National University, is a Vietnamese medical dataset containing 34,984 entities and 10,027 sentences, each with information related to patients' medical history, diagnosis, and treatment.

(7) acrDrAid [15] is a dataset for studying AI assisted diagnosis, which contains medical records, images, laboratory results, diagnostic reports, and other data from different hospitals and departments. The acrDraAid dataset provides a public, standardised and diversified data platform, where researchers and developers can develop and test various methods and applications of AI assisted diagnosis. acrDraAid dataset currently has three versions, namely acrDraAid-1.0, acrDraAid-2.0, and acrDraAid-3.0. Each of them contains different types and quantities of data.

11.7. Important research works

11.7.1. Summary

In 2021, Zhang et al. [41] developed SMEDBERT, integrating structured medical knowledge from UMLS and SNOMED CT into the BERT model. They pre-trained the SmedBERT model using a large corpus from medical text databases like PubMed and MIMIC-III, employing masked language modelling and entity prediction objectives. Subsequently, they fine-tuned the SmedBERT model on four downstream medical text mining tasks: entity recognition, relation extraction, concept normalisation, and questions answering. It was proven, through experiments, that the fine-tuned SmedBERT model performed well across all tasks and outperformed baseline models using BERT or BioBERT that do not require structured knowledge.

In 2022, Yalunin et al. [42] developed RuBioRoBERTa, a biomedical language model pre-trained on a large Russian biomedical text corpus. Based on the RoBERTa architecture, this model was trained using over 60 GB of data from biomedical sources. In evaluations on tasks such as NER, natural language inference, QA, and text classification, RuBioRoBERTa outperformed the multilingual BioBERT. It achieved a high F1 score of 13.5% (0.875 on the BIOMERT dataset) in NER and a 10% F1 score improvement in RE. The accuracy of biomedical QA exceeded 90%, and text classification accuracy also improved. These results established RuBioRoBERTa as a leading model in the Russian biomedical NLP field.

In 2022, Minh et al. [15] developed ViHealthBERT, the first domain-specific PLM focused on the Vietnamese medical field. Moreover, they built two new Vietnamese datasets for the medical field. One addresses the ambiguity of acronyms and another for summarising frequently asked questions. They evaluated ViHealthBERT on four health-related tasks: acronym ambiguity resolution, summarising frequent questions, entity recognition, and SA. The experiments demonstrated that ViHealthBERT outperformed general-domain language models like PhoBERT and mBERT across all tasks, showcasing the effectiveness of a specific PLM for the Vietnamese medical field.

11.7.2. Main similarities and differences

They share similarities in several respects as follows. (1) All their models are PLMs for biomedical text mining, subfields of NLP and bioinformatics. Their common goal is to provide better natural language understanding and information extraction capabilities for medical texts such as scientific papers, clinical records, health records. (2) They are all based on BERT or its variants, which are Transformer architecture models using self-attention mechanisms and masked language models for training on large-scale unlabelled text data. They all use supervised learning and task-specific datasets for fine-tuning downstream tasks, adapting to their domains' specific contexts and information needs. (3) They represent state-of-the-art or competitive models in their respective languages and task domains. (4) These methods have successfully demonstrated the effectiveness and practicality of PLMs in biomedical text mining, laying a strong foundation for further research and applications in this field.

Despite these similarities, these methods have notable differences as follows. (1) *Language*: SmedBERT is for Chinese, RuBioRoBERTa is for Russian, and ViHealthBERT is for Vietnamese. Each was pre-trained on different biomedical text corpora in their respective languages. (2) *Knowledge Graph*: SmedBERT integrated structured semantic knowledge from knowledge graphs, while RuBioRoBERTa and ViHealthBERT did not. SmedBERT used a hybrid attention mechanism, learning from the neighbours of linked entities in the knowledge graph. (3) *Model Architecture*: SmedBERT is based on BERT, RuBioRoBERTa on RoBERTa, and ViHealthBERT on ALBERT. RoBERTa and ALBERT are improved

Table 30
Comparison of text mining methods based on PLMs.

Reference	Publication venue	Focus	Pro	Con	Code download link
Zhang et al. (2021) [41]	arXiv Preprint	The vast medical terminology and its complex relationships are difficult to understand in text	Utilised large-scale corpora from medical text databases such as PubMed and MIMIC-III for pre-training	May limit the capture of more nuanced semantic information in text mentions	https://github.com/MatNLP/SMedBERT
Yalunin et al. (2022) [42]	arXiv Preprint	Optimising Russian text mining within the biomedical domain	Trained using over 60 GB of data from biomedical sources	Could limit its generalisation capabilities and coverage of domain-specific vocabulary	https://github.com/sberbank-ai-lab/RuBioRoBERTa
Minh et al. (2022) [15]	Language Resources and Evaluation Conference (Ordinary)	Optimising domain-specific PLMs for the Vietnamese healthcare sector	Effectively conducts text mining in Vietnamese	Could impact its expressiveness and representational capacity	https://github.com/demdecuong/vihealthbert

Table 31
Comparison of text mining methods' result based on PLMs.

Reference	Model	Evaluation dataset	F1	Accuracy	Precision	Recall	R-1	R-2	R-L
Zhang et al. (2021) [41]	SMedBERT	Homemade (NER)	83.85	–	–	–	–	–	–
		Homemade (RE)	91.42	–	–	–	–	–	–
		Homemade (QA)	78.86	–	–	–	–	–	–
		Homemade (NLI)	96.76	–	–	–	–	–	–
Yalunin et al. (2022) [42]	RuBioRoBERTa	RuMedBench	–	71.54	–	–	–	–	–
Minh et al. (2022) [15]	ViHealthBERT	COVID-19 (MLM)	96.77	–	–	–	–	–	–
		ViMQ (MLM)	86.6	–	–	–	–	–	–
		acrDrAid (MLM + NSP)	86.7	–	93.14	81.09	–	–	–
		acrDrAid (MLM + CP)	85.59	–	94.02	78.38	–	–	–
		FAQ (MLM)	–	–	–	–	50.45	31.35	43.85

Table 32
Datasets for medical non-English task.

Dataset	Used by	Language	Domain	Sample size	Download link
Medical website	[43]	Persian	Medical	200,000 medical Q&A pairs	–
MedSTS	[23]	Korean	Medical	3121 Korean sentence pairs	–
NACHOS	[44]	French	Medical	7.4G of notes	https://drbert.univ-avignon.fr/
NBDW	[44]	French	Medical	1.7 million de-identified hospitalisation report	–
CyberLeninka	[42]	Russian	Science	338,000 papers	https://cyberleninka.ru/
RuMedBench	[42]	Russian	Biomedical	1000 papers	https://github.com/sberbank-ai-lab/RuMedBench
ViMQ	[15]	Vietnamese	Medical	34,984 entities and 10,027 sentences	https://github.com/tadeephuy/vimq
ChiMed	[10]	Chinese	Medical	200,744 pages of Q&A	https://github.com/yuanheTian/ChiMed
Yidu-N7K	[28]	Chinese	Medical	–	http://openkg.cn/dataset/yidu-n7k
CCKS-2020-CEM	[28]	Chinese	Clinical	26,414/5706 texts of and training/test set	https://gitee.com/gongmingwang/ccks_ner

versions of BERT, with some modifications and optimisations in training and design. (4) *Evaluation Tasks*: SmedBERT was evaluated on five Chinese medical NLP tasks, RuBioRoBERTa on four Russian medical NLP tasks, and ViHealthBERT on three Vietnamese medical NLP tasks. These tasks included QA, question matching, natural language inference, NER, RE, and text classification.Comparisons of their methodologies and code download links can be found in Table 30, and the experimental evaluation results in Table 31.

11.7.3. Limitations

Some limitations of these methods are as follows. One limitation of SmedBERT is that it only uses a single knowledge graph, the Chinese Medical Knowledge Graph (CMeKG), to enhance its semantic representation capabilities. However, other valuable knowledge sources could be integrated into SmedBERT, such as biomedical ontologies, databases, and external knowledge graphs. Moreover, SmedBERT does not explicitly model the relationships between entities in the knowledge graph, which may limit the acquisition of more refined semantic information mentioned in the text.

A limitation of RuBioRoBERTa is that it was pre-trained using a relatively small Russian biomedical text corpus, which might restrict its generalisation ability and coverage of specific domain vocabulary. Furthermore, the corpus might contain noise and errors, such as spelling mistakes, inconsistent formatting, and low-quality translations. Another limitation is that its performance has not been evaluated on more complex tasks like RE and QA.

A limitation of ViHealthBERT is that it uses ALBERT as its base model, which has some constraints in terms of parameter sharing and factorised embeddings compared to BERT or RoBERTa. This could affect its expressive power and representational capacity. Another limitation is that ViHealthBERT has not compared its results with other Vietnamese-specific domain models (such as PhoBERT or VncoreNLP), which may limit our understanding of its performance in the Vietnamese NLP field.

12. Non-english

This section reviews research on using PLMs for NLP tasks involving medical documents not in English.

12.1. Datasets

Table 32 summarises the commonly used non-English medical datasets in terms of language, domain, sample quantity, and download links.

(1) Online Medical Consultation Website [43], a Persian medical dataset from an online medical consultation website, contains 200,000 medical question–answer pairs, each pair consisting of a medical question posed by a patient and an answer provided by a professional doctor.

(2) MedSTS [158], provided by Seoul National University's MedSTS project, is a Korean medical dataset consisting of 3121 English sentence pairs extracted from medical literature, each pair annotated with a manually assigned semantic similarity score ranging from 0 to 5, where 0 means completely unrelated, and 5 means exactly the same.

(3) NACHOS [44] (open crawled French Healthcare corpus), a French medical open source dataset compiled by crawling a variety of textual sources around the medical topic. It consists of more than one billion words, drawn from 24 French speaking high-quality websites. The corpus includes a wide range of medical information: descriptions of diseases and conditions, information on treatments and medications, general health-related advice, official scientific meeting reports, anonymised clinical cases, scientific literature, thesis, French translation pairs, university health courses and a large range of data obtained from raw textual sources, web scrapping, and optical character recognition (OCR).

(4) NBDW [44], from France's National Health Data Warehouse (NBDW), is a French medical dataset with 1.7 million anonymised hospital report samples, each report containing information about the patient's basic details, examination results, and diagnosis.

(5) CyberLeninka has been briefed in Section 11.6.

(6) RuMedBench has been briefed in Section 11.6.

(7) ViMQ has been briefed in Section 11.6.

(8) ChiMed has been briefed in Section 7.6.

(9) Yidu-N7K has been briefed in Section 7.6.

(10) CCKS-2020-CEMR has been briefed in Section 7.6.

12.2. Persian pre-trained language models

In 2021, Taghizadeh et al. [43] developed SINA-BERT, a PLM designed explicitly for Persian medical texts, aiming to address the lack of high-quality Persian PLMs in the medical field. Based on the BERT architecture, SINA-BERT was pre-trained on a large corpus of Persian medical texts from sources including medical journals and health news websites. The authors evaluated SINA-BERT on three tasks: medical question classification, medical SA, and medical question retrieval, creating Persian annotated datasets for training and evaluation. SINA-BERT outperformed previous BERT-based models on Persian medical texts for these tasks.

12.3. Korean pre-trained language models

In 2022, to address the lack of Korean-specific medical PLMs, Kim et al. [23] collected a Korean medical corpus for training a language model. They combined the BERT architecture with advanced Korean PLMs to build a pre-training framework for medical environments. In external evaluations, their model improved the Pearson correlation coefficient by 0.046 on the Korean Medical Semantic Text Similarity dataset and increased the F1 score by 0.053 on the Korean Medical NER task. Their experiments demonstrated the feasibility of using PLMs for biomedical text mining.

12.4. French pre-trained language models

In 2023, Labrak et al. [44] conducted original research on a French medical PLM. For the first time, they compared the performance of PLMs trained on public web data with those trained on private data from medical institutions. They also evaluated different learning strategies for various biomedical NLP tasks. A key finding of their study is that further pre-training on target data can more effectively use existing foreign biomedical PLMs. As a result of their research, they presented DrBERT, France's first French PLM specifically developed for the biomedical domain. They built a huge medical data corpus for training this model and shared it under a free license for public download and use.

12.5. Russian pre-trained language models

Yalunin et al. [42] research on a Russian PLM in the medical field has been detailed in Section 11.

12.6. Vietnamese pre-trained language models

Minh et al. [15] research on a Vietnamese PLM in the medical field has been detailed in Section 11.

12.7. Chinese pre-trained language models

Tan et al. [10] and Wen et al. [28] research on a Chinese PLM in the medical field has been detailed in Section 7.

12.8. Main similarities and differences

These models show similarities in the following aspects: (1) They all use BERT as the base model for their PLMs. (2) They all follow a similar pre-training process and objectives to BERT, namely predicting masked words and the next sentence from the input text. This objective helps the model learn general language patterns and representations that are transferable to downstream tasks. (3) They all use task-specific datasets and corresponding evaluation metrics to fine-tune the models for biomedical text mining tasks. (4) They all demonstrate the effectiveness and practicality of their models in various biomedical text mining tasks such as NER, RE, SA, QA, document classification, and document summarisation. These tasks are crucial for extracting valuable information from biomedical texts and supporting various applications such as clinical decision support, drug discovery, disease diagnosis, and health education.

The main differences among these non-English language models for the medical field are as follows. (1) They use different languages for the PLMs. Sina-BERT is in Persian, Kim et al.'s Korean model is in Korean, DrBERT is in French, RuBioRoBERTa is in Russian, ViHealthBERT is in Vietnamese, and Tan et al. [10] and Wen et al. [28] are in Chinese. (2) The sources of the corpora vary. SinaBERT uses Persian medical texts, KoBERT uses Korean medical corpora, DrBERT uses French medical texts, RuBioRoBERTa uses Russian biomedical data, ViHealthBERT uses Vietnamese health data, and Tan et al. [10] and Wen et al. [28] use Chinese electronic medical records. (3) *The model architectures differ*: SinaBERT is based on BERT-Base, KoBERT is derived from BERT-Base and BioBERT, DrBERT uses the CamemBERT framework, RuBioRoBERTa employs the RoBERTa architecture, and ViHealthBERT is based on BERT-Base. (4) *The pre-training strategies vary*: SinaBERT uses masked language modelling, KoBERT conducts masked pre-training language model training on medical texts, DrBERT employs MLM (Masked Language Model) and NSP (Next Sentence Prediction) tasks, RuBioRoBERTa relies mainly on MLM, and ViHealthBERT uses a masked pre-training language model. (5) *The evaluation methods differ*: SinaBERT mainly targets Persian medical text classification, KoBERT evaluates medical natural language inference and NER, DrBERT focuses on clinical text classification and NER, RuBioRoBERTa targets biomedical NER, and ViHealthBERT evaluates medical text classification and NER. (6) Only KoBERT and ViHealthBERT compare their models with other PLMs, other papers lack comparative evaluations. (7) Only KoBERT analyses the errors and limitations of their model. Other papers focus on quantitative analysis. Compared to other papers that focus on 1–2 tasks, KoBERT covers a wider range of medical NLP tasks.

Table 33
Comparison of non-English task method based on PLMs.

Reference	Publication venue	Focus	Pro	Con	Code download link
Taghizadeh et al. (2021) [43]	arXiv Preprint	Addressing the lack of high-quality Persian PLMs in the medical field	Capable of effectively processing Persian text	May limit its performance and generalisation capabilities	–
Kim et al. (2022) [23]	Scientific Reports (JCR Q2, IF 4.6)	Addressing the lack of Korean medical-specific PLMs	Can accurately judge the semantic text similarity of Korean medical text	Could result in limited generalisability of the model	https://github.com/KU-RIAS/KM-BERT-Korean-Medical-BERT
Labrek et al. (2023) [44]	bioRxiv Preprint	Addressing the lack of French PLMs in the medical field	Capable of effectively processing French text with existing foreign biomedical resources	May be challenging to generalise to other tasks	–
Yalunin et al. (2022) [42]	arXiv Preprint	Optimising Russian text mining within the biomedical domain	Trained using over 60 GB of data from biomedical sources	Could limit its generalisation capabilities and coverage of domain-specific vocabulary	https://github.com/sberbank-ai-lab/RuBioRoBERTa
Minh et al. (2022) [15]	Language Resources and Evaluation Conference (Ordinary)	Optimising domain-specific PLMs for the Vietnamese healthcare sector	Effectively conducts text mining in Vietnamese	Could impact its expressiveness and representational capacity	https://github.com/demdecuong/vihealthbert
Tan et al. (2021) [10]	International Conference on Asian Language (Ordinary)	Extract valuable information from a large amount of clinical medical texts	Build a Chinese medicine NER model based on different PLMs and aggregate the results	The scope of evaluation is limited	–
Wen et al. (2021) [28]	Data Intelligence (JCR Q1, IF 3.9)	Make fuller use of unlabelled medical text to improve medical NER	Use unlabelled medical text to train PLMs within the domain	Reliance on text quality and accuracy can lead to errors in the model	–

Table 34
Comparison of non-English task methods' result based on PLMs.

Reference	Model	Evaluation dataset	F1	Precision	Recall	Accuracy	R-1/-2/-L
Taghizadeh et al. (2021) [43]	SINA-BERT	Homemade (MAQ)	94.77	94.91	94.63	96.14	–
		Homemade (MSA)	94.49	95	94	94	–
Kim et al. (2022) [23]	KM-BERT	Homemade	86.6	–	–	95.5	–
Labrek et al. (2023) [44]	DrBERT	aHF NER	47.08 (Avg)	54.92 (Avg)	45.03 (Avg)	–	–
		aHF classification	80.56 (Avg)	80.59 (Avg)	80.56 (Avg)	–	–
		NER Medical Report	89.64 (Avg)	87.78 (Avg)	92.78 (Avg)	–	–
		Specialities Sorting	99.86 (Avg)	99.84 (Avg)	99.88 (Avg)	–	–
Yalunin et al. (2022) [42]	RuBioRoBERTa	RuMedBench	–	–	–	71.54	–
Minh et al. (2022) [15]	ViHealthBERT	COVID-19 (MLM)	96.77	–	–	–	–
		ViMQ (MLM)	86.6	–	–	–	–
		acrDrAid (MLM+NSP)	86.7	93.14	81.09	–	–
		acrDrAid (MLM+CP)	85.59	94.02	78.38	–	–
		FAQ (MLM)	–	–	–	–	50.45/31.35/43.85
Tan et al. (2021) [10]	ZEN2-large + BiLSTM + CRF + Radical features	ChiMed	84.43	–	–	–	–
	ZEN2-base + BiLSTM + CRF + Radical features		84.14	–	–	–	–
Wen et al. (2021) [28]	Fusion model	CCKS-2020-CEMR	88.7/95.3 (strict/relax)	–	–	–	–

Their methods comparison and code download links can be seen in Table 33, and the experimental evaluation results in Table 34. It is noted that Labrak et al. [44] evaluated models with different parameter sizes (large or small), and the results in Table 34 are the average scores of these models.

12.9. Limitations

These methods still have some limitations. The Sina-BERT model was only evaluated on Persian medical texts, so its applicability to other languages or domains remains unclear. Moreover, the relatively small scale of the datasets used for pre-training and evaluation might have impacted its performance and generalisation capabilities.

The Korean Medical BERT was trained on a smaller dataset, covering clinical records from only a few hospitals. This might limit its generalisation ability across different types of medical texts. Furthermore, the evaluation was limited to NER and RE tasks, which might not fully demonstrate its performance.

DrBERT might encounter difficulties when being extended to other tasks, which could limit its application in a broader range of medical text mining tasks.

RuBioRoBERTa, specifically designed for Russian, leaves its effectiveness in other languages, particularly related Slavic languages, uncertain. Moreover, since its pre-training data mainly comes from the biomedical field, the model might not fully capture the diversity of clinical medicine.

Table 35
Datasets for other medical task.

Dataset	Used by	Language	Domain	Sample size	Download link
Homemade	[22]	German	Radiology	380,000 unstructured radiology reports	https://github.com/fast-raidiology/bert-for-radiology
MEDIQA-2019	[16]	English	Biomedical	Depends on different subtasks	https://github.com/heyunh2015/diseaseBERT
TRECQA-2017	[16]	English	General	53,417/1117/1442 Q&A pairs of train/development/test version	https://trec.nist.gov/data/qa.html
MEDNLI	[16]	English	medical	14,049 clinician notes pairs	https://github.com/heyunh2015/diseaseBERT
BC5CDR	[16]	English	Biomedical	1500 PubMed papers	https://github.com/heyunh2015/diseaseBERT
NCBI	[16]	English	Biomedical	–	https://github.com/heyunh2015/diseaseBERT
SAHSU	[18]	Chinese	Biomedical	5090 electronic medical records (2M)	–
CCKS-2019-CMER	[18]	Chinese	General	5363/1816 training/test sets	https://gitee.com/gongmingwang/ccks_ner
NMEDW	[27]	English	Clinical	396 cases and 3261 controls	https://www.feinberg.northwestern.edu/research/cores/units/edw.html
WCM	[27]	English	Genomics	548 cases and 2015 controls	https://weill.cornell.edu/
MIMIC-CXR	[45]	English	Radiology	371,920 chest X-rays and 227,943 imaging studies	https://physionet.org/content/mimic-cxr/2.0.0/
OPEN-I	[45]	English	Biomedical	7471 chest X-ray images and 3955 reports	https://openi.nlm.nih.gov/faq#collection

ViHealthBERT was only pre-trained and evaluated for the Vietnamese language, with its pre-training data sourced from limited-domain health websites and excerpts from medical textbooks. This might adversely affect its performance in actual clinical texts.

13. Others

This section provides a broad overview of additional uses of PLMs within the medical domain.

13.1. Datasets

Table 35 summarises other commonly used medical datasets in terms of language, domain, sample count, and download links.

(1) Home-Made Dataset [22] is collected and annotated by German radiologists, comprising 3.8 million unstructured radiology reports covering various types of radiological examinations, results, and diagnoses. This dataset can be used for NLP tasks such as the automatic generation, classification, summarisation, and information extraction of radiology reports.

(2) MEDIQA-2019 has been briefed in Section 4.6.

(3) TRECQA-2017 has been briefed in Section 4.6.

(4) MEDNLI [159] is a Natural Language Inference (NLI) dataset in the medical domain, developed by doctors from the history section of MIMIC-III clinical records, annotating the inference relationship (entailment, neutral, or contradiction) between two sentences. The MEDIQA-NLI test set consists of 405 text and hypothesis pairs. The training set is the MedNLI dataset, which includes 14,049 clinical sentence pairs extracted from the MIMIC-III database.

(5) BC5CDR has been briefed in Section 7.6.

(6) NCBI has been briefed in Section 4.6.

(7) SAHSU [18] includes 5090 electronic medical records from the departments of General Surgery, Intervention and Oncology from the last 2 years.

(8) CCKS-2019-CMER [160] is from the China Conference on Knowledge Graph and Semantic Computing (CCKS) 2019, focusing on Chinese Medical Entity Recognition in medical texts. This task is crucial for advancing NLP in Mandarin, particularly for identifying and classifying medical entities like diseases, symptoms, and treatments within unstructured text. The dataset provided includes annotated Chinese medical texts.

(9) NMEDW [161] is a significant resource at Northwestern Medicine, a leading healthcare provider and medical research institution. Northwestern Medicine Enterprise Data Warehouse (NMEDW) integrates clinical, operational, and financial data from across the Northwestern Medicine healthcare system. It serves as a centralised repository of patient records, treatment data, laboratory results, and other critical healthcare information. Researchers and healthcare professionals use the NMEDW for a wide range of purposes, including clinical research, patient care quality improvement, operational efficiency analysis, and population health management. The data in NMEDW is often leveraged to support clinical trials, epidemiological studies, and health services research, providing insights that can lead to improved patient outcomes and healthcare practices.

(10) WCM (Weill Cornell Medicine) [162] contains extensive health data repositories to support a wide range of medical research, from basic science to clinical trials and population health studies.

(11) MIMIC-CXR has been briefed in Section 3.6.

(12) MIMIC-III has been briefed in Section 3.6.

13.2. Important research works

13.2.1. Summary

In 2020, Bressem et al. [22] used the BERT model to classify radiology reports. By pre-training the model on 3.9 million reports and fine-tuning it on approximately 7000 reports, they achieved automatic classification of key lesions (such as pneumonia and pleural effusion) in chest X-ray reports, with an area under the ROC curve reaching 0.97–0.99. This significantly outperformed rule-based and other deep learning models with minimal annotation, demonstrating BERT's efficiency and accuracy in handling medical text information and providing an effective approach to extracting structured information from large-scale reports.

In 2021, Lu et al. [16] developed an efficient method to integrate domain knowledge from diverse sources such as ontologies, knowledge graphs, and texts into biomedical PLMs. They used adapter modules and gating mechanisms to selectively aggregate representations from knowledge sources into PLMs like SciBERT, avoiding retraining. Experimental results showed that integrating complementary knowledge from multiple sources effectively improved downstream performance in biomedical NLP tasks. This approach balanced gains with training efficiency through adapters and gating mechanisms. Analysis indicated that the model could selectively use the most relevant knowledge according to the demands of the final task. Overall, this work developed an effective method to integrate multi-source heterogeneous biomedical knowledge to enhance PLMs while minimising the computational costs of retraining.

In 2021, Ni et al. [18] developed a model called Stacked Residual GRU-CNN with CMEDLMS (SRGC-CMEDLMS), aimed at intelligently predicting various conditions in healthcare. They encoded clinical records using the biomedical language model ClinicalBERT and extracted temporal and spatial features of vital signs using stacked residual GRU-CNN. This method used a Multi-Layer Perception (MLP)

Table 36
Comparison of other medical task method based on PLMs.

Reference	Publication venue	Focus	Pro	Con	Code download link
Bressem et al. (2020) [22]	Bioinformatics (JCR Q1, IF 5.8)	Accurate categorisation of unstructured radiological report texts	Can aid in improving information extraction from free-text medical reports	Did not provide explanations or visualisations of the categorisation results	https://github.com/fast-raidiology/bert-for-radiology
Lu et al. (2021) [16]	Findings of the Association for Computational Linguistics: EMNLP (Top)	Domain-specific PLMs without explicitly integrating domain-specific knowledge	Can integrating domain knowledge into biomedical PLMs	The model requires to train a separate adapter for each domain knowledge source	–
Ni et al. (2021) [18]	Applied Soft Computing (JCR Q1, IF 8.7)	Enhancing the performance of biomedical NLP tasks, especially predictive tasks	Capable of predicting risk scores for various adverse events, such as sepsis, acute kidney injury	The model is unable to recognise polysemous words	–
Mao et al. (2023) [27]	Journal of Biomedical Informatics (JCR Q1, IF 4.5)	Predicting the risk of progression from Mild Cognitive Impairment to Alzheimer's Disease	Improved language understanding of AD-related texts	Requires substantial computational resources and time for pre-training and fine-tuning	–
Thawkar et al. (2023) [45]	arXiv Preprint	Enhancing the application of large visual language models in radiology	Can generating summaries of chest X-ray imaging results using a visual PLM	May be affected by the quality and completeness of the chest X-ray data	https://github.com/mbzuai-oryx/XrayGPT

to combine the outputs of both models, predicting scores for multiple adverse events such as sepsis, acute kidney injury, and death risk.

In 2023, Mao et al. [27] developed a PLM named AD-BERT for predicting the progression from Mild Cognitive Impairment (MCI) to Alzheimer's Disease (AD). They used a large clinical record dataset from the Alzheimer's Disease Neuroimaging Initiative (ADNI), labelled as MCI or AD, based on clinical diagnoses and cognitive scores. They fine-tuned the AD-BERT model on the dataset for a binary classification task and compared it with several baseline models, such as BERT, RoBERTa, and XLNet.

In 2023, Thawkar et al. [45] explored using a visual PLM to generate summaries for chest X-ray imaging. They developed a conversational medical visual language model named XrayGPT, which is capable of analysing and answering open-ended questions about chest X-rays and producing text summaries given new chest X-ray inputs. Evaluations showed that their model achieved higher ROUGE scores than purely textual methods, indicating it could generate more accurate and detailed summaries. This suggests that joint pre-training with visual data and language understanding can better generate models for summarising radiographic findings. The study highlighted the potential of multimodal visual language AI in assisting radiological reporting. More broadly, this proves that unified cross-modal models may help advance medical summarisation and explanation tasks.

13.2.2. Main similarities and differences

Similarities are as follows. (1) All applied deep learning to biomedical NLP tasks, demonstrating its utility in the field. (2) Most methods used PLMs like BERT or SciBERT (Lu et al. Mao et al. Thawkar et al.), highlighting the advantages of transfer learning. (3) Some focused on NLP of clinical texts like radiology reports or annotated data (Bressem et al. Mao et al. Thawkar et al.), with some specialising in chest X-rays (Bressem et al. Thawkar et al.), reflecting the importance of this imaging modality. (4) Some combined image data (X-rays or MRI) with textual data (Bressem et al. Thawkar et al. Ni et al.), emphasising the importance of multimodal approaches. (5) Most aimed to advance clinical applications such as diagnostics, predictions, classifications, or summaries. (6) Many integrated specific domain knowledge to enhance their models (Lu et al. Ni et al. Mao et al.). (7) Most used standard deep learning structures like CNN, GRU, and BERT.

The main differences between these methods are as follows. (1) Bressem et al. [22] mainly used deep learning natural language models pre-trained on radiology reports for classifying chest X-rays, while Lu et al. [16] focused on integrating multiple domain knowledge sources into biomedical language models. (2) Unlike Bressem et al. Lu et al.'s method used adapter modules and gating to effectively integrate knowledge into models like SciBERT. (3) Ni et al. [18] introduced a stacked residual GRU-CNN model incorporating a biomedical PLM for predicting medical events, significantly different in structure from Lu and Bressem's models. (4) Mao et al. [27] fine-tuned BERT on clinical records to predict Alzheimer's disease progression. Their main aim was disease prediction, not knowledge integration or report classification. (5) Thawkar et al. [45] used a visual PLM to generate summaries for chest X-rays, differing from the objectives of classification or prediction. (6) Both Bressem et al. and Thawkar et al. dealt with chest X-rays, but their research goals differed. Mao et al. and Lu et al. used biomedical PLMs, but their applications varied. (7) Lu et al.'s knowledge integration approach was the most unique, aiming to integrate different domain knowledge into PLMs effectively. Other methods differed in application scenarios or model structures. Their comparisons, code download links, and experimental evaluations are displayed in Tables 36 and 37, respectively.

13.2.3. Limitations

Some limitations of these methods are as follows. Bressem et al. only use chest X-ray reports for classification, potentially missing other patterns and observations in radiology. They also did not account for variability among subjects and report quality, which could affect the consistency and reliability of their model. They could expand their data sources (e.g., CT scans, MRI scans, ultrasound images), incorporate variability among subjects, and report quality assessment features into their model.

Lu et al.'s relatively small dataset may limit their data's generalisability. Their technique has not been integrated into real-world clinical workflows or systems. Their experiments lacked detailed analysis of model failures, and static evaluation methods might not fully reflect real clinical scenarios.

Ni et al. only used electronic health records as a source for predictive intelligence, potentially missing factors and outcomes related to chronic diseases. They also did not consider temporal and causal relationships between features and labels, affecting accuracy and interpretability. They could expand their work by using other predictive intelligence sources (e.g., genomic, proteomic, metabolomic data) and incorporating temporal and causal modelling techniques.

Mao et al. only used clinical records for disease progression prediction, potentially missing all possible symptoms and biomarkers for

Table 37
Comparison of other medical task methods' result based on PLMs.

Reference	Model	Evaluation dataset	Accuracy	MRR	Precision	Recall	F1	AUC	R-1/-2/-3
Bressem et al. (2020) [22]	BERT	Homemade	–	–	–	–	–	98	–
Lu et al. (2021) [16]	DAKI-ALBERT	MEDIQA-2019	80.22	91.22	84.36	–	–	–	–
		TRECQA-2017	80.33	58.65	62.31	–	–	–	–
		MEDNLI	86.85	–	–	–	–	–	–
	DAKI-BioBERT	BC5CDR	–	–	–	–	86.51	–	–
		NCBI	–	–	–	–	89.01	–	–
		MEDIQA-2019	80.22	91.22	–	–	–	–	–
Ni et al. (2021) [18]	CMed-BERT (StaResGRU-CNN_CRF)	CCKS 2019 (NER)	–	–	72.4	80.72	–	–	–
		SAHSU (NER)	–	–	86.69	90.74	–	–	–
	CMed-BERT	Homemade (Text Classification)	–	–	96.39	96.16	–	–	–
	CMed-GloVe	SAHSU-CCKS	–	–	80.19	76.47	–	–	–
Mao et al. (2023) [27]	AD-BERT	NMEDW	–	–	–	–	44	84.9	–
		WCM	–	–	–	–	68	88.3	–
Thawkar et al. (2023) [45]	XrayGPT	MIMIC-CXR	–	–	–	–	–	–	32.13/9.12/19.97

Table 38
Usage of evaluate metric.

Evaluate metric	Medical NLP tasks									
	TS	QAS	MT	SA	NER	Information extraction	Relation extraction	Text mining	Non-Eng	Others
Recall	–	–	–	✓	✓	✓	✓	✓	✓	✓
Precision	–	✓	–	✓	✓	✓	✓	✓	✓	✓
F1	✓	✓	–	✓	✓	✓	✓	✓	✓	✓
Accuracy	–	✓	–	✓	–	✓	–	✓	✓	✓
AUC-ROC	–	–	–	✓	–	✓	–	–	–	✓
BERTScore	✓	–	–	–	–	–	–	–	–	–
BLEU	✓	–	✓	–	✓	–	–	–	–	–
COMET	–	–	✓	–	–	–	–	–	–	–
Exact match	–	✓	–	–	–	–	–	–	–	–
Fleiss's kappa coefficient	–	–	–	✓	–	–	–	–	–	–
METEOR	–	–	✓	–	–	–	–	–	–	–
MRR	–	✓	–	–	–	–	–	–	–	–
ROUGE	✓	–	✓	–	–	–	–	✓	✓	✓
Human evaluation	✓	✓	✓	–	✓	–	–	–	–	–
Error rate	–	✓	✓	–	–	–	✓	✓	–	–
Model efficiency	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Qualitative analysis	–	–	–	–	–	–	–	✓	–	–

Alzheimer's disease. They did not evaluate their model's prediction efficacy for other types of dementia or cognitive impairments, affecting generality and applicability. To improve their research, they could consider introducing other data sources for disease progression prediction (e.g., neuroimaging, neuropsychological, genetic data) and evaluate the model's predictive efficacy for other types of dementia or cognitive impairments (e.g., vascular dementia or mild cognitive impairment).

Thawkar et al. used a small dataset of chest X-rays and their corresponding reports, potentially limiting the generalisability and diversity of the generated summaries. They did not compare the performance of XrayGPT with other medical visual language models or human experts, potentially limiting the validity and reliability of the results. They also did not assess the clinical relevance and usefulness of the generated summaries, which might limit the applicability and impact of the method in real-world settings. They did not analyse errors and failures in XrayGPT, potentially limiting understanding and improvement of the method.

14. Challenges and future research

Despite significant achievements of PLMs in medical NLP tasks, future work with PLMs still faces a range of significant challenges and opportunities:

1. *Reliability*: Before the widespread deployment of PLMs in clinical practice, a comprehensive validation of their ability to handle heterogeneous medical texts across various specialities, demographics, and data sources is required. This entails rigorous testing of the models on noisy inputs and adversarial samples to ensure their reliability.
2. *Interpretability*: To establish trust among medical professionals in model predictions and attention weights, interpretable methods are needed to explain the decision-making of models. This means that the decision process of the model must be transparent for ease of understanding and interpretation.
3. *Bias and Fairness*: Measures should be taken to reduce demographic biases inherent in PLMs to ensure fair performance across different patient groups and avoid potential bias issues.
4. *Efficiency*: The PLMs need to be compressed and accelerated for lower latency clinical reasoning. Researchers should explore various alternatives, such as model distillation, to address this issue.
5. *Modality*: Integrating and fusing other data modalities, such as medical imaging, sensor data, genomics, with text is expected to enhance the performance of PLMs in integration and patient condition analysis.
6. *Clinical Assessment*: Rigorous clinical studies are necessary to accurately assess the effect of PLMs in real clinical settings, ensuring their actual benefit on patient outcomes.
7. *Domain Adaptability*: The challenge of domain adaptability in applying PLMs to medicine is significant, given medical knowledge's specialised vocabulary and evolving nature. PLMs trained

on general language corpora often struggle to interpret medical terminologies accurately and to adapt to new information, leading to potential errors in critical medical tasks. Addressing this requires continuous learning, domain-specific pre-training, and integrating medical knowledge into PLMs. Enhancing PLMs' adaptability to the medical domain is crucial for their practical application in improving patient care, advancing medical research, and supporting healthcare professionals.

15. Conclusion

PLMs are powerful tools with the potential to transform the medical field fundamentally. In this survey, we have examined the latest advancements in applying PLMs in medical NLP tasks. Specifically, we summarised key research using PLMs such as BERT, RoBERTa, BioBERT, and ChatGPT in information extraction, question-answering, sentiment analysis, medical named entity recognition, medical text mining, medical examinations, and other application areas. These studies indicate that PLMs tailored for the medical field capture domain-specific semantic and syntactic knowledge more effectively than models trained on general text. These models have achieved state-of-the-art performance in various medical NLP tasks with limited domain-specific training data through transfer learning and efficient fine-tuning strategies. Table 38 summarises the usage of various metrics for evaluating PLM-based methods of various NLP tasks in medicine.

However, future work remains full of opportunities and challenges, needing to focus on reliability, interpretability, bias elimination, and clinical integration. As the importance of PLMs in medical NLP continues to grow, verifying their reliability across different datasets and patient groups becomes crucial. Developing methods to explain the behaviour and predictions of PLMs is vital for building trust and extending applications. Moreover, there is still significant room for improvement in model efficiency to accelerate real-time inference for immediate application.

Overall, PLMs demonstrate tremendous potential in assisting clinicians by mining information from text-based medical data. The key to realising this potential lies in close collaboration among NLP researchers, healthcare professionals, and clinical institutions to transition these models from the laboratory to clinical practice. We hope this survey inspires and guides future efforts to fully unleash the potential of PLMs in the medical field and ultimately improve healthcare delivery.

CRediT authorship contribution statement

Xudong Luo: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Zhiqi Deng:** Writing – review & editing, Writing – original draft, Visualization, Investigation. **Binxia Yang:** Writing – original draft, Resources. **Michael Y. Luo:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work we used ChatGPT and Grammarly in order to improve our English and used Bing Chat to help us to collect relevant references. After using this tool/service, we reviewed and edited the content very carefully as needed and take full responsibility for the content of the publication.

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